

PHILIPPINE ARMY MANUAL 3-0502-01

TANK OPERATIONS

Proponent: Mechanized Infantry Division, Philippine Army

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LT GEN ROLANDO JOSELITO D BAUTISTA AFP
Commanding General, Philippine Army

Preface

1. This manual establishes the doctrine for planning, preparing, executing and assessing and evaluating the conduct of tank operations within the Philippine Army and joint operations with other services.

2. This manual provides principles, techniques, tactics, and procedures for employment of tank units and an authoritative reference for doctrine, materiel, force structure, training and SOP development.

3. It covers the operational principles in the employment of tank assets. It also provides commanders and staff with guidance in planning and executing tank operations.

4. The proponent of this manual is Mechanized Infantry Division, Philippine Army. However, Doctrine Center is responsible for the review and updating of this publication. Send comments or recommendations to the the Director, Doctrine Center, TRADOC, PA at Camp O'Donnell, Sta Lucia, Capas, Tarlac 2315 or via email at dcannex45@gmail.com. Feedbacks to this publication are also welcome telephonically via the Bonifacio Telephone Exchange trunk line number 845-9555 or military local line number 765 extension 4971.

5. References

   The following references may be used for further reading:

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   f. PAM 3-041 Artillery Fire Support to Brigade Operations, 03 March 2010.
   g. PAM 4-00 Philippine Army Logistics Manual, 13 November 2007.
   h. PAM 3-06 Philippine Army Engineers Operations Manual, 13 November 2007.
   i. PAM 3-062 Engineer Counter Mobility Operations, 03 March 2010.
   j. FM 3-90.1 Tank and Mechanized Infantry Team, 09 December 2002.
   k. ADRP 3-07 Stability, August 2012
   l. ADRP 3-90 Offense and Defense, August 2012
   m. ATP 3-20.15 Tank Platoon , August 2012
   n. FM 17-15 Tank Platoon, April 1996

6. All publications, manuals and directives inconsistent with this manual are hereby rescinded.

7. Unless this publication states otherwise, masculine nouns and pronouns do not refer exclusively to men.

Content **Page**

Chapter I- Introduction……..……..... 1
The Operational Environment……. 1
The Potential Threat Array…….. 3
Chapter II – Fundamentals of Tank Operation..... 4
Roles of Tank Units..... 4
Mission……..... 4
Basic Tasks for Tank Units……..…… 5
Tank Capabilities and Limitations..... 5
Tank Organization……..…..…… 7
Tank Employment..... 11
Tank Formations..... 16
Direct Fire Control……... 22
Position Location Devices..... 29
Readiness Conditions..... 31
Operations Security…….. 32
Chapter III- Offensive Operations...…….…..……... 36
Battalion in the Offense……..…… 37
Company in the Offense……...……... 43
Tank Platoon in Offensive Operations……. 54
Methods of Tank/Infantry Movement……. 57
Fire and Maneuver……... 59
Battle Drills…….. 61
Action at Danger Areas…… 69
Types of Danger Areas…… 69
Movement To Contact……. 72
Methods of Attack…… 75
Type of Attack……. 76
Offensive Operation in Built-up Areas…… 78
Raid…….. 86
Bypass…….. 87
Consolidation and Reorganization……... 87
Tank Platoon during Limited Visibility…..……. 88
Chapter IV –Defensive Operations……..……... 91
Purposes of Defense...…..... 91
Defensive Tasks…..... 92
Tank battalion in Defensive Operations…… 96
Tank Company in Defensive Operation…….. 99
Tank Platoon in Defensive Operations…… 110
Direct Fire Planning……. 110
Defense Preparation……. 112
Defense of a Battle Position……. 114
Defense in Sector……. 115
Reverse Slope Defense…… 116
Perimeter Defense…… 119
Strongpoint Defense……. 120
Defense in Built-up Areas…… 121
Reserve Unit……. 123
Counterattack Unit…… 124

Tank Platoon Defense during Limited Visibility……. 125
Glossary……..... 127
Abbreviation and Acronym……..…… 132
Figure Lists...……...
Figure 2.1 – The Tank Platoon…… 8
Figure 2.2 – The Tank Company …… 8
Figure 2.3 – Tank Battalion…… 10
Figure 2.4 – Headquarters Company……... 10
Figure 2.5 – Scout Platoon…….. 11
Figure 2.6 – Mortar Platoon……. 11
Figure 2.7 – Wedge Formation…… 17
Figure 2.8 – Vee Formation…… 18
Figure 2.9 – Echelon left and echelon right formation…… 19
Figure 2.10 – Line formation……... 20
Figure 2.11 – Column and Staggered column formation …… 21
Figure 2.12 – Coil Formation……... 21
Figure 2.13 – Herringbone formation…….. 22
Figure 2.14 – Frontal Fire…… 23
Figure 2.15 – Cross Fire……... 24
Figure 2.16 – Depth Fire……. 24
Figure 2.17 – Combat column formation……. 26
Figure 2.18 – Wedge Formation…….. 26
Figure 2.19 – Modified wedge formation…… 27
Figure 2.20 – Vee formation (infantry truck mounted)…… 27
Figure 2.21 – Line Formation…….. 28
Figure 2.22 – Column…….. 29
Figure 2.23 – Wedge…… 29
Figure 2.24 – Vee……. 29
Figure 2.25 – Readiness condition……... 32
Figure 3.1 – BCT/TF/BDE movement to contact as lead for a division attack…… 38
Figure 3.2 – Tank task force in a hasty attack…….. 39
Figure 3.3 – Task force in a deliberate attack(breach)……. 40
Figure 3.4 – Task force in a deliberate attack(assault)……. 41
Figure 3.5 – Raid with Tanks……... 43
Figure 3.6 – Company movement with escape routes……. 44
Figure 3.7 – MTC Tanks company as part of a brigade Vee……... 45
Figure 3.8 – MTC Tanks company as part of a brigade Column……. 46
Figure 3.9 – Example of clover leaf technique…… 46
Figure 3.10 – Infantry brigade/Tank company hasty attack example1…… 47
Figure 3.11 – Infantry brigade/Tank company hasty attack example 2…... 47
Figure 3.12 - Infantry brigade/Tank company hasty attack example3……. 47
Figure 3.13 – Infantry brigade/Tank company deliberate attack example1. 48
Figure 3.14 – Infantry brigade/Tank company deliberate attack example2. 49
Figure 3.15 – Tank company enveloping a built up area……. 50
Figure 3.16 – Blocking positions around a built up area……. 51
Figure 3.17 – Search for potential Target…… 57
Figure 3.18 – Tank formation with mounted infantry……. 58
Figure 3.19 – Fire and Maneuver……. 59
Figure 3.20 – Attack by Fire…… 61

Figure 3.21 – Change of formation drill…….. 62
Figure 3.22 – Action Left Drills……... 63
Figure 3.23 – Action right Drills…….. 63
Figure 3.24 – Action rear Drills……... 64
Figure 3.25.a – Use of action drill to react to enemy contact (action front from a column)…… 64
Figure 3.25.b - Use of action drill to react to enemy contact (action rear from a wedge or line)……. 64
Figure 3.25.c - Use of action drill to react to enemy contact (action left from a wedge or line)……... 64
Figure 3.25.d - Use of action drill to react to enemy contact (action fright from a staggered column)…… 64
Figure 3.25.e - Use of action drill to react to enemy contact (action right from a Vee)…….. 65
Figure 3.26 – React to indirect fire drill…….. 65
Figure 3.27 – React to air attack drill…….. 66
Figure 3.28 – Aiming point for engaging enemy aircraft…… 67
Figure 3.29 – Break contact drill…… 68
Figure 3.30 – Attack an objective drill……. 69
Figure 3.31 – Example of action at a danger area with tank overwatching infantry…… 70
Figure 3.32 – Breaching an obstacle…… 71
Figure 3.33 – Example of a tank platoon conducting a hasty attack on a fixed enemy force during search and attack operation……. 73
Figure 3.34 – Movement to conduct using the approach march technique (battalion vee)…….. 73
Figure 3.35 – Movement to condact (wedge)…….. 73
Figure 3.36 – Movement to contact (column)…….. 74
Figure 3.37 – Tank platoon attacking by fire……... 75
Figure 3.38 – Tank platoon and infantry advance together……. 76
Figure 3.39 – Isolating the built up area…….. 80
Figure 3.40 – Concentrate fire to gain a foothold…… 80
Figure 3.41 – Enveloping the assault forces…… 81
Figure 3.42 – Tank advancing with infantry…… 82
Figure 3.43 – Isolating and seizing a building……. 82
Figure 3.44 – Example of tank dead space on narrow streets…….. 85
Figure 3.45 – Bypass of an enemy position……. 87
Figure 4.1 – Delay from successive position……... 93
Figure 4.2 – Delay from alternate position…….. 94
Figure 4.3 – Battalion defending in sector……... 97
Figure 4.4 – Battle position defense…… 98
Figure 4.5 – Company direct fire plan……. 101
Figure 4.6 – Infantry brigade sectors showing example of tank…….. 103
Figure 4.7 – Tank company supporting a strong point defense……... 104
Figure 4.8 – Tank company supporting a perimeter defense……... 105
Figure 4.9 – Infantry brigade anti-armor defense……..…….. 106
Figure 4.10 – Counter attack by fire…… 109
Figure 4.11 – Counter attack by fire and maneuver…… 109
Figure 4.12 – Tank platoon sector sketch…… 111

Figure 4.13 – Tank platoon in a battle position……... 114
Figure.4.14 – Infantry battalion anti-tank defense…….. 115
Figure 4.15 – Organization of a reverse slope defense…… 116
Figure 4.16 – Adaptation of the reverse slope defense…… 119
Figure 4.17 – Example of infantry urban defense with tanks…….. 122
Figure 4.18 – Counter attack by fire…… 124
Figure 4.19 – Counter attack by fire and maneuver…… 125
Table Lists.……...
Table 2.1 – Infantry battalion mission and Tank Platoon Task……... 12
Table 2.2 – Infantry Brigade mission and Tank company Task…….. 14
Table 4.1 – Factors in battle position and sector defense……...… 96

Chapter I
Introduction

0101. The development of tanks dates back from the days of chariots and siege engines that relied on animal or human power for their propulsion. These types of machines eventually became obsolete due to more sophisticated tactics in wars and battles and there was very little invention in terms of propulsion for hundreds of years. However, the advent of steam inspired the creation of some military vehicles.

0102. One example was James Cowan's steam 'battle car' of 1855 which never gained any popularity because it was deemed uncivilized. Of course, despite strong ideas of the correct way to fight, the mechanization of the military was inevitable. The impact of technology to the military started to become clear as more and more armies began using steam tractors to transport troops and weapons. However, petrol/gas engines were more efficient and versatile and, by the start of the Twentieth Century, guns and armor were being added to commercial automobiles.

0103. Armored cars were developed during the years leading to the First World War. After a year of fighting it became obsolete because of trench warfare making the war a very defensive affair that these cars were useless. Something better needed to be invented. Armies wanted new ways of traversing trenches, shell holes, and barbed wires as the infantry endured heavy losses fighting in these circumstances.

0104. Tank evolution in the Philippine Army (PA) started from a simple M1 Scout car to Renault FT 17 which was intended only for training.The earliest tank battles occurred in the Philippines on December 1941 in Bataan and Corregidor where M3s put up a fierce resistance with the 192nd and 194th Light Tank Battalions, mostly against Ha-Go tanks and Shinhoto Chi-Ha Tanks of Japan. For the first time, tanks and armored vehicles were used in the Philippines replacing the horses. Units using them also included the 17th Ordnance Company (Armored), United States Army, which later became the Provisional Tank Group, USSAFE using M3 Stuarts**,** Universal Carriers and M3 Gun Motor Carriages and the 2nd Provisional Tank Battalion of the 2nd Regular Division, PA, USSAFE also using the Bren gun carriers.

0105. After the war, M4 Shermans, M41 Tanks, and Scorpion Tanks among other vehicles, formed the basis of the armored and mechanized forces of the Philippine Army. The 1st and 2nd Light Tank Companies were raised in 1946 to provide the starting blocks for armored warfare in the country. Later, a tank training company was formed in Pampanga to train PA servicemen in armored warfare techniques.

The Operational Environment

0106. The future Operational Environment (OE) presents Army maneuver forces with complex and challenging conditions. The OE remains difficult to predict and is subject to rapid and even radical changes. The Philippines, as an area of operations, poses eight (8) operational environment conditions that will influence the success of tank force’s mission.

0107. The archipelagic characteristic of the physical operating environment dictates that most military operations will likely happen across a non-contiguous land domain that will require a capability to operate in this kind of condition.

0108. The geographic location of the Philippine territory along the typhoon belt and ring of fire makes the Philippines a consistent witness to severe weather conditions, volcanic and seismic activities that bring calamities and disasters. Therefore, there is a need for armor forces to be trained and equipped for Humanitarian Assistance and Disaster Response Operation.

0109. The high probability of operating in highly urbanized areas such as metropolitan centers requires a large number of tank units supported with high level of armor protection, firepower, and mobility complemented with enhanced situational awareness for day and night operations whether in normal or extreme weather conditions. These should be interfaced with communication, target acquisition, and fire control systems to identify focus areas and minimize casualties and collateral damage.

0110. A universally-connected environment allows greater interactions and information that could instantly reach people. This has influenced moods and swing in the dynamic of geo-politics and the socio-cultural landscape therefore, the action and behavior of nations has become more integrated and dependent on each other which makes the maintenance of national and regional security more precarious and demanding.

0111. A strength as well as a vulnerability, the Philippine Army Tank units now and in the future faces a great challenge in increasingly asymmetrical environment of the information age thus require to build a posture of strength through acquisition and upgrade of war machines and sub-systems to deter ground invasion. Being in a politically, economically, and socially constrained environment, tank units considers the effects of economic capacity, political will and social support in the exercise of its functions to attain military objectives.

0112. A collective environment which is a shared responsibility among the Government Agencies (GA), Non-Government Organizations (NGO) and partner nation, to preserve a territory is the most potent weapon in the full spectrum operation. The strength of tanks will depend on the level of support and cooperation provided by other actors in the operational area. In the same end, the projection of force relies on how unified a nation, an armed forces, an army or a maneuver unit is.

0113. The country and its neighbors have diverse culture unique in every region. The culture of people in a particular region impacts on the conduct of military operations within and beyond Philippine shores. Generating popular support is a challenge to Tank units in this kind of environment.

0114. The OE includes threats ranging from internal destabilizing threats of local dissidents and private militias, to a non-state asymmetric threats of local and foreign terrorists and to a technologically advance conventional threats of a foreign armed forces that might develop into a hybrid threats when terrorists supported insurgency, and terrorism is sponsored by another state. This may transform all this threats into a military formations experienced in close fighting who are well equipped, well led, well

trained, and dedicated to their cause. They routinely employ regular conventional armies supported by irregular forces and an ever-changing variety of conventional and unconventional tactics.

0115. Threats may possess weapons of mass destruction and technology allowing them to be disruptive over widespread areas. At the same time, the indigenous local population may remain uncommitted to the Philippine government until the National government through the military's strength, assures the population's security. This combination of fighting a determined enemy while securing the population will challenge future Philippine Army especially with regard to the tank formations and capabilities.

Potential Threat Array

0116. Because of its versatility, tank units can expect to conduct combat operations across the spectrum of contingencies against determined enemies equipped with modern weapon systems. These may include nuclear, biological, or chemical (NBC) weapons, forcing tank units to conduct operations in contaminated environments. In restrictive terrain, mine warfare becomes a likely threat especially with the wide variety of antitank mines available in the world market today. Additionally, use of scatter-able mines may constitute a significant advantage.

0117. Most engagements will be most likely at close range (300 to 800 meters) because of the types of restrictive terrain in which tanks will operate. The following is a list of potential common threats that tank units must be prepared to engage:

a. Non-armored targets: 1) Bunkers, Thick-walled Building occupied by the enemy, 2) Automatic weapons positions, 3) Antitank guided missiles (ATGM), 4) Enemy Troops, 5) Artillery, 6) Motor vehicles, and 7) Low flying Aircrafts or Helicopters.

b. Lightly armored vehicles

c. Tanks

Chapter II
Fundamentals of Tank Operations

Role of Tank Units

0201. Tank units will use their unique capabilities to conduct combat operations, often in support of contingency plans, across the operational continuum (peacetime, conflict, and war). They will be required to operate in a wide range of political, military, and geographical environments. Their tactical missions include providing security, reconnaissance, and anti-armor firepower to the division, as well as standard tank operations to engage and destroy enemy forces using mobility, firepower, and shock effect in coordination with other combat arms. These missions may require rapid strategic and tactical deployment nationwide.

0202. Tank units are most likely to be employed as part of contingency operations which dictate an increased role for tanks operating with infantry and mechanized infantry forces. Tanks significantly reduce the risks infantry and mechanized infantry forces face during contingency operations.

0203. Tanks increase the operational force's mobility, protection, and lethality immediately upon deployment. They provide accurate, destructive fires that the operational commander can use to shape the battlefield, defeat the enemy, or fill the gap until other tank units arrive. Tank forces support the ground force commander's requirement to conduct operations in wartime as well as in operations other than war. They can execute peacetime contingency operations aimed at influencing enemy decisions through political and psychological shock effect, or they can engage and defeat the enemy in combat.

Mission

0204. The mission of tank units is to close with the enemy and neutralize them by fire, maneuver, and shock effect in coordination with other arms. They may also conduct Internal Security Operations (ISO) activities as part of a task force in peacetime environment as directed.

0205. Tank units may be required to support the following missions:

a. Operations other than war, such as Counter-insurgency

b. Combating terrorism

c. Shows of force and demonstrations

d. Strikes and raids

e. Peacekeeping and peace enforcement

f. Rapid reinforcement of forward-deployed forces

g. Wartime contingency operations

h. Humanitarian Assistance and Disaster Relief operations

i. Noncombatant evacuation operations (NEO)

j. Security operations

k. Other operations as specified by higher commanders

Basic Tasks for Tank Units

0206. These missions will require separately supportable tank units to conduct a variety of tasks, including:

a. Close assaults with infantry.

b. Reduction of strongpoints, destruction of bunkers and other hard targets, and roadblocks.

c. Operations in built-up areas (BUA).

d. Defense with infantry

e. Force security.

f. Flexible, mobile reserve operations for the Brigades or BCTs, Division, and Unified Commands to provide rapid response to enemy mounted forces.

g. Rear area operations.

Tank Capabilities and Limitations

0207. Tank units have the following capabilities:

a. Support the close fight as part of a combined arms force using accurate anti-tank fires, direct and indirect fires.

b. Excellent cross-country mobility, sophisticated communications, enhanced target acquisition, lethal firepower, and effective armor protection produce high degree of firepower, mobility, armor protection, and shock effect.

c. Operate on a conventional or NBC battlefield.

d. Use thermal sights to greatly enhance the night fighting capabilities of the combined arms team.

e. Operate in an opposed entry role.

f. Use strategic and tactical mobility to advantage.

g. Provide armor protection against small arms, machine gun, and overhead artillery fire.

h. Detach quickly from their parent unit and be employed during initial stages of contingency or reinforcing operations.

i. Accept attachment of operational control (OPCON) of combat engineer units and self-propelled artillery units.

j. Deploy tailored tank and/or reconnaissance packages with inherent C2 and logistical support. This includes requirements to pre-position units in contingency areas of operations.

k. Accomplish rapid movement and limited penetrations.

l. Exploit success and pursue defeated enemy elements as part of a larger force.

m. Some support equipment, such as recovery and fuel vehicles, can deploy together with the support teams.

n. Attack or defend under hostile fire and during limited visibility condition.

o. Provide mobility, armor protection, firepower, and flexible communications to successfully exploit the integrated use of indirect fires.

p. Destroy enemy mounted or dismounted forces.

q. When reinforced, acts as a Division covering force in offensive or defensive operations.

r. Provide mortar fire support for organic and attached units.

s. On-board optics and sighting systems enable the crews to acquire and destroy enemy tanks, armored vehicles, and fortifications using the main gun or to use machine guns to suppress enemy positions, personnel, and lightly armored targets.

t. Use of visual signals and the single channel ground/airborne radio system (SINCGARS) facilitates rapid and secure communication of orders and instructions to allow tank crews to quickly mass the effects of their weapon systems while remaining dispersed to limit the effects of the enemy's weapons. The tank's armor protects crewmembers from small arms fire, most artillery, and some anti-armor systems.

u. Global positioning systems (GPS) and inertial position navigation (POSNAV) systems allow today's tanks to move to virtually any designated location with greater speed and accuracy than ever before.

v. Tanks can move rapidly under a variety of terrain conditions, negotiating soft ground, trenches, small trees, and limited obstacles. In addition,

0208. The limitations of tank units include the following:

a. The light and medium tanks do not possess the level of protection in the main battle tank.

b. Tanks are vulnerable to enemy infantry with anti-armor weapons and may not withstand the impact of enemy tank fires, attack helicopters (AH), missiles o antitank guided missiles (ATGM), anti-tank guns, and antitank mines.

c. Tanks are dependent on flatbeds for deployment to the combat theater of operations. They heavy equipment requires a large number of flatbeds during the strategic deployment of an entire battalion.

d. Consumption of supply items is moderate to high, especially in POL (Class III), Ammunition (Class V), and Repair parts (Class IX).

e. Infantry units do not have the organic transportation assets to support the tanks CSS needs, especially in Classes III, V, and IX.

f. Mobility and firepower are restricted in built-up areas, dense woods, and other extremely close terrain.

g. Reduced visibility leaves tanks vulnerable to dismounted infantry attacks.

h. Existing or reinforcing obstacles can also restrict or stop tank movement.

Tank Organization

0209. **Tank Platoon.** Although the tank platoon is an integral part of the tank company, it is capable of detaching from the company and then operating with infantry battalion. It is important for the tank platoon leader to become familiar with the organization and operations of the infantry. When the platoon operates with its parent tank company, its fundamental employment is similar to tank platoon. This chapter describes the employment of the tank platoon as it would apply to the support of infantry.

0210. The tank platoon consists of four Light to Medium or Main Battle tanks and its crew. Figure 2.1 shows platoon organization.

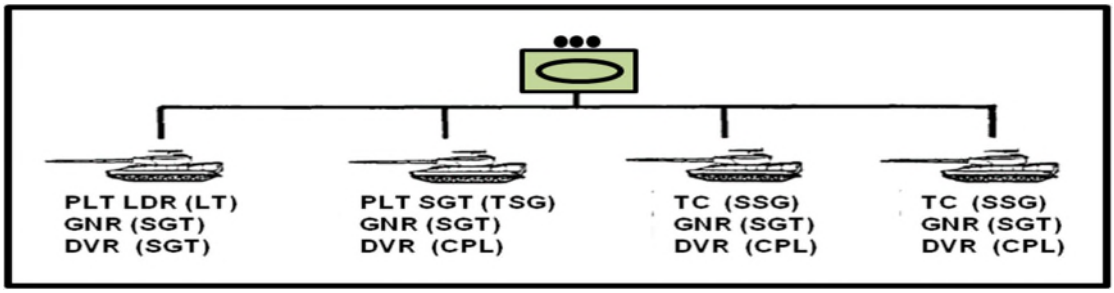


Figure 2.1 - Tank Platoon

0211. The tank platoon requires nearly the same CS and CSS as the tank company. Because this may not be routinely possible, tank platoons may have to fight with degraded maintenance, medical evacuation, and Classes III and V support. When task organized to an infantry battalion, the tank platoon should be augmented in those areas, based on the battalion's augmentation such as Maintenance team, Ammunition section, Fuel section and Supply section w/ water trailer.

0212. **Tank Company** is subordinate to the tank battalion, but it has the flexibility to task organize to an infantry brigade. When the company is employed with its parent battalion, its basic employment is similar to the tank company. This chapter describes employment of the tank company when it operates with an infantry brigade.

0213. The tank company's combat vehicles consist of 14 Tanks. The company is organized into three platoons of four Tanks each and a company headquarters consisting of two Tanks. See Figure 2.2 for a diagram of company organization.

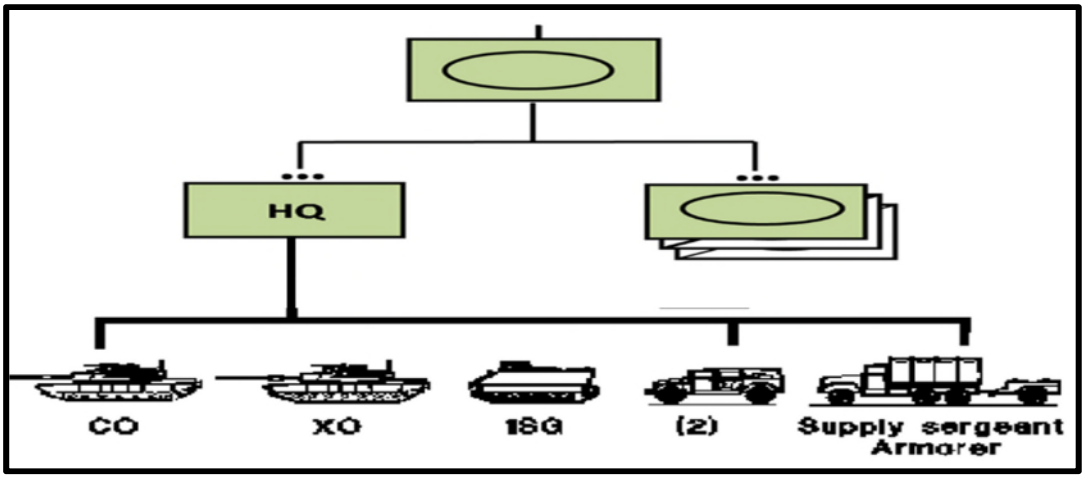


Figure 2.2 – Tank Company

0214. The task organization of a tank company to an infantry brigade is the norm. The brigade commander has the option to employ the tank company as a separate combat element or further task-organize tank platoons to infantry battalions. This sub-section will cover the employment of the tank company as a company or team operating with its parent tank battalion or task organized to the infantry brigade as a single combat element. The tank company can be organized as follows:

a. As a company under OPCON to an infantry battalion, or under brigade control, or under tank battalion control.

b. Partially task organized (one or two platoons) to battalions with the company headquarters and remaining platoon(s) retained at brigade control or task organized to a battalion.

c. Providing platoons to each of the three infantry battalions. The company commander may go to the battalion with the most critical mission. He can receive attachment of infantry and/or anti-armor platoons along with the tank platoon to form a team organization.

0215. The CSS capability of both tanks and infantry forces begins to degrade after one to three days. Infantry units have difficulty operating with tank units because the logistics support structure of infantry is generally austere. Tank units operating with infantry forces for more than three days require more detailed CSS planning and augmentation at all levels.

0216. Attachment requires the infantry brigade to operate with the tank company. To do this, the infantry brigade must receive attachments of tank battalion CSS assets. These should include Classes III and V, transportation assets, and tank maintenance assets.

0217. Normally, the tank company will come with CS and CSS assets. Much of this support can be standardized with the final determination based on the tactical situation. Commonly these are, but not limited to; Self-Propelled Fire Support Team (FIST), Armored Combat Engineer Platoon, Maintenance Team, Ammo Section, and Fuel Section.

0218. The Integrity of the tank company may be maintained to achieve mass, to facilitate C2, and to provide CSS. In some situations however, it will be necessary to attach tank platoons to infantry battalions. It will be prudent under most circumstances to maintain a tank reserve at brigade level.

0219. **Tank Battalion**. The tank battalion is supporting an infantry division. The Tank battalion can deploy as an organic unit, as companies, or platoons task organized to infantry brigades. The battalion may move into a theater as the majority of an infantry division is deployed. The Tank units may be organized as Task Force or Task Group when a large enemy armor threat exists and/or escalation of hostilities is expected. When the battalion functions as a TF, its basic employment is to counter enemy armor threats until heavier forces arrive in the area of operation (AO).

0220. The Tank battalion is organized into four Tank companies and a headquarters company. Figure 2.3 shows the battalion organization.

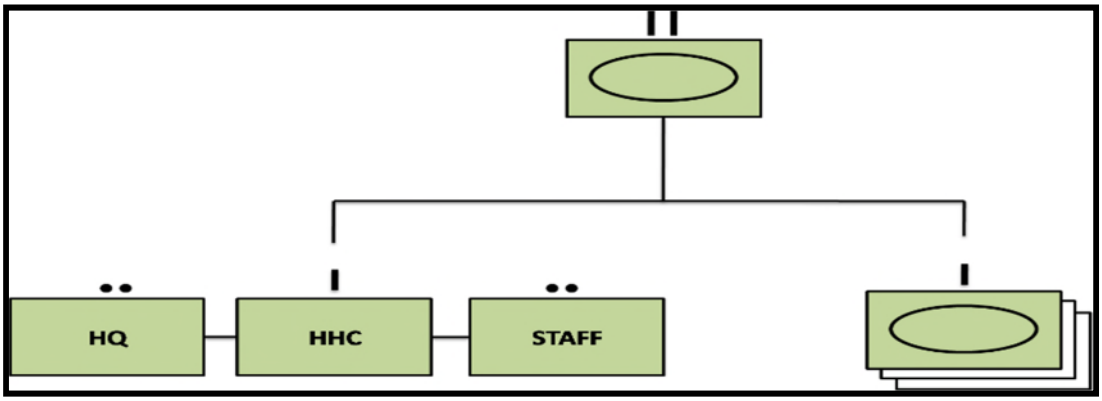


Figure 2.3 – Tank battalion

0221. **The Headquarters and Headquarters Company** (HHC) in a Tank battalion is organized as shown in Figure 2.4. Unlike the tank battalion, it must support four Tank companies that usually support three different infantry brigades. Many times, the platoons within the companies are supporting nine different infantry battalions within the brigades. Consequently, control of the support assets within the HHC are decentralized; strong leadership, flexibility, and effective planning are necessary to accomplish the support mission.

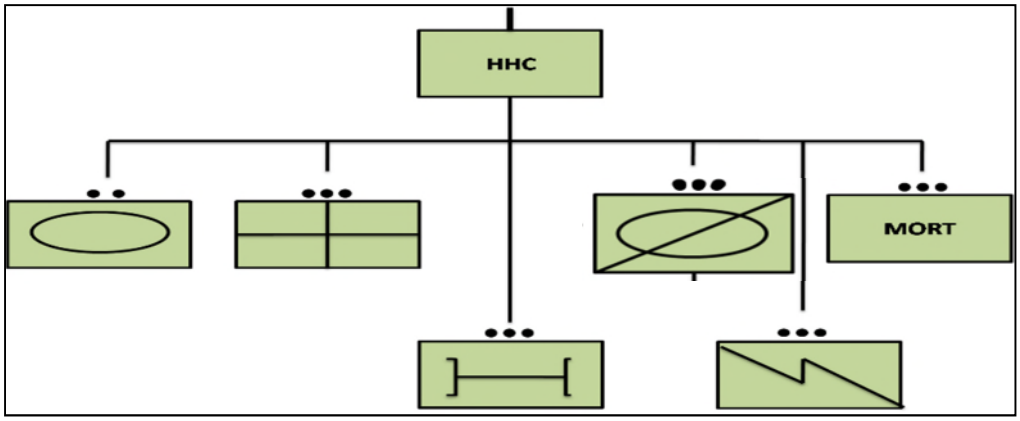


Figure 2.4 – Headquarters Company

0222. **Scout and Mortar Platoon.** The battalion scout platoon is organize as shown in Figure 2.5, performs reconnaissance, provides limited security, and assists in controlling movement of the battalion TF. The scout platoon is one of the commander's primary sources of combat intelligence before the battle and his eyes and ears during the battle. The platoon is not organized or equipped to conduct independent offensive, defensive, or retrograde operations. It operates as part of the battalion and should be assigned missions that capitalize on its reconnaissance capabilities. The mortar platoon is organized as shown in Figure 2.6 it's also organic to the battalion. It has high-angle, relatively short-range area fire weapons, and is well suited for providing close indirect Fire Support with a minimum of 120mm diameter projectile.

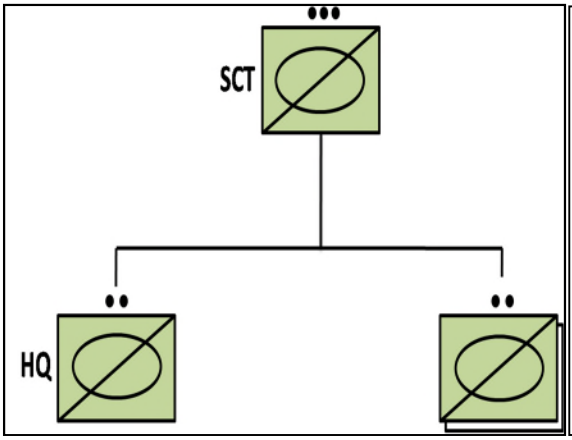


Figure 2.5 – Scout platoon.

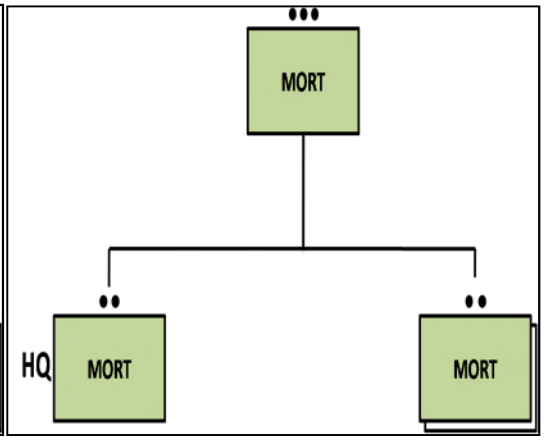


Figure 2.6 – Mortar platoon.

Tank Employment

0223. **Fundamentals.** Tank forces will provide combat capability throughout the battlefield and will be an integral part of the joint combined-arms contingency force. Tank platoons and companies may be employed in conjunction with non-mounted forces of squad through brigade size as mission requirements dictate. Tank battalions may also be employed as a division maneuver force and receive support from combat support (CS) and combat service support (CSS) units on the battlefield. Infantry can employ tank units as part of its security operations. Tanks and infantry can work together to conduct effective screening force operations, both offensive and defensive, to slow and direct the flow of enemy forces. Tank, even when outnumbered, can be used to shape the battlefield, causing the enemy to deploy its tank into engagement areas (EA) that can be targeted with aviation attack aircrafts. Tank units are also appropriate as reserves. Their mobility and firepower allow them to be used to strike the enemy at the critical time and place to seize or regain the initiative and ensure the destruction of the enemy force. Tank may also be required to perform standard security and reconnaissance missions. These may be conducted within the division AO alone when cavalry is not available or is not in sufficient quantities. These missions include:

a. Guard (with reinforcement).

b. Screen.

c. Zone or area reconnaissance.

d. Reconnaissance in force.

e. Route reconnaissance and security.

0224. **Tank Platoon**. The primary missions of the tank platoon are to move, attack, and defend. It may take on-additional tasks related to accomplishing its primary missions. The platoon operates as part of a tank company or task organized to an infantry battalion. Table 2.1 shows the infantry battalions' typical missions and the tank tasks that maybe required. When task organized to infantry, the tank platoon generally performs in two ways. First, the platoon may be employed as the primary maneuver element. Second, it may be in a direct FS role when infantry is the primary maneuver

element. The decision of which role is used depends on METT-TC. The tank platoon may also be used as a separate special platoon, or it may be attached to one of the infantry companies in direct support (DS). This chapter will address the concerns of the tank platoon leader under such conditions.

0225. The platoon is the first level where the tank unit leader must be trained to interact with an infantry controlling headquarters staff. Further, the platoon leader must simultaneously act as the tank unit advisor to the battalion commander; he must rely on the staff for immediate CS and CSS. If the tank platoon's company commander or XO is in the vicinity of the sector or zone, some assistance may be coordinated through that commander; however, this is not a certainty.

Table 2.1– Infantry Battalion missions and Tank platoon tasks.

Infantry Battalion Mission / Task	**Tank Platoon Tasks**
Movement to Contact	Overwatch Attack by fire Provide mutually supporting fire Screen Provide direct fire suppression Reserve Counter-attack
Attack	Maneuver as lead unit Provide suppressive fires Counter-attack Attack by fire Isolate an objective Security during consolidation Deceive enemy Screen Support or assault during breach Exploitation unit
Defend	Deceive enemy Screening unit Security unit Reserve Counter-attack Cover obstacles with long range direct fires
Delay	Overwatch Counter-attack by fire Deception Reinforce Reserve Counter-attack unit
Withdrawal	Screening unit Deceive the enemy Detachment left in contact (DLC) Rear guard Reserve

Raid	Deception Attack by fire Secure exit routes Fix enemy unit
Passage of lines	Overwatch Reserve

0226. **Tank Company**. The infantry brigade commander will establish the role and position of the tank company based on the factors of METT-TC and the current situation. The use of the tank company can vary based on the commander's estimated situation. The tank company may be employed as a maneuver, over-watch fire, or reserve force.

0227. As a maneuver element, the company is normally given the mission of seizing an objective. The objective may be undefended or occupied by an enemy force. The tank company should lead the attack against automatic weapons, anti-personnel mines, wire entanglements, and enemy tank or motorized forces. Tank and infantry can move together in the assault against entrenched infantry, jungle positions, heavily fortified areas, towns and villages, and during periods of low visibility.

0228. The Tank Company (or portions of the company) can provide over-watch or direct fires for the infantry maneuver or assault. The brigade or battalion (if task organized to battalion level) gives the company a support by fire or over-watch position for this mission. The company observes the enemy and provides information to the maneuver element. It places destructive or suppressive fires on known and suspected enemy positions, adjusts indirect fires to support the maneuver force, protects the maneuver force against counterattacks, and provides other assistance. Tank over-watches and supports by fire while infantry assaults:

a. Against constructed anti-tank defenses such as ditches, abatis, and minefield.

b. Through heavy woods.

c. Within cities, villages, and towns.

d. In mountainous terrain.

e. When the terrain is extremely restrictive along the approach axis to the objective.

f. When fields of fire and observation are adequate for long-range main gun and machine gun fires.

g. When enemy tanks on the objective must be suppressed and the only tank approach is frontal.

0229. The company may be designated as the brigade reserve and move in depth of the brigade formation. Its general location and possible missions are normally specified by the commander based on METT-TC. The commitment of this reserve is the most

critical decision of the brigade commander. The reserve may be assigned one or more of the following missions or tasks:

a. Assume the mission of the attacking unit.

b. Attack from a different direction.

c. Support the attacking unit by fire.

d. Provide flank security against an armored threat or on tank avenues of approach.

e. Protect or assist during consolidation on the objective.

f. Protect key intersections and bridges.

g. Block a counterattack.

0230. Detailed infantry missions and the corresponding tank company tasks that may be executed when operating with the infantry brigade are listed in the succeeding table 2.2

Table 2.2 – Infantry Brigade missions and Tank Company tasks

Infantry Brigade	Tank company
Movement to contact	Overwatch enemy avenues of approach; attack by fire; provide mutual supporting fire; provide direct-fire suppression on prepared positions; overwatch/assist in reducing obstacles; serve as guard, reserve or counter-attack force.
Hasty attack	Attack by fire; conduct an assault breach; serve as reserve; conduct exploitation.
Deliberate attack	Isolate objective; attack by fire; provide direct fire; deceive the enemy concerning the main effort; overwatch counter-attack routes; serve as reserve; conduct exploitation.
Exploitation	Lead the exploitation; overwatch; serve as reserve or security force.
Pursuit	Lead the direct-pressure force; overwatch; serve as reserve, enveloping, or security force.
Follow and Support	Provide direct FS; reduce obstacles; clear bypassed forces; secure key terrain and LOCs; provide rear area security.
Cover	Overwatch; screen and guard.
Defend in sector	Defend a BP; cover obstacles with long range fires; serve as security of reserve force; deceive enemy about main defense.
Breakout from encirclement	Serve as rupture force or rear guard.
Link-up	Serve as contact element or security. Overwatch or reserve force.
Delay	Overwatch; counter-attack by fire; conduct deception, reinforce; serve as reserve, or counter attack force.
Withdrawal	Serve as DILC; rear guard; reserve or covering forces; overwatch; fix enemy attack; conduct deception; occupy position in depth.
Retirement	Serve as security force

0231. **Tank Battalion.** The tank battalion can operate as a pure battalion or deploy in echelons as companies and platoons attached to infantry brigades and battalions respectively. If the entire infantry division deploys, part of the battalion may consolidate once in AO. The battalion headquarters then forms the nucleus of a tank battalion task group.

0118. The tank battalion provides the division commander with the following:

a. A highly mobile, protected potent ground combat force.

b. Shock effect.

c. Effective anti-armor capability.

d. Bunker- and building-busting capability and direct, close-in FS for dismounted infantry.

0232. The tank battalion may conduct operations as a battalion Task Group under the following conditions:

a. The enemy has a considerable mechanized or armor force.

b. A contingency mission has matured to a level in which the entire division has deployed to include the tank battalion.

c. Terrain favors the use of a larger maneuver force and/or supports long-range fires.

0233. The tank battalion can expect to execute the following missions:

a. Attack

b. Defend

c. Screen/Guard

0234. **Attack.** The battalion provides the infantry division commander with his most maneuverable, survivable, and potent ground force on the battlefield. He may use this capability either to lead the division, to fix the enemy force or as his maneuver force. Keep in mind that the battalion's speed is in relation to the speed of the predominantly dismounted force. The division commander can use this to his advantage by buying time to maneuver his infantry units. Infantry moving with the tank unit must be mobile. Truck transportation assets are available in the brigade however, if more than one company requires transportation, trucks must come from PA forward support.

0235. **Defend**. In open terrain, the battalion is most effective when operating as a total unit. In closed terrain, better suited for infantry, the battalion is most effective operating alongside the infantry as platoons and companies OPCON to the infantry battalions and brigades defending in depth.

0236. **Screen/Guard.** The division commander may give the tank battalion deployment priority if the enemy threat is predominantly mechanized or is in wide-open terrain allowing long-range fires. The battalion in this circumstance may be tasked to establish a screening force to provide security for the division as it flows into the AO. As the division builds in theater, the battalion's screening mission may end, and the companies and platoons within the battalion may augment the infantry brigades once again. In a major force buildup involving the arrival of cavalry, armored, or mechanized forces, the battalion may be relieved.

Tank Formations

0237. Formations are used to establish tank positions and sectors of responsibility during operations. Formations facilitate control; increase protection, speed, and fire effectiveness; and alleviate confusion. Formations are not intended to be rigid; tanks are not expected to be a specific distance apart. Position of each tank in the formation depends on the terrain and the wingman driver's ability to see the lead vehicle.

Tank Platoon Formations

0238. Individual vehicles should move in the same relative position within the formation. This will ensure that each crew knows where to move, who is behind them, and where to observe and direct fires. Gun tube orientation for rear and flank tanks should be modified to ensure 360-degree security based on the position of the platoon within the parent unit formation. The seven basic formations for the platoon are:

a. Wedge

b. Vee

c. Echelon

d. Line

e. Column

f. Coil

g. Herringbone

0239. **Wedge.** The wedge formation is employed when a platoon is provided over-watch by another element and terrain is open or rolling (see Figure 2.7). It is normally used when the enemy situation is vague and contact is imminent.

0240. The wedge has these advantages:

a. It permits excellent fire to the front and good fire to each flank when leading infantry formations.

b. It allows platoon leaders excellent observation up front while being covered by their wingmen.

c. It enhances control since leaders are in close contact and can easily relay hand-and-arm signals to each other and any following infantry formations.

d. The wedge has these disadvantages:

e. It requires lateral space for movement; therefore, it is difficult to use in closed terrain with dismounted infantry.

f. It may expose the entire platoon to enemy fire simultaneously.

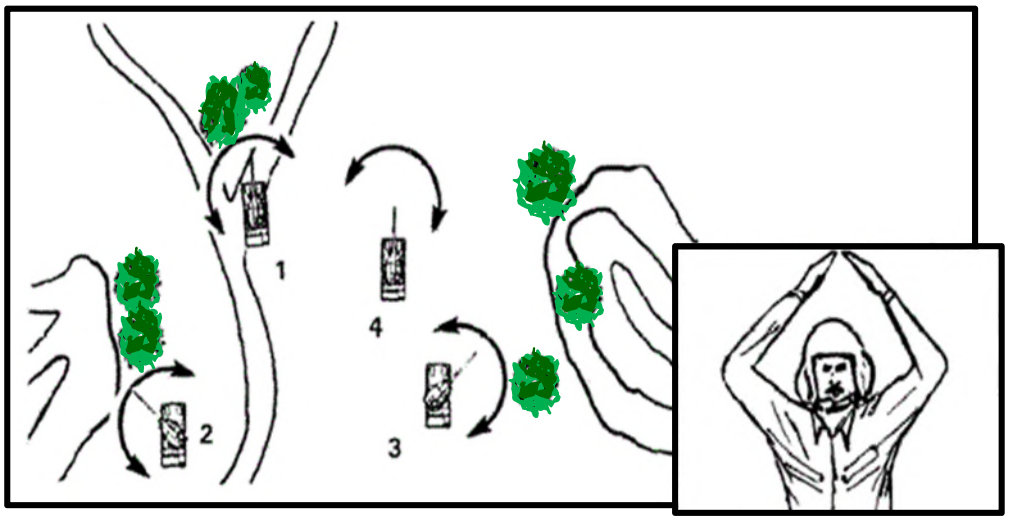


Figure 2.7 – Wedge formation.

0241. **Vee.** The platoon leader and his wingman lead, followed by the platoon sergeant's section. This formation is used when weather or terrain restricts movement or when overwatch within the platoon is required (see Figure 2.8). The vee has these advantages:

a. It provides excellent protection.

b. It provides excellent control.

c. It facilitates rapid deployment to any other formation.

0242. The vee formation has these disadvantages:

a. It limits fires to the front.

b. It is more difficult to maintain orientation than in a wedge.

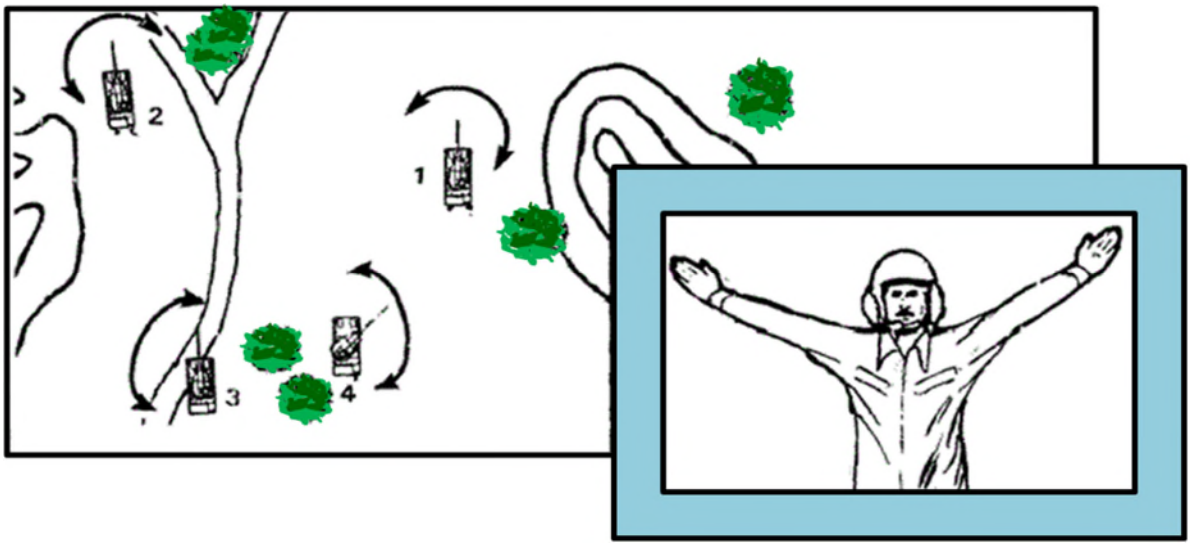


Figure 2-8. Vee formation.

0243. **Echelon**. The echelon formation is used to screen an exposed flank of a larger moving unit or the platoon (see Figure 2.9). It is also used when an infantry unit faces a significant flank threat, such as when it is bypassing a strongpoint or BUA. Echelons has these advantages:

a. They provide excellent firepower to the front and one flank.

b. They provide the best security to the higher unit formation.

c. They facilitate rapid deployment perpendicular to the direction of movement.

0244. The echelon formation has these disadvantages:

a. It is difficult to control in wooded terrain.

b. It is difficult to integrate into an infantry formation.

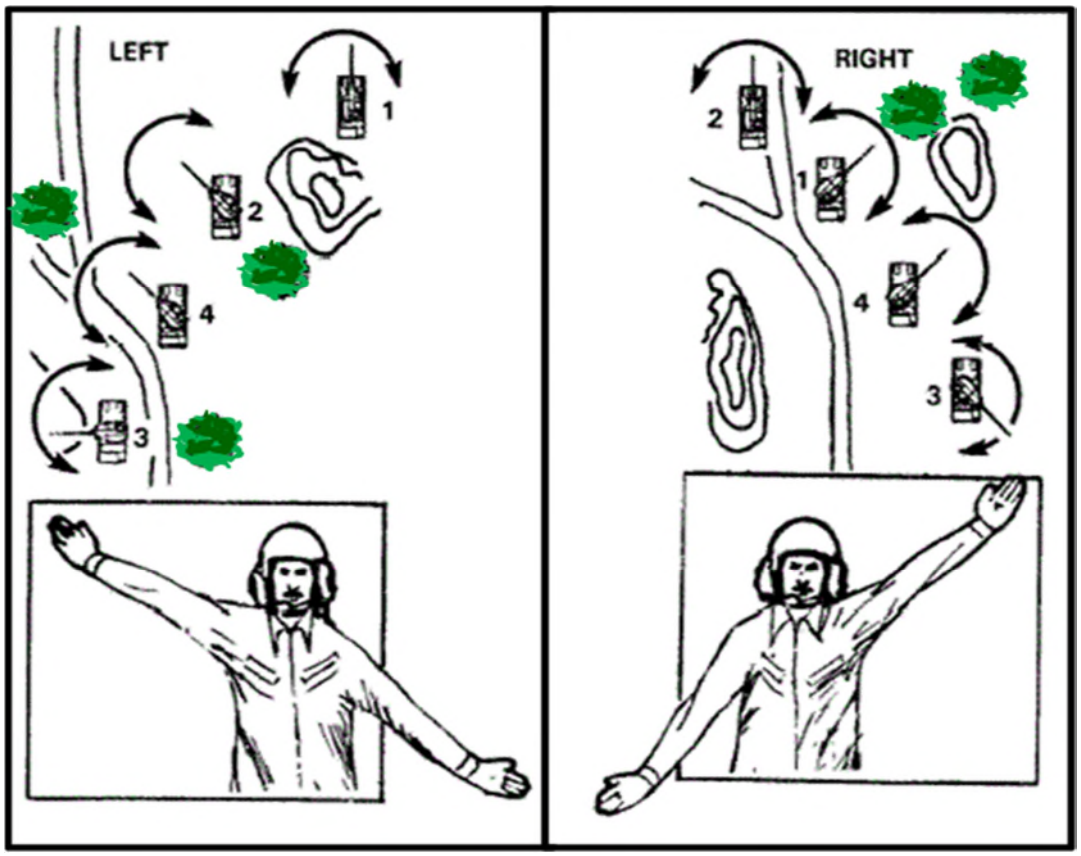


Figure 2.9 – Echelon left and echelon right formations.

0245. **Line.** A line formation is used to cross dangerous areas and assault a position (see Figure 2.10). It also facilitates mutual support when emerging from limited visibility conditions such as smoke or heavy woods. The line has these advantages:

a. It provides excellent firepower forward.

b. It provides protection to dismounted infantry.

c. Maximum vehicles can close on an objective in minimum time.

0246. The line formation has these disadvantages:

a. It provides minimum fire to flanks.

b. It is less secure than other formations because of lack of depth.

c. It is the most difficult to control.

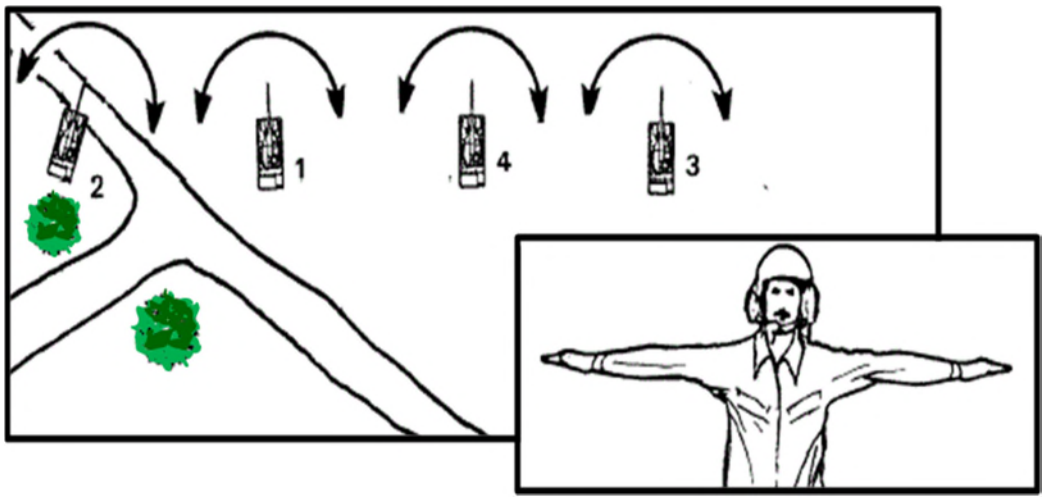


Figure 2.10 – Line formation.

0247. **Column/Staggered Column.** In these formations, the platoon leader positions himself where he can best control his elements. If he does not lead, he must ensure the lead tank commander is thoroughly familiar with the route of march and direction of travel. The column is used in night movements, in fog, when passing through defiles or dense woods, and during road marches when speed is required. The staggered column is used when terrain allows for dispersion (see Figure 2.11).

0248. The column or staggered column has these advantages:

a. It provides excellent control.

b. It provides excellent firepower to the flanks.

c. It facilitates rapid deployment to other formations.

d. It facilitates rapid movement.

e. The dispersion increases security against enemy air and artillery attack.

0249. A disadvantage of the column and staggered column is that they allow little fire to the front.

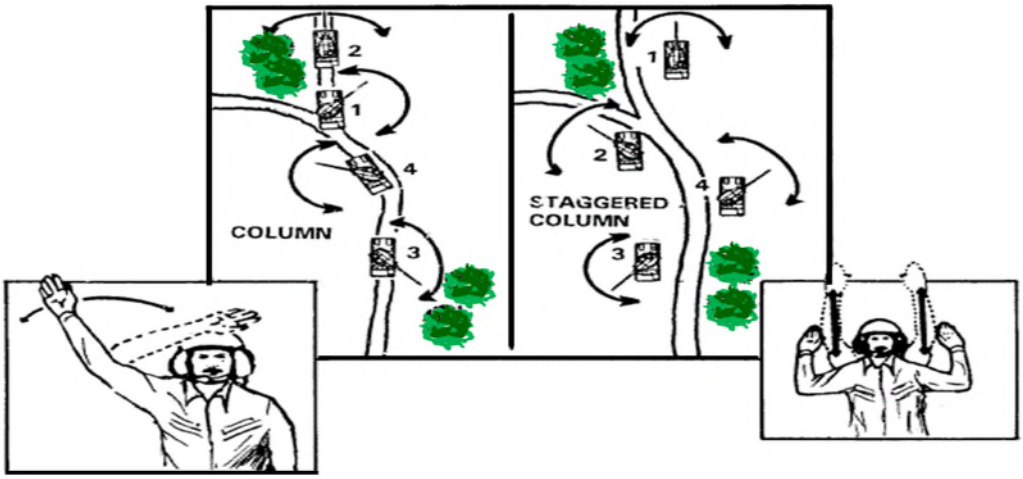


Figure 2.11 – Column and staggered column formation.

0250. **Coil.** The platoon can employ this formation when it is operating independently and experiences extended halts or lulls in combat (see Figure 2.12). The platoon leader positions his tank, and the remaining tank commanders position their tanks based on the terrain.

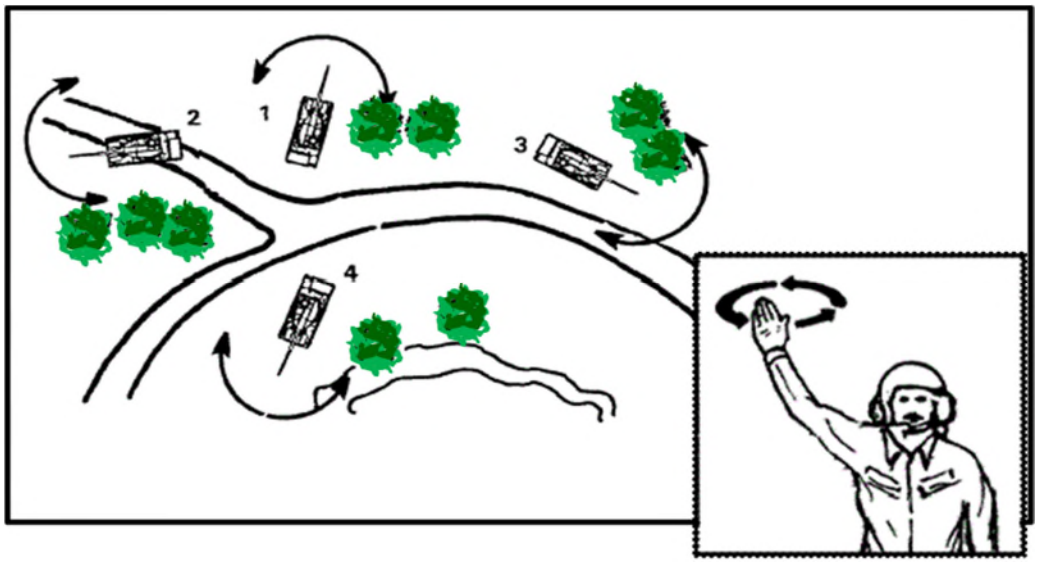


Figure 2.12 – Coil formation.

0251. The coil formation has these advantages:

a. It provides good all-around security.

b. It facilitates expanding to a perimeter defense.

c. It provides protection to infantry in open terrain.

0252. A disadvantage of the coil is that without infantry it offers only limited security in heavily wooded terrain.

0253. **Herringbone.** The platoon uses this formation when it assumes a hasty defensive posture or temporary halt on a road where terrain does not allow adequate

off-road dispersion. Tanks move off the road if terrain permits (see Figure 2.13). Infantry should dismount and seek cover and concealment while providing additional security and observation.

0254. The herringbone formation has these advantages:

a. Tanks can pass through the center of the formation.

b. It facilitates dismounting of infantrymen prior to further deployment of the tank platoon.

c. The herringbone has these disadvantages:

d. Tanks may be vulnerable to enemy air attack.

e. Security is limited (without infantry) and terrain does not allow for dispersion.

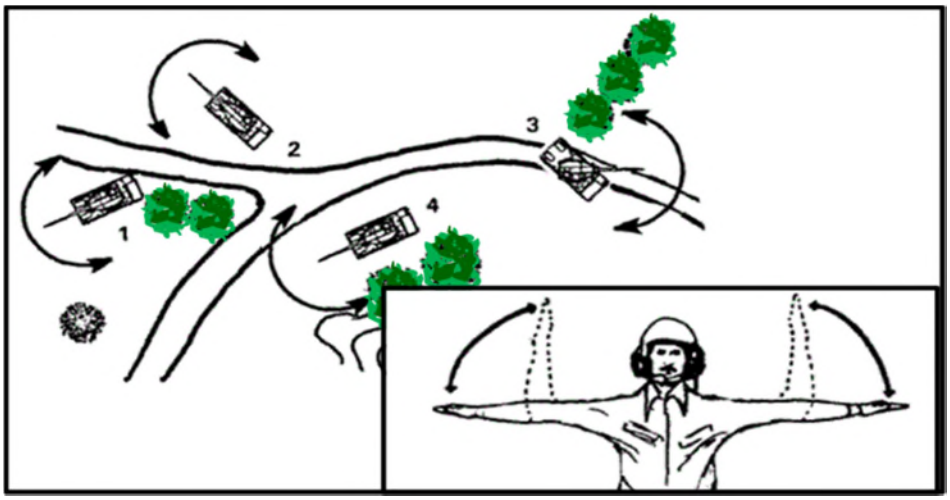


Figure 2.13 – Herringbone formation

Direct Fire Control

0255. Tank fires must be properly distributed and controlled to effectively support any operations, especially in close proximity to infantry. In most cases, this is accomplished by the vehicle commanders following established unit SOPs and direct fire plans. Occasionally, the platoon leader will be in position to direct the fires of the entire platoon.

0256. **Platoon Fire Plan.** The platoon fire plan provides the platoon leader with the necessary information to distribute and control the fire of all available direct and indirect weapons. The fire plan assists the platoon leader in determining how well the platoon has covered its assigned sector and in deciding which vehicle should move if shifts are required.

0257. **Control Measures**. The dynamics of the offense will normally require that fires can be controlled using the radio. Alerts and commands must be brief. Engagements will normally be initiated by the tank crew that sees the enemy. During defensive

operations, visual control measures can be used to start and stop engagements, shift fires, and signal prearranged actions.

0258. **Engagement Priorities**. A tank should always engage the most dangerous threat first. During offensive operations, the most dangerous threat will be the enemy antitank weapon systems. During defensive operations, platoons have freedom of choice concerning what to engage and when to begin engagements. The platoon leader should control tank fires so enemy C2 tank are engaged first. He can do this by assigning a tank or section the mission of observing for the C2 tank. Once they are acquired, the platoon leader initiates the platoon engagement.

0259. **Fire Patterns**. Three basic fire patterns will cover most situations and provide fast, effective fire distribution. No matter which method of engagement is used, the goal is to engage near and flank targets first, then shift fires to the center and far targets. Fire patterns are particularly effective when all tanks can see all of the enemy in the EA. The basic fire patterns are described in the following paragraphs.

a. **Frontal Fire**. Frontal fire is used when the enemy is dispersed laterally in relation to the platoon and all tanks are firing to the front. The left-most tank engages the left-most target, and the right-most tank engages the right-most target. The two center tanks engage targets to their direct front. When targets are destroyed, fires are shifted toward the center of the enemy formation (see Figure 2.14).

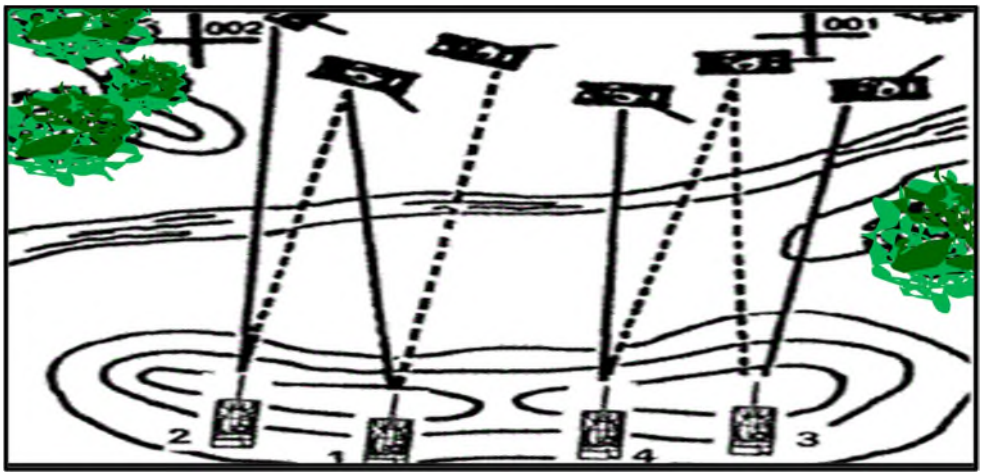


Figure 2.14 – Frontal fire.

b. **Cross Fire.** Cross fire is employed when the enemy is exposed laterally but obstructions prevent all tanks from firing to the front. The left-most tank engages the right-most target, and the right-most tank engages the left-most target. The two center tanks engage targets diagonal to their own positions. When targets are destroyed, fires are shifted toward the center of the enemy formation (see Figure 2.15)

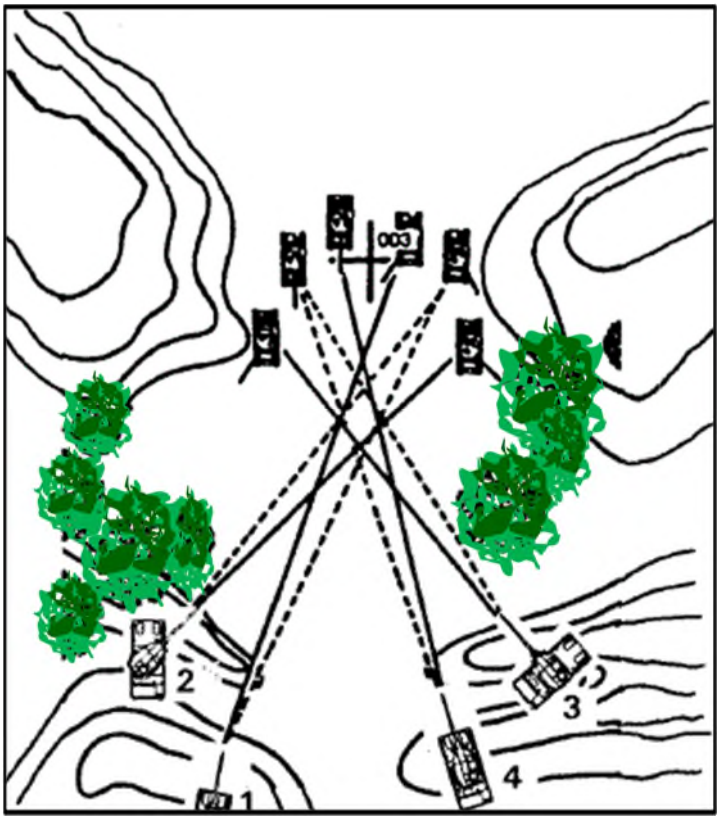


Figure 2.15 – Cross fire

c. **Depth Fire**. Depth fire is used when the enemy is exposed in column. The left-most tank engages the rear target, then shifts toward the center. The tank second from left engages a center target, then shifts toward the rear. The right-most tank engages the front target, then shifts toward the center. The tank second from right engages a target in the center and shifts toward the front of the enemy (see Figure 2.16).

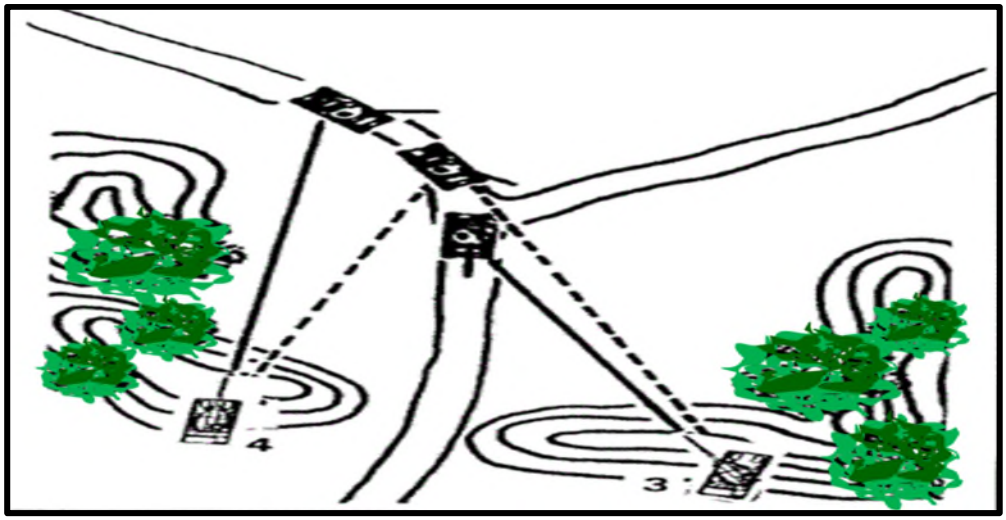


Figure 2.16 – Depth fire

0260. **Techniques of Fire Control**. In addition to employing fire patterns to distribute fires, platoon leaders may choose one of three firing techniques to control the direct fires of their units:

a. **Simultaneous fire.** This is the primary firing technique used by the platoon. It is employed during most offensive engagements when the unit encounters surprise targets. It is also used in most defensive engagements when the enemy array is numerous enough to require multiple engagements by each tank in the unit.

b. **Alternating fire.** Alternating fire is normally used when the platoon is in a defensive position or is undetected. Each tank alternates firing and observing with its wingman. Subsequent fire, by command, is simultaneous. During alternating fire, the wingmen are normally the first tank to fire. The platoon leader and platoon sergeant provide observation, then fire. The process continues until all targets are destroyed or the leader switches to simultaneous fire.

c. **Observed fire.** This is normally used when the platoon is in protective defensive positions and engagement ranges are in excess of 2,000 meters. The first tank to fire engages while the wingman observes. This technique allows for maximum observation and assistance while protecting the location of the observing tank. The observed tank must remain prepared to engage in the event, the firing tank consistently misses, experiences a malfunction, or runs low on ammunition.

Tank Company Formations

0261. The company commander uses formations to control movement. SOPs should standardize the company's use of reporting procedures, formations, movement, and C2 techniques. Basic battle drills and formations can be found in this Chapter. The company has five basic formations: combat column, wedge, vee, line, and echelon. When designating the company formation, the company commander must:

a. Establishes the relationship of one platoon to another on the ground.

b. Expresses where he envisions the enemy to be and how he intends to react to contact.

c. Establishes where firepower is needed.

d. Establishes how (mounted or dismounted) and where infantry travels in relation to the vehicles.

e. Establishes the degree of security desired.

0262. A combat formation is not rigid. Terrain and common sense will frequently dictate needed changes. Unless directed, it is not necessary for the company to use the same formation as the infantry brigade or battalion.

0263. **Column** (see Figure 2.17). This formation is used when speed is critical and the possibility of enemy contact is not likely. Infantrymen are usually mounted on light tanks or trucks (if available). A modified version, the combat column, can be used if the terrain allows for dispersion. The combat column can be used when contact is likely and speed is important. Infantryman, if APC-mounted, should not travel on the lead platoons vehicles.

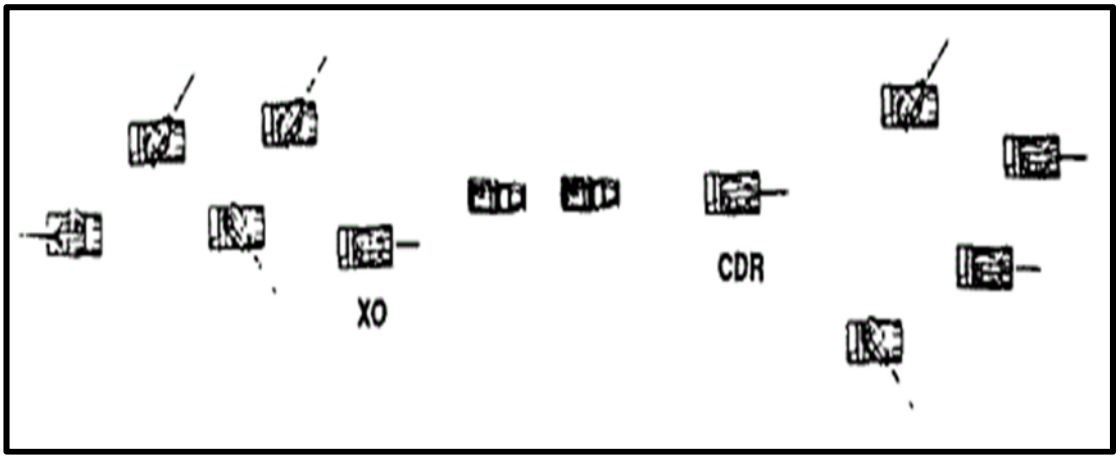


Figure 2.17 – Combat column formation

0264. **Wedge** (see Figure 2.18). The wedge is used when the situation is vague and contact with the enemy is imminent. This formation can be used to move in relatively open terrain to provide some protection to dismounted infantrymen. See Figure 2.19 for an example of a modified wedge formation.

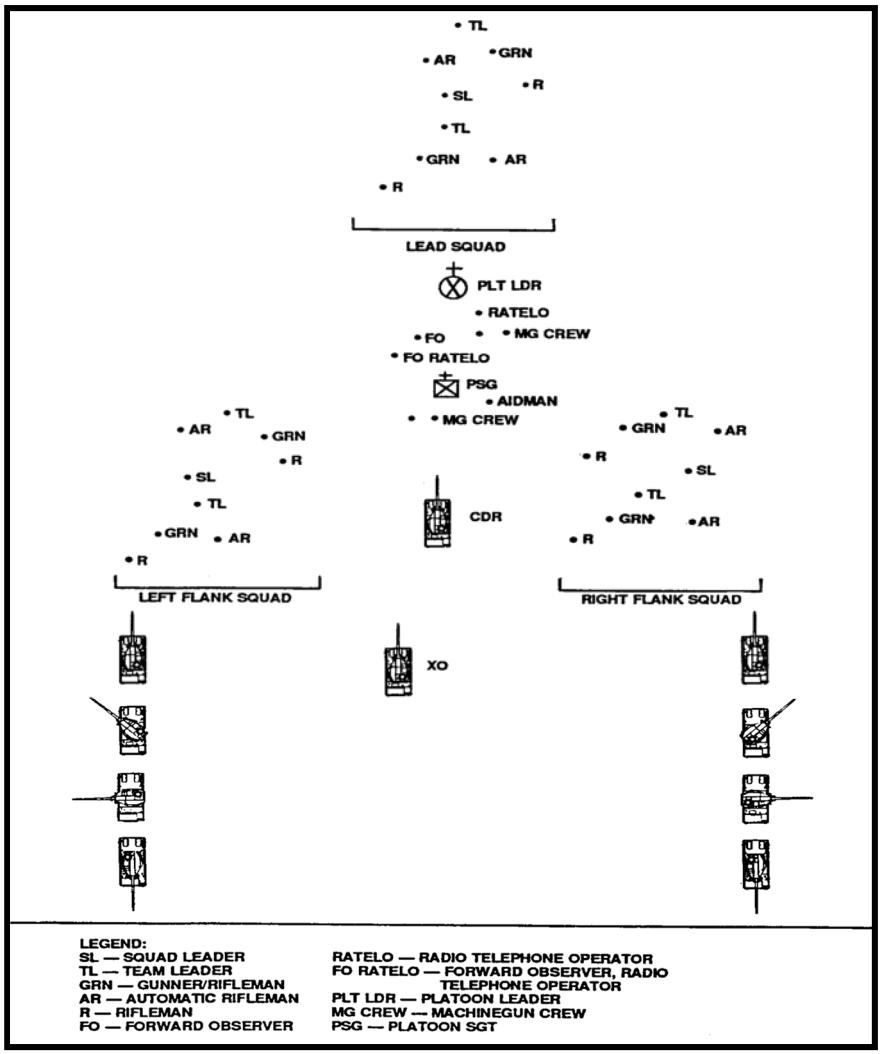


Figure 2.18 – Wedge formation

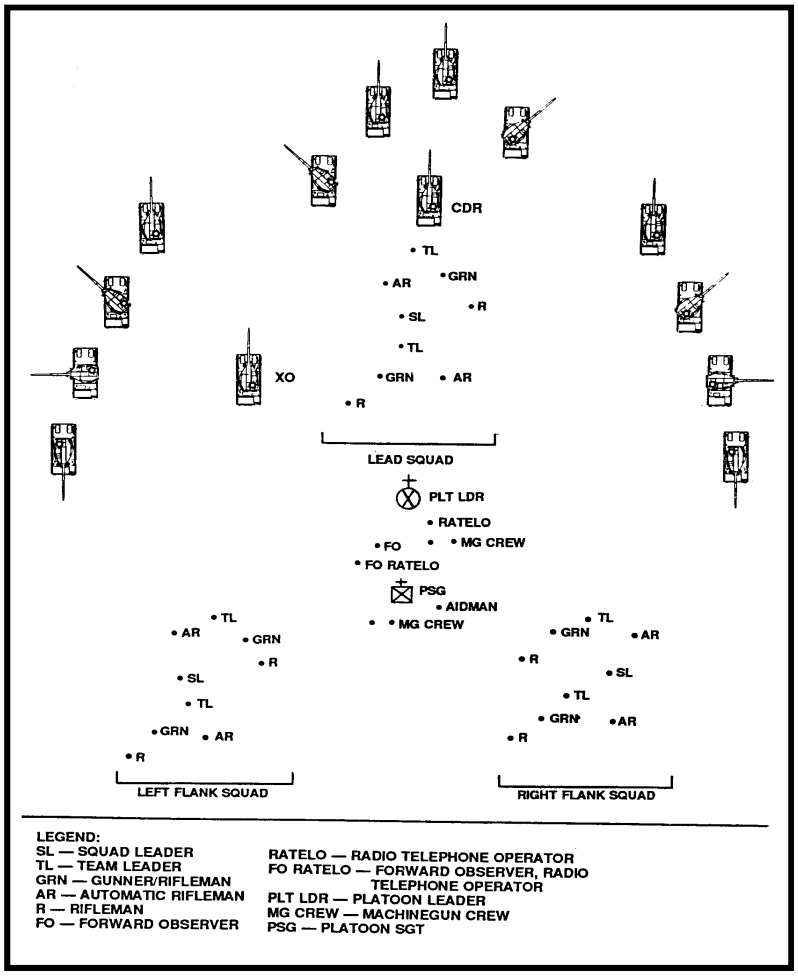


Figure 2.19 – Modified wedge formation

0265. **Vee** (see Figure 2.20). This formation is used when probability of enemy contact is high. It provides maximum firepower up front while leaving a platoon in depth to maneuver. This is a difficult formation to use when task organized as a team of two tank platoons and one infantry platoon.

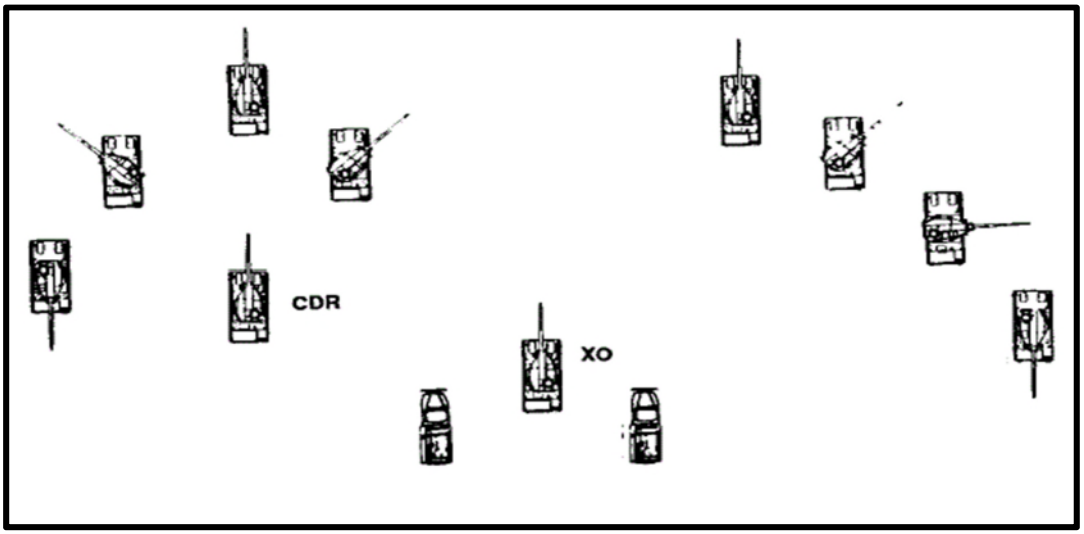


Figure 2.20 – Vee formation (infantry truck mounted).

0266. **Line** (see Figure 2.21). This formation is normally used, if terrain permits, when assaulting an objective. It can also be used to emerge from smoke or wood lines. In this case, platoons should use over-watch or support by fire techniques to cover each other. Infantry cannot move forward of the tanks when assaulting due to the danger of discarding-sabot ammunition.

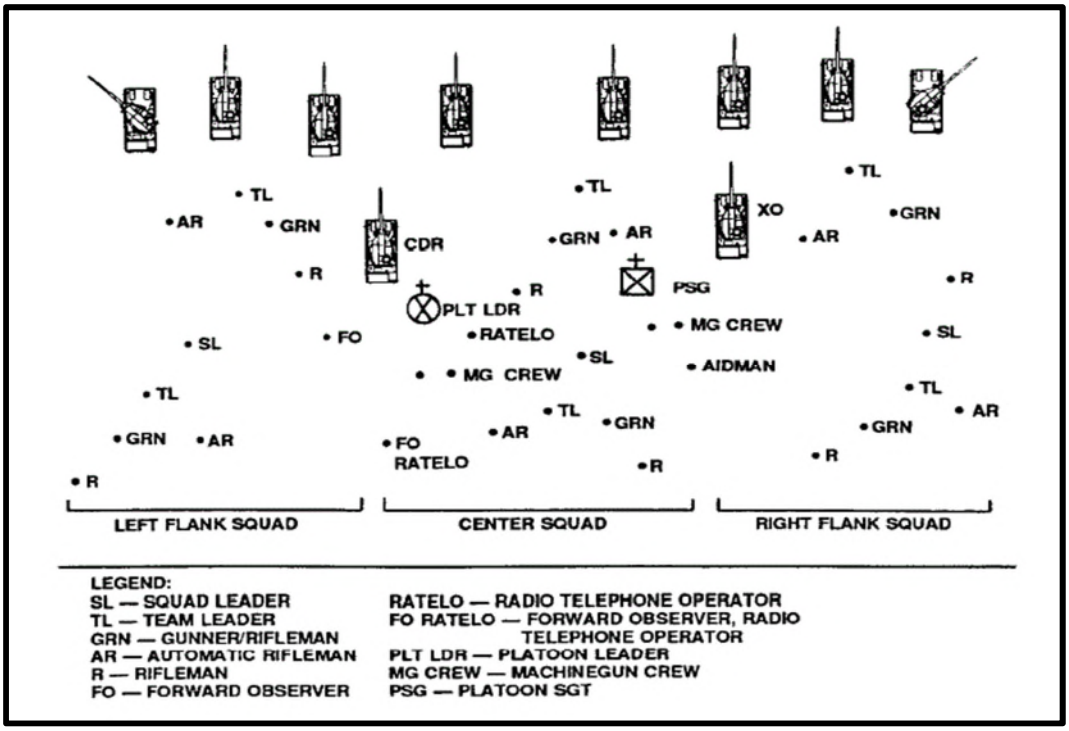


Figure 2.21 – Line formation.

0267. **Echelon.** The echelon is used when one flank faces a significant threat. Usually used in pure tanks. It is extremely difficult to use with dismounted infantry if there is contact to the opposite side of the echelon. If that happens, the tanks will not be able to conduct action drills without endangering infantrymen.

Tank Battalion Formations

0268. The tank battalion combined with other maneuver forces may move in any of the basic formations. It may use more than one formation in a given movement if the terrain changes. Infantry should be mounted on trucks (if available) or on follow-on forces company team vehicles if speed is important or the movement is long. Other factors such as distance between units are dependent on METT-TC. Figures 2.22 through 2.24 are examples of tank formations in a combined arms configuration of brigade or BCT or formerly known as the Task Force.

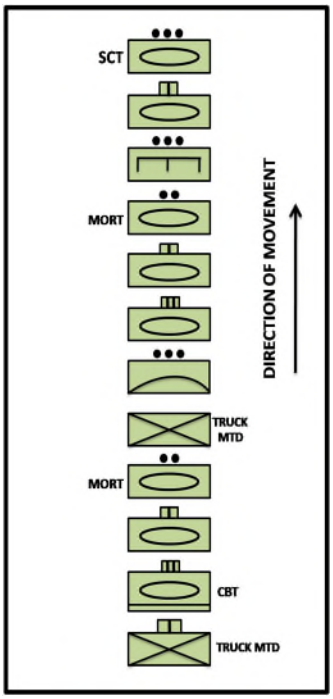


Figure 2.22 – Column.

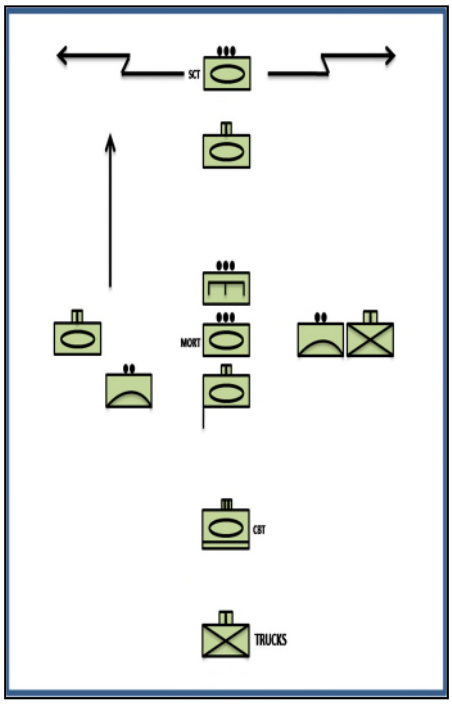


Figure 2.23 – Wedge.

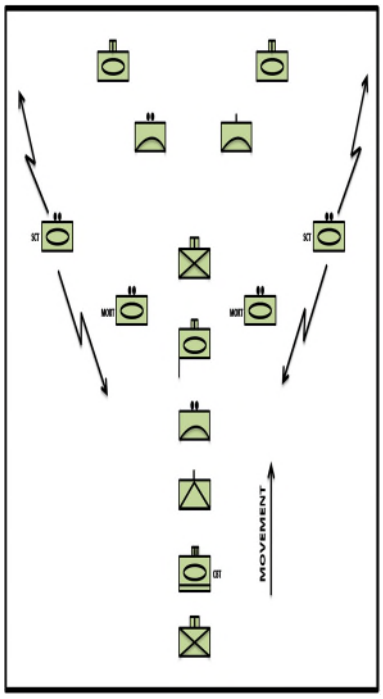


Figure 2.24 – Vee.

Position Location Devices

0269. Position location systems are an aid in navigation and should not take the place of map and compass navigation. Leaders must continue to use map and compass as a primary means in the event, position location system signals are interrupted by interference from vegetation, weather, or other masking features. There may also be times when satellites or land-based emitters are inoperative, causing lapses in signals. Several types of position location systems are available to receive signals from satellites or land-based emitters and then calculate and display the position of the user in military grid coordinates and/or latitude/longitude degrees. Some of the techniques described here refer to specific features that most devices will have. Some devices, however, may not be capable of all of the functions described.

0270. **Waypoints.** The navigational functions of position location systems are based on way-points. A waypoint is a known position entered into the system's memory by the user. Waypoints can be entered either as degrees of latitude and longitude or as military grid coordinates. The user can either enter waypoints or store the unit's position at various times.

0271. **Navigation.** Position location systems can be used to assist in navigation. To navigate, identify points along your route you wish to cross or at your destination. Next, enter these points as waypoints. You can then move from waypoint to waypoint and arrive at your final destination with great accuracy. Distance and direction (range and bearing) between two known points will be accurate only if they are first entered as waypoints. The system will store your present position and then compute the distance and direction to the known point. When you are ready to start your movement, get the range and bearing to the first waypoint. As you approach within a given distance of the waypoint, an alarm will sound, indicating that you have reached the waypoint. Display the range and bearing to the next waypoint.

0272. **Cross-country Navigation.** When navigating cross country, the system will direct movement from point to point. However, obstacles en-route may force detours from time to time. For detours of more than a few meters, the system can assist you in getting back on track. Some systems will display the distance you are off course, a new course to the waypoint, and an estimated arrival time based on your speed for the last two minutes. A left or right arrow shows the direction to the original track and a new bearing to the waypoint. If you need to reach the desired point and the route taken is of little importance, then you have merely to follow the ideated course. The course shown is the new direction to the way-point and will not return you to the original path. If you must return to the original path, you will turn in the direction of the arrow and travel the indicated distance until the system shows a correct reading.

0273. **Road March.** Position location systems can be useful on road marches in identifying checkpoints or coordination points on long routes that lack distinctive features. You enter a waypoint for the position of the checkpoint, select the range and bearing display to the waypoint, and drive until the alarm sounds. In this case, the bearing to the waypoint will be of little use since you will be following a road and can expect to make numerous deviations from the straight-line bearing. If you are entering a road at an unknown point, the bearing will provide a quick way to determine if you need to turn left or right to get to the waypoint.

0274. Uses in Offensive Operations. Position location devices can provide valuable assistance in a variety of situations during offensive operations, including the following:

a. **Assembly area to LD**. Position location systems can locate a unit's position within an assembly area. A waypoint with the grid location of the center unit's area would ensure proper placement within the assembly area. Waypoints at the start point (SP), the release point (RP), and perhaps along the route would help to guide the unit to the LD.

b. **LD to objective**. After crossing the LD, the unit could enter key points along the axis of advance as waypoints. Additional waypoints on checkpoints or coordination points would help to positively identify the unit's location.

c. **Phase lines (PL) and limit of advance (LOA).** PLs are necessary in coordination of the attack, but the terrain does not always lend itself to easily identifiable PLs. With the global positioning system (GPS), PLs can be placed without reliance on terrain features. The border alert feature will sound an alarm when the unit reaches a designated line on the ground. The same method can be used for locating an LOA line.

d. **On the objective**. Once on the objective, platoons will consolidate and reorganize. In directing the platoons' defensive fire orientation, a distant point, such as a target reference point (TRP), can be selected and entered as a waypoint. The platoon would then take a range/bearing to the point and use that bearing as their orientation. If fuel and ammunition resupply is not performed at the platoon location, the resupply site can be entered as a waypoint to help subordinate units pinpoint its location. The same process can be used to locate collection points for maintenance, EPWs, and wounded.

0275. Uses in Defensive Operations. Uses of position location devices during defensive operations include the following:

a. **Battle position.** In establishing a BP, a unit can use the anchor watch feature of most position location systems to ensure that all elements are within the proper area. The anchor watch sounds an alarm whenever the unit moves too far from a designated center point. Once a vehicle has established its location on primary and subsequent BPs, the location can then be saved as a waypoint to aid in finding it again later.

b. **Position location systems** can also assist movement from one BP to another, particularly during limited visibility. During the reconnaissance/rehearsal of the route, stop and enter your current position as a waypoint at all critical locations such as trail crossings, fords, obstacles, or turns. Each point will be saved in sequence. You can then follow the sequence of waypoints between the BPs.

c. **Fire planning**. Waypoints can also be used to ensure proper orientation of fires using the range and bearing feature.

d. **Movement.** In a passage of lines, the unit that establishes the passage can give grid coordinates of entry points, release points (RP), and critical turns to the passing unit. These coordinates are entered as waypoints. The passing unit can follow the way-points and ensure a safe passage.

Readiness Conditions

0276. A readiness condition (REDCON) establishes the time necessary for the unit to move or be combat ready. It reflects the commander’s expectation of how ready the unit is for combat. The unit SOP should describe REDCON in terms of the critical tasks of preparation that the unit has completed and the time available to prepare. Figure 2.25 is a sample list of REDCON levels with an explanation of each level.

REDCON 1: Full alert, ready to move and fight.
-All personnel alert and prepared for action
-Vehicles loaded and secured, and weapons manned.
-Platoons ready to move on notification
-Engines started
REDCON 2: Stand to complete, ready to fight.
-Pre-combat checks complete
-Equipment stowed (except for wire, NBC alarms, and commo gear)
-Prepared to move or fight in position, infantry is dismounted in positions.
-Status reports submitted to company team CP.
-Vehicles ready to move in 15minutes, if class IV (mines, concertina, overhead cover, sand bags) are abandoned.
REDCON 3: Reduced alert.
-Fifty percent of each crew stand down for feeding, rest and maintenance.
-Remaining personnel man vehicles and Ops (including air guard weapons), prepare position and obstacles and monitor radios and telephones.
-Vehicles ready to move within 30 minutes of notification. Most class IV can be recovered.
REDCON 4: Minimum alert.
-Ops manned and one alert per platoon, monitor radios and man turret weapons.

Figure 2.25 – Readiness Condition

Operations Security

0277. OPSEC is all measures taken to maintain security and achieve tactical surprise. Units must deny the enemy information about planned, ongoing, and after-operation activity until it is too late for the enemy to react effectively. This section will summarize OPSEC in terms of counter surveillance, information security, signal security, and physical security.

0278. Tank units are extremely vulnerable to surveillance. The AO, especially in operations other than war, will consist predominantly of dismounted infantry and wheeled utility vehicles.

0279. Counter-surveillance involves taking measures to protect friendly activities from being observed or detected (visually, electronically, or seismologically) by the enemy. Examples of counter surveillance measures and guidelines that apply to them:

0280. **Noise and light discipline** follow these guidelines:

a. Shield all light sources from enemy view.

b. Move only when necessary.

c. Use headsets or combat vehicle crewman (CVC) helmets to avoid radio noise.

d. Do not slam hatches or doors.

e. Use the short count method to start engines simultaneously.

f. Perform resupply and maintenance in areas that are masked by terrain.

g. Sandbag or shield generators.

h. Use hand-and-arm signals when possible.

i. Use extra precaution at night when noise and light carry farther.

j. Do not allow smoking outdoors at night.

0281. **Camouflage** follow these guidelines:

a. Place tree branches or other vegetation on vehicles; hold them in place with commo wire. Drape camouflage nets over turrets.

b. Park vehicles in natural concealment and shadows.

c. Cover all headlights, mirrors, and optics when possible.

d. Enhance camouflage paint with white wash or mud when possible to break up the outline of the vehicle.

e. Consider the effects of dust and smoke when moving.

f. When possible, ensure vehicles drive in the tracks created by the vehicles ahead of them or in previously created tracks.

g. Blend vehicles with other objects having a thermal emission.

0282. **Concealment** follow these guidelines:

a. Disperse vehicles and personnel under foliage or inside structures when possible.

b. Conceal vehicles behind objects that can block the thermal line of sight.

c. Keep vehicles travelling on existing tracks or roadways in heavily used areas, such as CPs and trains.

0283. **Challenge and Password.** Ensure these are used and enforced.

0284. **Air Defense.** Consider ADA coverage against enemy air reconnaissance.

0285. **Jamming** use the following procedures:

a. Consider jamming to disrupt enemy communications.

b. Destroy enemy jamming, direction finding, and intercept equipment.

0286. **Smoke. S**creen enemy observers and friendly movement with smoke.

0287. **Deception**. In most instances, it is impossible to keep tanks from being detected or observed by enemy forces because of their large signature. Deception plans should be employed so that friendly routine actions are conducted with greater uncertainty. Deception can play a significant role in masking the movement of formations, and inducing the enemy to miscalculate friendly objectives or weaknesses.

0288. **Information security** is the protection of friendly information. Leaders can help prevent compromising sensitive information:

a. Ensuring soldiers to not send critical information in the snail mail, e-mail and text messages.

b. Policing areas to ensure nothing value is left behind.

c. Destroying overlays, orders, and other documents after use or when they are no longer necessary.

0289. **Signal security** is a vital component of OPSEC procedures. Take the following steps to protect friendly communications:

a. Use secure communications means or operational and numerical codes.

b. Use low-power transmissions and terrain to mask signals from enemy direction-finding equipment.

c. Keep radio transmissions short. Messengers or wire should be used for lengthy messages.

d. Units must practice using communication and electronic operation instructions (CEOI), SOPs, and operational terms.

e. Protect cryptographic systems and classified documents from capture or loss. Before an area is vacated, inspect it for any materials that could provide friendly information to the enemy.

f. Patrol wire lines to prevent enemy tapping.

g. When CEOI codes or cryptographic equipment is lost or captured, report the facts promptly to the next higher command. The unit SOP must contain instructions for destruction of equipment and classified documents to prevent capture or use by the enemy.

0290. **Physical security** is the protection of material and equipment. Some examples of physical security include:

a. Employing guards, OPs, and patrols at all unit and CP sites.

b. Employing anti-intrusion devices, such as platoon early warning systems (PEWS) and trip flares, when stationary.

Chapter III
Offensive Operations

0301. In the offense, tank units should be assign missions that capitalize on their maneuver and firepower capabilities. These include missions to destroy enemy forces, develop intelligence about the enemy, seize or control terrain, deceive and divert the enemy, deprive the enemy of resources to demoralize him, hold the enemy in position, and destroy and disrupt enemy command, control, and logistics facilities. Tank is used to defeat a defending enemy force by seeking decisive results in the enemy's rear and flank areas. Close combat and assaults against enemy tank units fail to exploit tank strength of speed, mobility, and agility. This does not mean tank cannot be used during assaults. In some situations, there may be no alternative; Commanders should carefully weigh the factors of METT-TC when making these decisions.

0302. Tank is employed using two distinct methods: as the direct or semi-indirect fire support element or as the maneuver element. Timing, protection, mobility, and speed are the key elements that must be synchronized between the mounted element and the dismounted element. Unit leaders must understand and employ the following fundamentals of offensive operations:

a. **Fight as a combined arms team**. Tank always fights as part of a combined arms team. Capabilities and limitations of tank and infantry make them complementary when employed as a team.

b. **Know the enemy**. The tank unit leader must know and understand the capabilities of the enemy's weapon systems and defensive doctrine, including the enemy's capability to conduct ambushes.

c. **See the battlefield**. The leader must know and be able to identify key terrain. He should also learn to identify covered and concealed routes during movement. The tank unit leader must anticipate how the enemy will use the terrain and then determine how to counter it.

d. **Use weapon systems to best advantage**. The tank leader must know the capabilities and limitations of his own weapon systems. Knowing the best kill probability ranges of all weapons (main gun, machine guns, and infantry weapons) is a key. Leaders can improve kill probabilities by engaging enemy vehicle flanks.

e. **Concentrate combat power**. The tank unit leader must be able to control and concentrate weapon systems. To do this, he uses proper C2 techniques and trains his unit to shoot, move, and communicate effectively under all conditions. The leader also makes maximum use of available indirect fires.

f. **Use maneuver to best advantage**. Tanks must move rapidly, strike first, and maintain the momentum until the enemy has been killed or captured.

g. **Coordinate continuous support**. Tank unit leaders must always be aware of their logistical status. Logistics support is extremely difficult in the austere

environment of light forces. Leaders must understand the procedures for critical support, such as evacuation of personnel and equipment or resupply of ammunition and fuel. This is important because task organization changes are more frequent when tank operates with a contingency JTF.

h. **Be flexible**. Tank unit leaders achieve flexibility by ensuring units are properly trained, by adhering to standing operating procedures (SOP) and battle drills, and by becoming tactically proficient. They must understand the commander's intent and anticipate changes in the situation that will help complete the mission.

Battalion in the Offense

Movement to Contact

0303. The battalion conducts a movement to make or regain contact with the enemy and to develop the situation. The battalion should be given a movement to contact mission as the lead element of a brigade or division attack, or as a counterattack element (see Figure 3.1). Movement to contact terminates with the occupation of an assigned objective or when enemy resistance requires the battalion to deploy and conduct an attack to continue forward movement.

0304. Key planning considerations for a tank battalion movement to contact are:

a. **Speed required by the division.** The rate of march in the tank battalion can easily exceed the rate of march in the other infantry units.

b. **Available avenues of approach.** The tank battalion should move along an axis of advance that maximizes its firepower and maneuverability.

c. **Requirements to maintain mutual support and synchronization between maneuver units, security, and FS.** The BCT/TF/Bde is organized with a security force, advance guard, main body, and flank and rear guards.

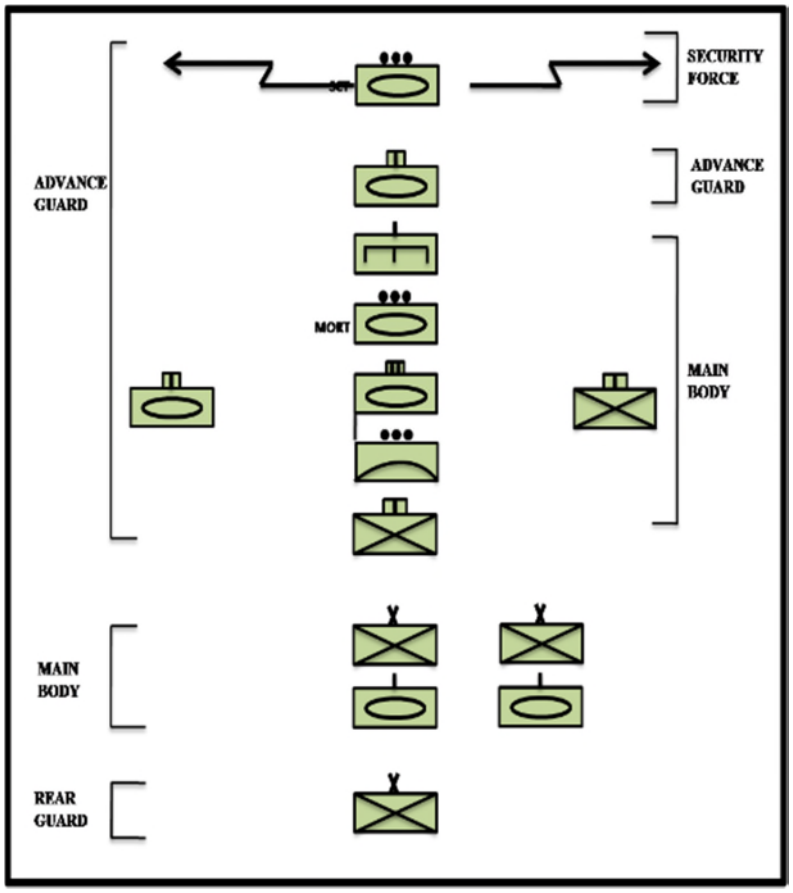


Figure 3.1 – BCT/TF/Bde movement to contact as lead for a division attack.

0305. The following considerations apply:

a. The security force is normally established with the tank battalion scout platoon who is equipped with ATGM to provide long-range anti-tank capability in the security force. Planners must consider trafficability of the terrain since the wheeled scouts are not as maneuverable as the tracked Tanks.

b. The advance guard is usually a company team. Company teams are task organized, with tank and infantry platoons forming a mutually supporting team. Attached infantry may move with the company team by a variety of means. The infantry may move dismounted, mounted in tanks, or mounted in trucks until contact is made or the situation requires the infantry to dismount.

c. The company team may be augmented from the infantry battalions. Other attachments may include an engineer platoon and an ADA section.

d. The main body remains behind the advance guard lead element, keying its movement to the advance guard. It is flexible enough to maneuver rapidly and provide responsive support when committed. The main body contains the bulk of the TF's combat force; it may be augmented with anti-tank assets, the infantry battalions' antitank platoons, engineers, ADA sections, and GSR. Infantry in the main body moves in trucks if possible.

e. Flank security is normally accomplished with platoon-size elements (under company control) from one or more of the companies in the main body. The flank guard in a light environment may be antitank-equipped elements from the battalion's scout platoon or attached antitank platoons.

f. The trailing company of the main body provides a rear guard to protect the TF's rear.

g. CSS assets move with the TF as part of the combat trains or field trains.

Hasty Attack

0306. The hasty attack is conducted either as a result of a meeting engagement or when a bypass has not been authorized and the enemy force is in a vulnerable position. Hasty attacks are initiated and controlled with FRAGOs and are usually indicated when a movement to contact results in enemy contact. Figure 3.2 shows a tank hasty attack. Hasty attacks at the TF level are conducted, with the following planning considerations:

a. Commanders must carefully consider where and when to dismount the infantry to maximize its effectiveness. The infantry should be transported as close to the battle as possible before dismounting, taking security into consideration. Tanks cannot fight with infantry riding on top of them; however, Tanks are vulnerable to enemy dismounted forces if they outrun their own supporting infantry. Tank is most effective when working as a team with its infantry. That team must be preserved during maneuver.

b. Anti-tank equipped armored vehicles are best deployed in overwatch. During hasty attacks, the maneuverability and survivability of the tanks should be maximized with the less maneuverable and survivable anti-tank AVs in overwatch.

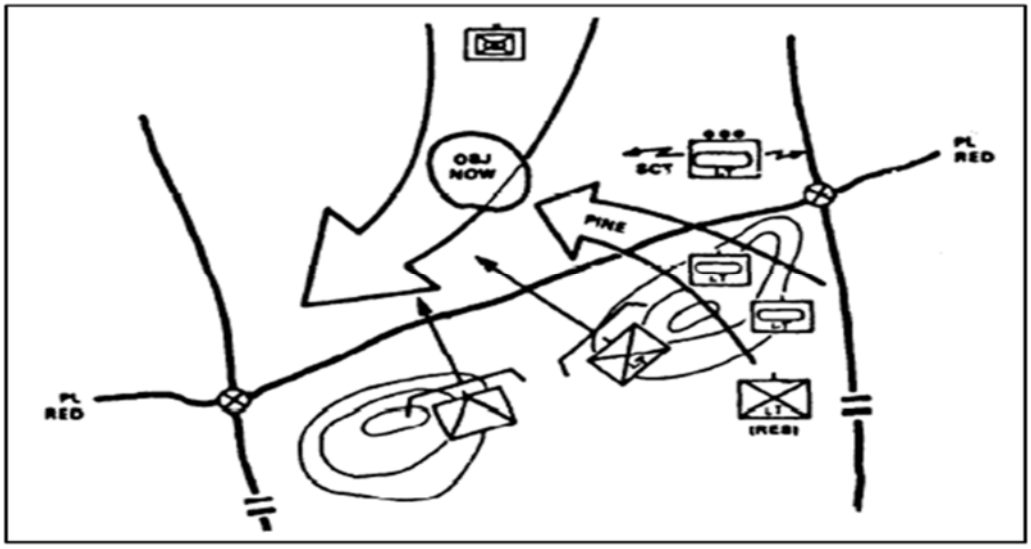


Figure 3.2 – Tank task force in a hasty attack.

Deliberate Attack

0307. TF deliberate attacks differ from the hasty attack in characterized by precise planning based on detailed information, thorough preparation, and rehearsals. A deliberate attack normally includes large volumes of supporting fires, main and supporting attacks, and deception measures.

0308. A Tank battalion TF will usually be the division's main attack element against an enemy force consisting of mechanized forces. Figures 3.3 and 3.4 illustrate Tank deliberate attacks in breaching and assault operations. Planners should consider the following factors:

a. The timing of the attack must be synchronized with the other elements of the division. The rest of the division may be moving dismounted or truck-mounted.

b. The division possesses limited mechanical obstacle breaching systems. The breaching of obstacles in most cases will be conducted manually with dismounted engineers or with the aid of light dozers or SEEs, requiring suppression from tanks and smoke for protection and concealment. Obstacle breaching therefore requires extensive planning and rehearsals.

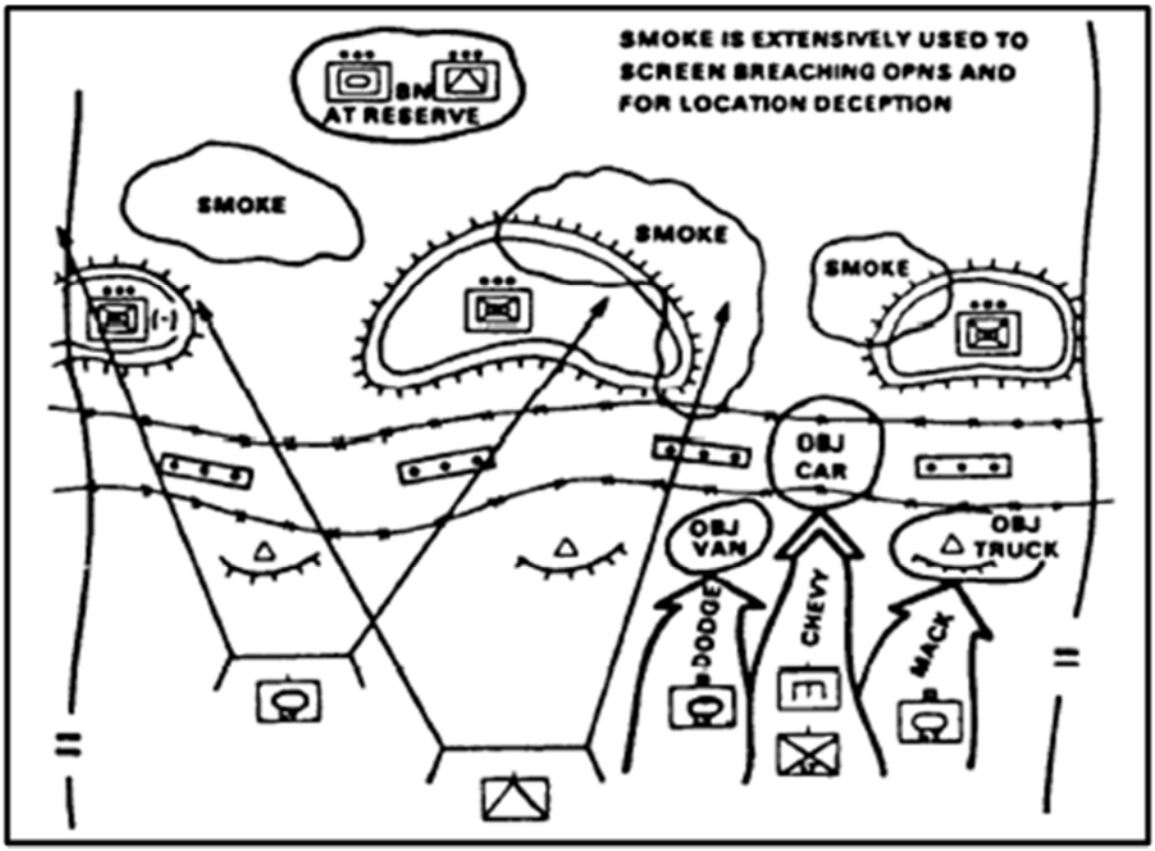


Figure 3.3 – Task Force in a deliberate attack (breach).

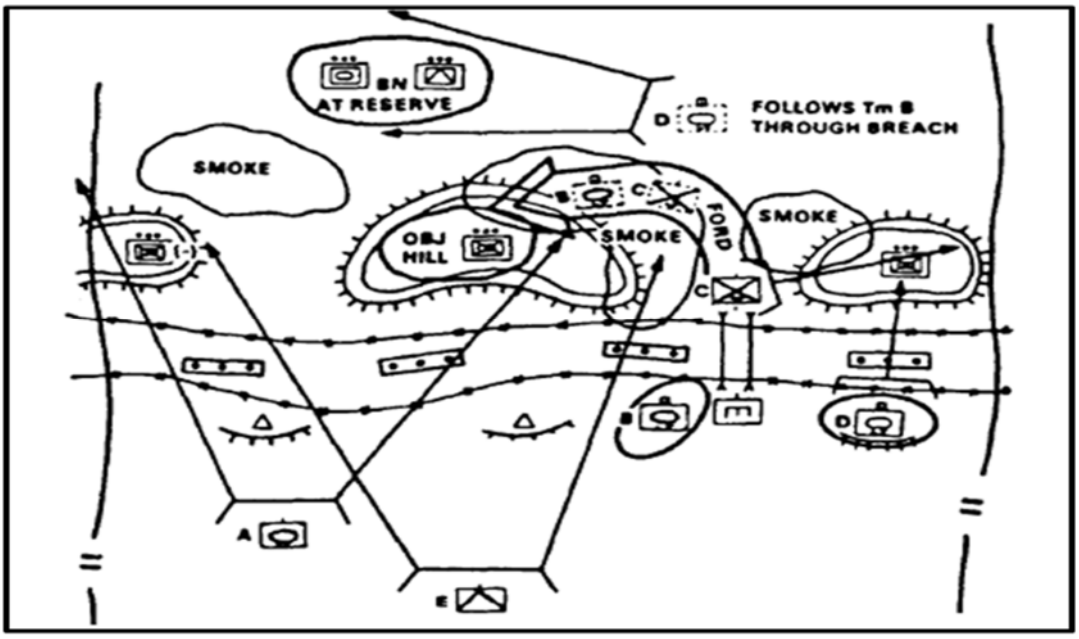


Figure 3.4 - Task Force in a deliberate attack (assault).

Exploitation

0309. The exploitation is conducted to take advantage of success. It prevents the enemy from reconstituting an organized defense or conducting an orderly withdrawal. It may follow any successful attack. The keys to successful exploitation are speed in execution and pressure on the enemy. Exploitations are usually oriented on a terrain objective containing enemy's reserves, CS, CSS, and C2 facilities. The Tank battalion is the division commander's exploitation force. It is the most mobile ground element and can regain and maintain contact with the withdrawing enemy force. Planners must prepare for support during exploitation, with considerations for refueling and rearming. Air resupply may be used to support the force during exploitation.

Pursuit

0310. The pursuit normally follows a successful exploitation. It differs from an exploitation, in that a pursuit is oriented primarily on the enemy force rather than on terrain objectives. While a terrain objective may be designated, the enemy force is the primary objective. The purpose of the pursuit is to overrun the enemy and destroy him.

0311. If the enemy is predominantly dismounted, the division commander will probably use his infantry in the pursuit. The Tank TF may be designated as the division reserve during the pursuit of a dismounted enemy force; it may become the main effort if the fleeing enemy is mounted.

Raid

0312. A raid is an attack into enemy territory to accomplish a specific purpose, with no intention of gaining or holding terrain. Raids may be conducted to:

a. Capture prisoners.
b. Capture or destroy specific enemy material.

c. Destroy logistical installations.

d. Obtain information concerning enemy locations, dispositions, strength, intentions, or methods of operation.

e. Disrupt enemy plans.

0313. The Tank TF may conduct, or direct subordinate elements to conduct a raid. Raids may be conducted mounted or dismounted. A mounted raid is normally conducted as an exploitation with a limit of advance or as an attack with a limited-depth objective.

0314. Raids may be conducted in daylight or darkness, within or beyond supporting distance of the parent unit. When the area to be raided is beyond supporting distance of friendly lines, the raiding party operates as a separate force. Raiding force security is vital because the raiding party is vulnerable to attack from all directions. Raids should be timed so that the raiding force arrives at the objective area at dawn, twilight, or other times of low visibility. FS, if in range, should be planned.

0315. During movement in daylight, the raiding force uses covered and concealed routes of approach. During reduced visibility, advance and flank security detachments precede the raiding force. They prevent premature discovery of the raid by locating enemy security detachments and directing the raiding party around them.
The withdrawal is usually made over a different route from the approach. Security detachments are employed to ensure that the routes of withdrawal are open. Protective fires are planned along the axes of advance and withdrawal. Rally points are planned for units to assemble after they have completed the mission and are ready to withdraw.

0316. Planning considerations for raid includes the type and number of vehicles and weapons that the raiding force will have movement distance and length of time. The raiding party will operate in enemy territory and expected enemy resistance. The raiding force usually carries everything required to sustain itself during the operation. Resupply of the raiding force, if required, is by aircraft. Figure 3.5 illustrates a raid with tank.

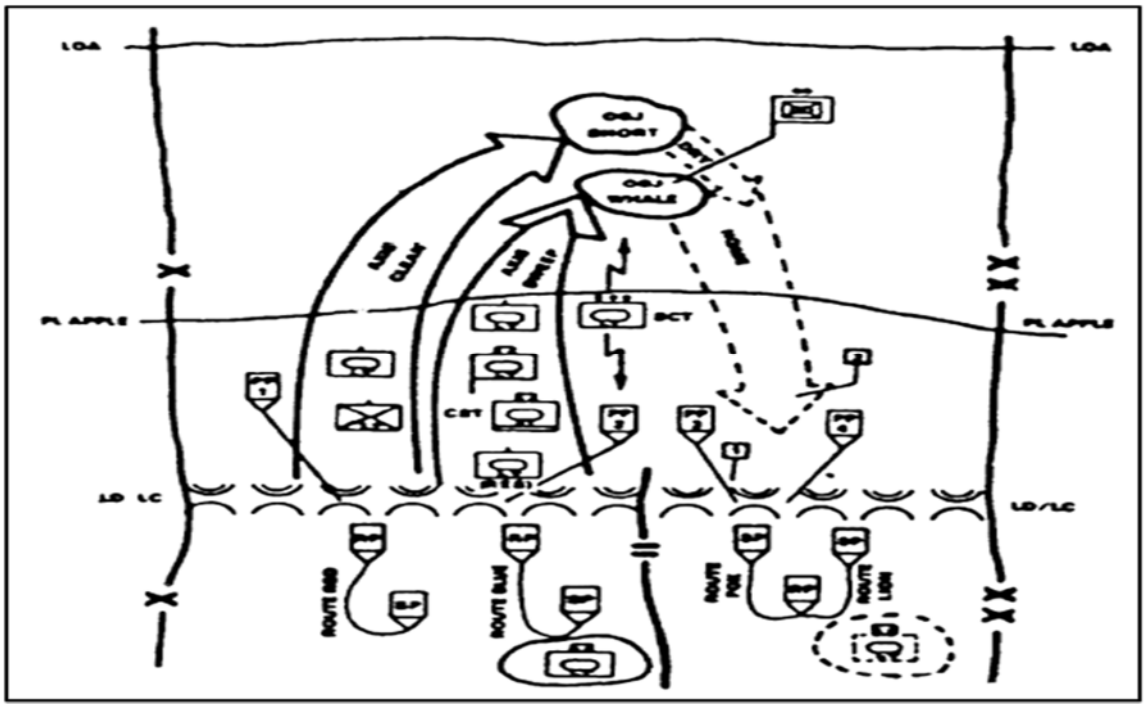


Figure 3.5 – Raid with Tanks

Company in the Offense

0317. When operating with the infantry brigade, the tank company can be employed in one of four methods:

a. As part of the main attack.

b. As a reserve or follow-on in support of the main attack.

c. As reconnaissance and security forces forward, on the flanks and on the rear of the main and supporting attacks.

d. As part of the supporting attack.

0318. The tank company participates in two types of offensive missions—movement to contact and attack (hasty and deliberate). The company may participate in an exploitation or pursuit. It will normally participate in these missions as part of a larger infantry force (brigade) but on some occasions it may operate as part of an Armor Task Group.

0319. Maneuver. The infantry brigade uses the same basic forms of maneuver as the infantry battalion. Avoiding Enemy Anti-armor Fire. When the terrain allows the tank company to maneuver in mass, it presents an easily identifiable target and is vulnerable to enemy anti-armor and tank fires. The company should use all available cover and concealment; the enemy cannot mass fires against the company if they cannot see it. Despite its obvious advantage, however moving along covered and concealed routes costs the company in terms of speed, control, and vulnerability to short-range, handheld anti-armor weapons. Smoke is used to defeat enemy observation and target acquisition.

0320. Tanks must never in a skyline or move directly forward from a defilade position. Avoid dusty terrain when possible; it betrays the tank. Before crossing open areas, use attached infantry or dismount personnel with binoculars to observe the area carefully for possible enemy positions prior to emerging from cover. If enemy locations are identified or suspected, suppress them or use smoke on them before crossing. Cross open areas as quickly as possible. Use rapid rushes from one covered position to another. If a tank is exposed for less than 30 seconds, it will be extremely difficult for an enemy ATGM gunner to acquire, fire, track, and hit it at long range. Use 15- to 20-second rushes when suspected ATGM positions are close.

0321. In situations where use of covered routes is too time-consuming, plan for escape routes (see Figure 3.6).

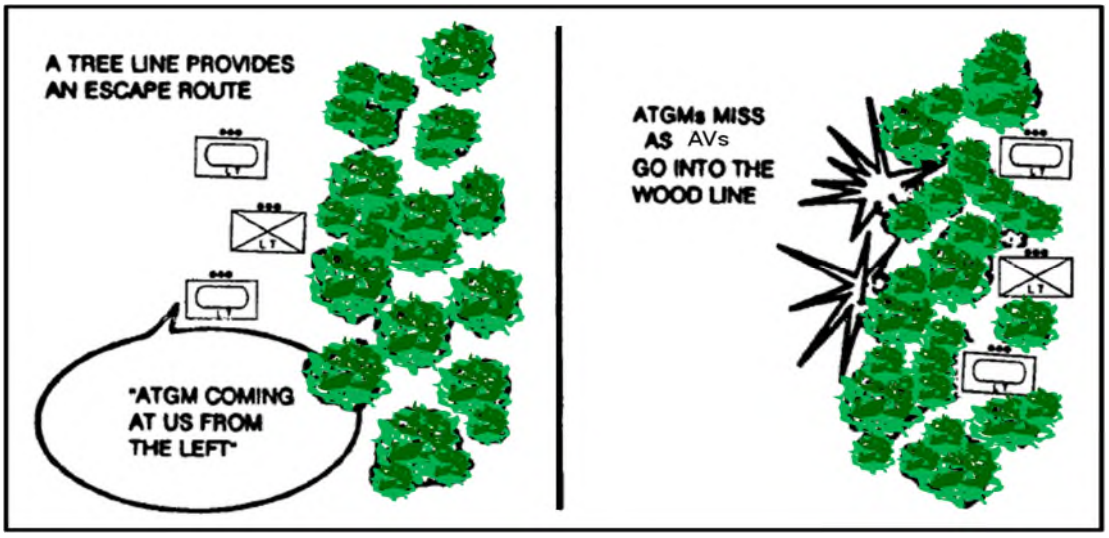


Figure 3.6 – Company movement with escape routes

Movement to Contact

0322. A movement to contact (MTC) is an offensive operation designed to gain contact with the enemy. A tank company can organize one or two of its platoons to infantry battalions during a movement to contact. The remaining platoon(s) will be retained under the company headquarters with infantry attached as the brigade reserve. The company commander will designate specific movement techniques and formations to reduce danger to the unit while moving. Once contact is made, the company uses fire and movement to develop the situation. In the absence of command guidance, the company commander employs the factors of METT-TC to specify movement techniques and formations.

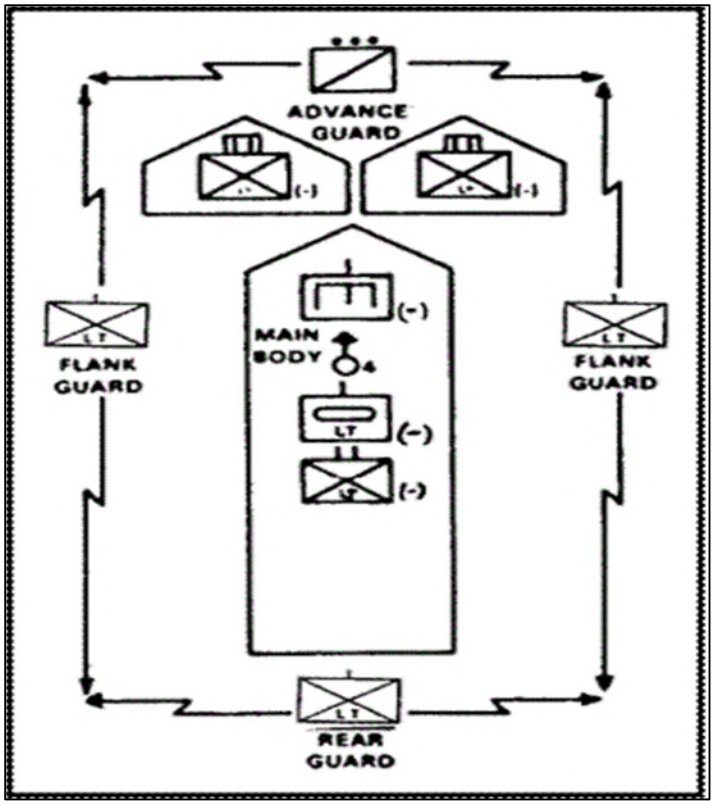


Figure 3.7 – Example of a tank company with infantry brigade
MTC: Tank Company as part of a Brigade Vee.

0323. The brigade will conduct a movement to contact in two general situations. The most common situation has the brigade operating independently and attacking with one or two lead battalions (see Figure 3.7). The other situation is when the entire division is out of contact, with a great distance between friendly and enemy AOs. The brigade either moves as the main divisional body or provides unit for security, guard, or security forces (see Figure 4.8). The brigade commander assesses the situation once contact is made and decides to attack, bypass, or defend based on how the situation develops.

0324. In the examples illustrated in Figures 3.7 and 3.8, the tank company has detached platoons to the lead battalion(s) and retained the remaining platoons under company control as the brigade reserve. This gives the lead battalions a small, mobile, self-contained force to locate and fix the enemy. The example in Figure 3-8 shows the tank company retained as a unit for ease of movement. The lead battalion receives an attachment of a tank platoon while the tank company (-) moves with an anti-tank platoon as a reserve. This may be done if there is a potential enemy mechanized threat. In some cases, the tank company can be used as forward or flank security for movement.

0325. The example in Figure 3.9 shows the cloverleaf technique when the tank company is given a large designated area to search rapidly.

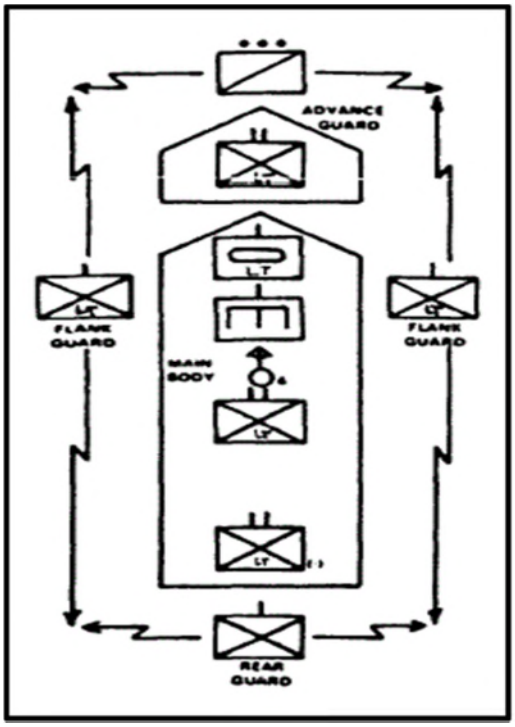


Figure 3.8 – Example of a tank Company with infantry brigade MTC: Tank Company as part of a brigade column.

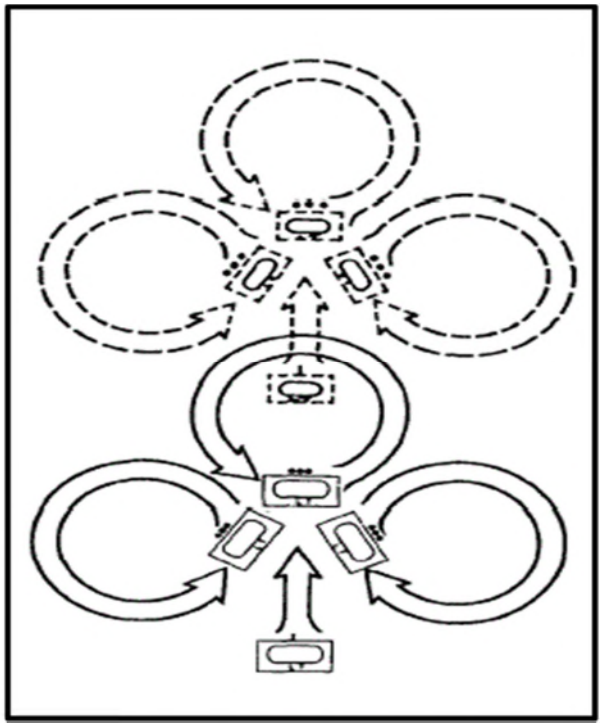


Figure 3.9 – Example of clover leaf technique.

Hasty Attack

0326. The hasty attack is used to exploit an opportunity to gain or maintain the initiative at company or higher level. It is characterized by quick planning and coordination. A hasty attack can develop when:

a. A movement to contact results in contact.

b. A deliberate attack is changed after it is under way.

c. Further advance is ordered after securing an objective.

d. A counterattack is ordered in the defense.

0327. A MTC may terminate in a hasty attack, during which the company may be assigned the mission of securing a terrain feature or destroying an enemy force. If enemy contact is made en-route to securing an objective, the unit may:

a. Fix and bypass the enemy, depending on enemy strength and the unit’s orders.

b. Attack by fire to destroy the enemy.

c. Conduct a hasty attack to kill the enemy and continue the attack to the objective.

0328. Depending on his orders and the size and location of the enemy, the infantry TF commander develops a plan to conduct a hasty attack when enemy contact is made. He designates an objective, a support by fire element, a support by fire element's overwatch positions, an assault element, covered and concealed routes into the flanks of the enemy. He then issues a fragmentary order to his company commanders.

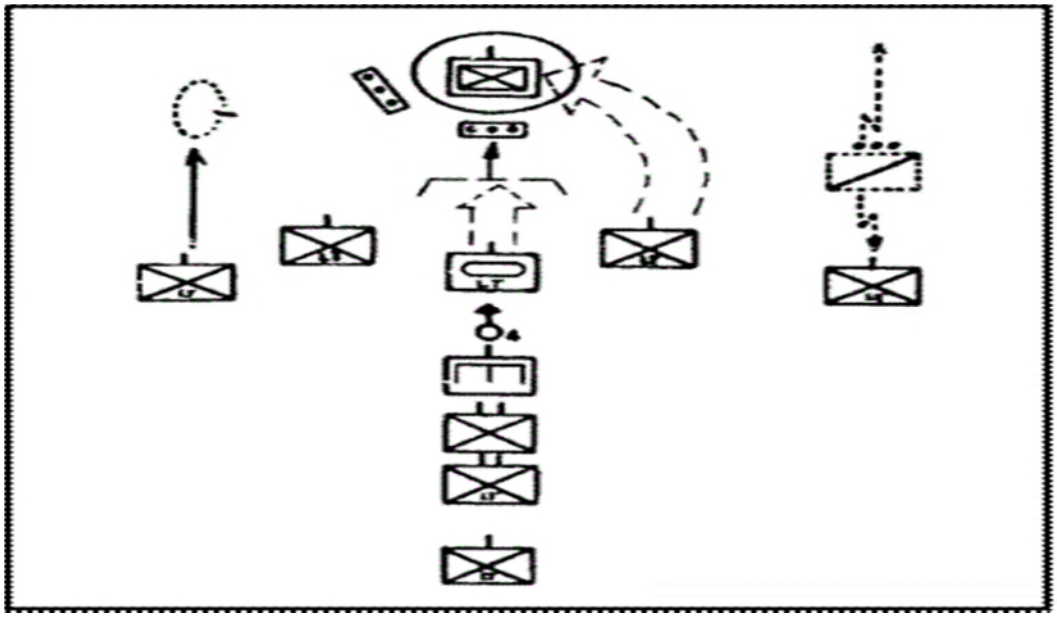


Figure 3.10 – Infantry brigade/tank company hasty attack example 1.

0329. The first infantry BCT/TF/Bde to make contact with the enemy conduct actions on contact, develops the situation, destroys the enemy if possible, and reports the situation to the brigade. The brigade commander may employ the tank company as an assault force. Figures 3.10 through 3.12 show examples of infantry brigade hasty attacks with a tank company.

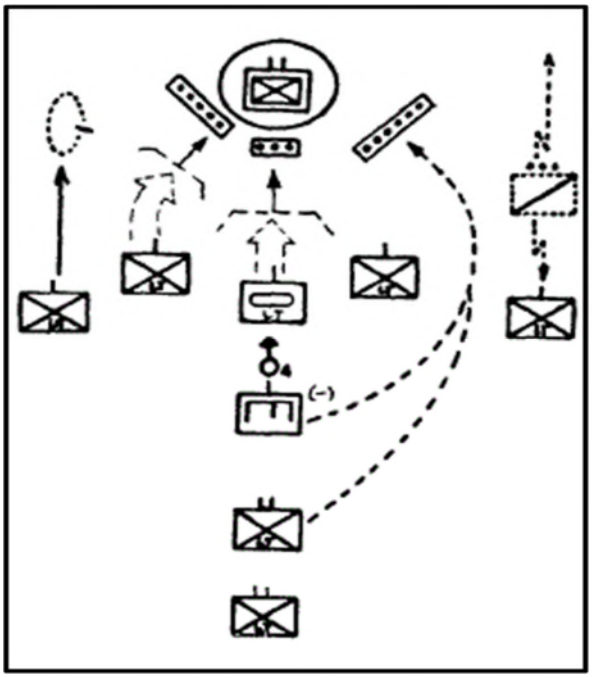


Figure 3.11 – Infantry brigade/tank company hasty attack example 2.

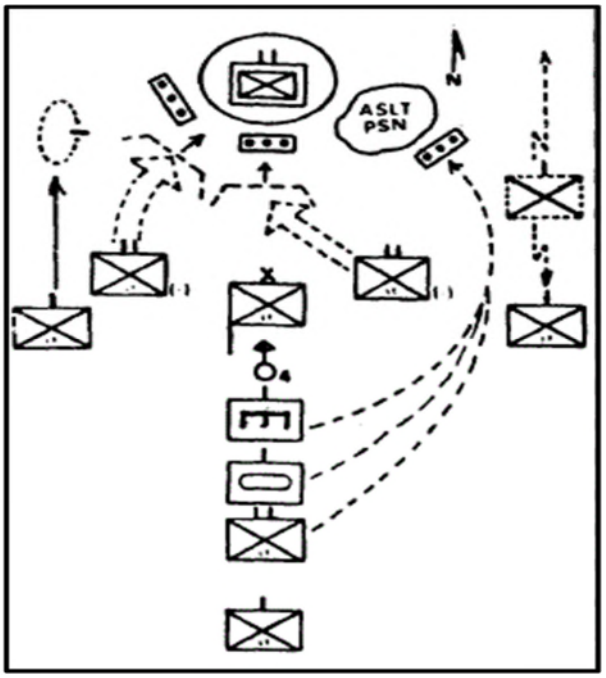


Figure 3.12 – Infantry brigade/tank company hasty attack example 3.

Deliberate Attack

0330. A deliberate attack is usually necessary against a well-organized, well-prepared enemy that cannot be turned nor bypassed. The infantry commander will normally have time to collect detailed information about the enemy. The brigade commander may consider using the tank company to assist in one or more of the following:

a. Isolating the penetration or objective area with direct fire to prevent enemy reinforcement or supporting fires.

b. Committing a strong force to overwhelm the enemy at an identified or created weak point.

c. Striking the enemy on the flanks or in the rear.

d. Using a form of maneuver that avoids the enemy's main strength.

e. Fixing other positions to minimize the enemy's capability to react.

f. Developing a deception plan.

0331. Figures 3.13 and 3.14 show examples of an infantry brigade deliberate attack with a tank company.

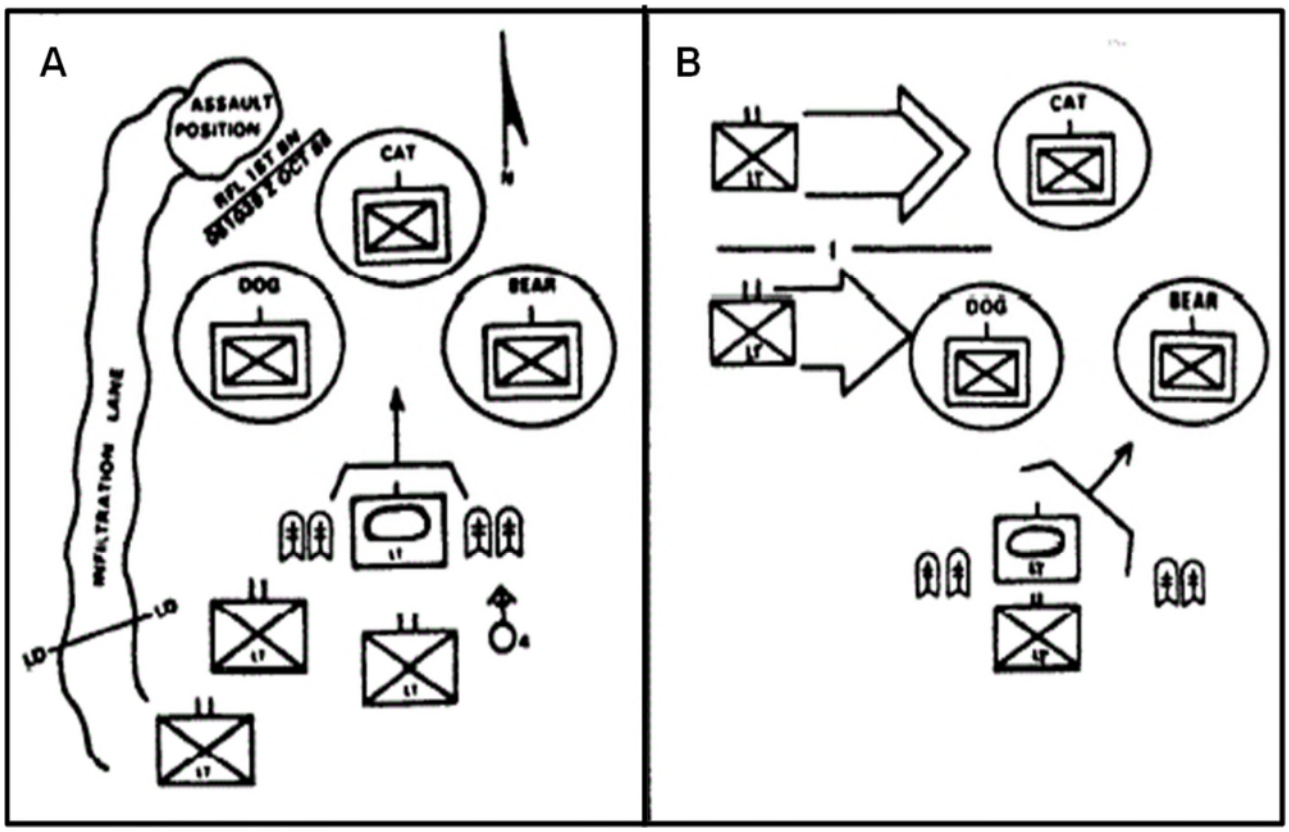


Figure 3.13 – Infantry brigade/tank company deliberate attack example 1.

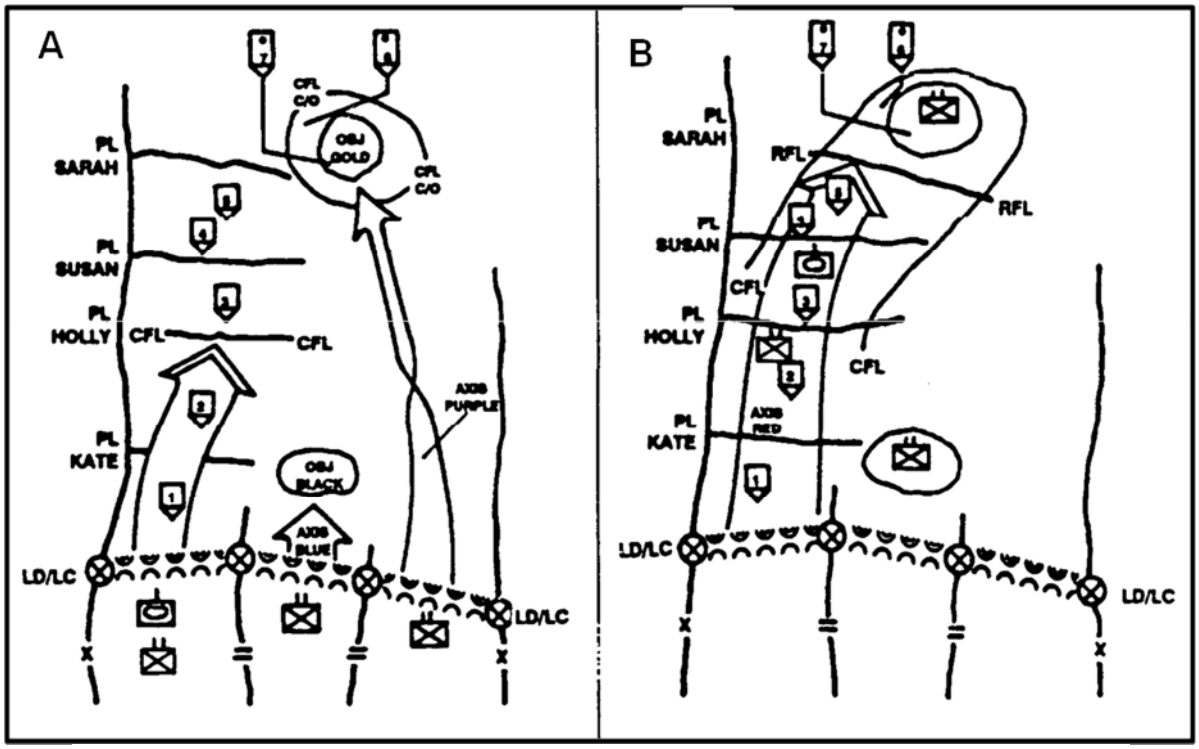


Figure 3.14 – Infantry brigade/tank company deliberate attack example 2.

Exploitation

0332. Exploitation is a continuation of an attack to take advantage of and maximize success. It prevents the enemy from reconstituting an organized defense or conducting an orderly withdrawal. It normally follows a successful attack that weakens or collapses enemy defenses. The exploitation focuses on the enemy and destroys his will to resist. Speed is essential. Mobility of the exploiting force must be equal to or greater than the enemy force.

0333. The tank company normally participates in the exploitation as part of a much larger force. The exploitation is executed as a movement to contact or hasty attack. Exploiting force missions include securing objectives deep in the enemy rear, cutting LOC, surrounding and destroying enemy units, denying escape routes to an encircled force, and destroying enemy reserves.

0334. The exploiting unit advances rapidly to the enemy rear area, destroying enemy combat units and lightly defended and undefended CS and CSS activities. Bypassed enemy forces are reported to higher headquarters for reduction by follow-and-support forces. The successful exploitation can turn into a pursuit, with the end goal being total destruction of the enemy's ability to fight.

Pursuit

0335. A pursuit normally follows a successful exploitation. The objective of a pursuit is to maintain pressure on the enemy and intercept, capture, or completely destroy him. Infantry units conducting a pursuit should maximize the use of tank assets to slow, disrupt, and confuse the enemy withdrawal. Again, the company would usually participate in a pursuit as part of a larger force in a movement to contact or hasty attack.

The pursuit is conducted using a direct pressure force, an encircling force, and a follow and-support force. The tank company may be part of these forces, which have the following functions:

a. The direct pressure force denies enemy units the opportunity to rest, regroup, or resupply by staying in direct contact with them and forcing them to stay on the move. At every opportunity, the direct pressure force envelops, cuts off, and destroys enemy elements.

b. The encircling force moves to get in the rear of the enemy and blocks his escape; in conjunction with the direct pressure force, it attacks to destroy the enemy force. It advances along routes parallel to the enemy's line of retreat to reach defiles, communication centers, bridges, and other key terrain ahead of the enemy main force.

c. The follow-and-support force is not a reserve, but it is committed to destroy bypassed enemy units, relieve direct pressure force elements (which have halted to contain enemy forces), secure LOC, secure terrain, or guard prisoners at key installations.

Offensive Operations in Built-Up Areas

0336. Tank units will not normally fight as a company within BUAs. However, the company may have sections or platoons task organized to infantry teams or TFs to provide direct FS for urban fighting. The phases of attacks of a BUA are also explained in this Chapter. The fight for urban terrain begins outside the BUA. The tank company can be used to isolate the area and provide direct FS for the assault force to gain a foothold.

0337. The tank company isolates the BUA by enveloping it and seizing key terrain that dominates approach and escape routes (see Figure 3.15). The isolation serves to prevent enemy reinforcement and/or contain the enemy within the BUA. The orientation of the tank platoons is important. Companies and platoons may be required to orient tanks toward the BUA as well as outward along mounted avenues of approach (see Figure 3.16).

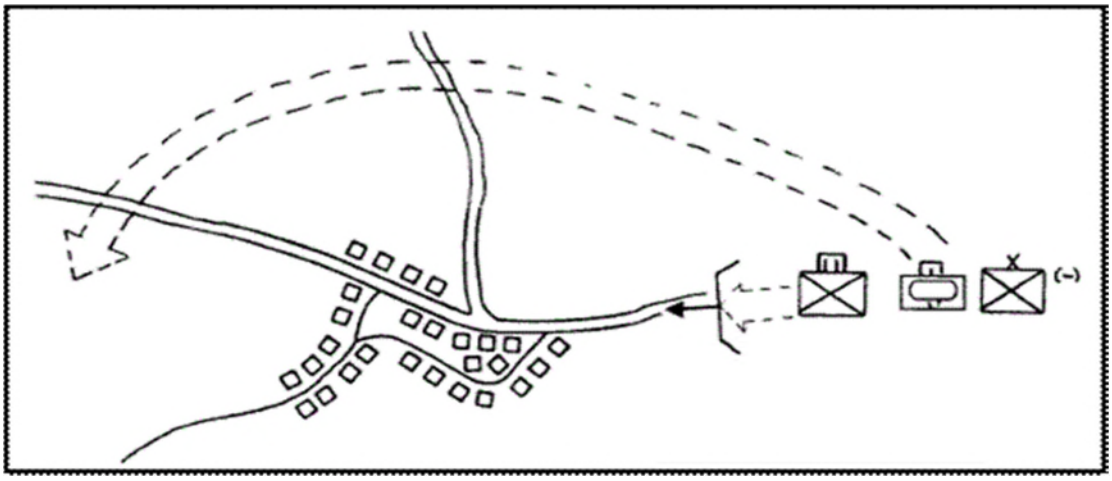


Figure 3.15 – Tank Company enveloping a built-up area.

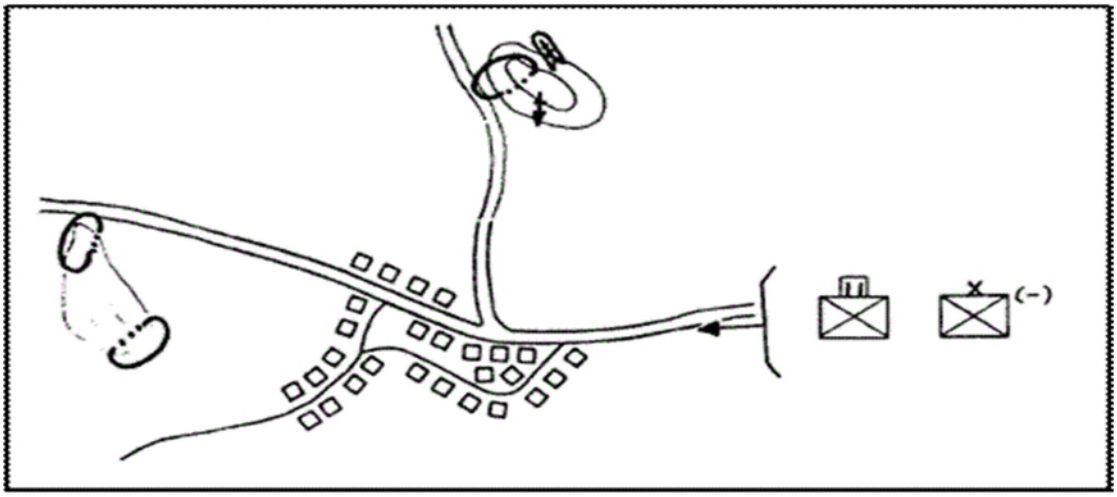


Figure 3.16 – Blocking positions around a built-up area.

0338. Tank companies may be required to move in or through BUA when:

a. There is no bypass.

b. They are required to seize key positions during exploitation or when the town is lightly or undefended.

c. They are escorting a convoy through a BUA that presents little or no threat.

d. There are two techniques for conducting a tank attack in or through a BUA.

0339. The tank company can penetrate poorly defended or unalarmed urban areas to occupy key objectives, holding them for a limited time. When the enemy defenses are weak or no organized enemy resistance is present, the lead armor unit can immediately attack. The force may not be predominantly infantry, but some infantry is required. The assaulting force drives directly to the center of the town, then proceeds to attack outward.

0340. During exploitation, when surprise has been achieved or when tactical (or terrain) condition makes an enveloping attack difficult; it may be possible to secure the main route through the town by a sudden attack. The attack is made at top speed, with all weapons employed against any enemy personnel encountered. The move continues through the town in dominating terrain on the other side. As the attack continues, flank guards strike quickly up the side streets if necessary to protect the main body. Complete occupation of the town is left to follow-on infantry if needed. In this method, the lead tank commander must be certain of his route and act boldly. If halted by antitank fire, he must notify the commander immediately to keep the following units from jamming up the streets behind him. He may request assistance, continue the assault, or reconnoiter for a bypass.

Other Offensive Tasks

0341. The following discussion describes other tactical tasks that may be required of the tank company during offensive operations.

0342. **Obstacle Breaching**. During offensive operations, the tank company may be required to assist the infantry brigade in breaching obstacles. The brigade has limited engineer support that is capable of breaching simple obstacles. The tank company plays a key part in isolating the breach point and providing direct fire support to the breach and assault forces.

0343. When it encounters obstacles, the infantry BCT/TF/Bde will initially reconnoiter to find a bypass. If no bypass is found and the obstacle is not complex, the BCT/TF/Bde commander may attempt an in-stride breach. If the obstacle is complex, the brigade may require augmentation from army combat engineers. The unit will be organized into support, breach, and assault forces with the following functions:

a. The support force should primarily consist of Tanks to provide long-range direct fire suppression of enemy defensive positions.

b. The breach force consists of the engineers and equipment necessary to clear a lane through the barrier. Infantry and, in some cases, tank will accompany the breach force to provide close-in security and supporting fires. The breach force usually operates under obscuration to protect it from enemy observation. The breach force may be required to follow and support the assault force.

c. The assault force attacks to destroy enemy forces defending the obstacle. Depending on the terrain and type of defense being assaulted, it should consist of infantry and tank.

0344. The breach operation is executed through suppression, obscuration, security, and reduction using all available infantry, tanks, engineers, artillery, and mortars. The considerations in the following paragraphs apply.

0345. **Suppression**. The support force suppresses enemy direct fire defenses. It must put direct fire on enemy’s position and adjust indirect tires and smoke from artillery and mortars.

0346. **Obscuration**. Use of smoke or naturally obscuring terrain is a must. Special engineer equipment, engineers, and infantrymen are extremely vulnerable during the breaching phase if the enemy cannot be destroyed, suppressed, or obscured. Obscuration can also hamper C2 during the breach. Training, preparation, and rehearsals can overcome these problems.

0347. **Security**. In some instances, infantrymen can secure the far side of the obstacle prior to fill lane clearance through infiltration, bypass, air assault, or minor breach. Elements on the far side can help to secure the exit points of the breach lane as well as provide suppressive fire on the enemy.

0348. **Reduction**. Reduction of the obstacle is the most complicated part of the assault breach. The support force shifts, the obscuration and suppressive fires as the breaching force moves forward. Special engineer equipment that may be required includes ACEs, MICLICs, dozers, scoop loaders, and armored vehicle launched bridges (AVLBs).

0349. **Consolidation**. Occupation of an objective is a critical task. Enemy forces on the objective must be eliminated or captured. The company must immediately prepare for a counterattack by the enemy. The company is most vulnerable and unorganized during consolidation, and the enemy will attempt to capitalize on this. During consolidation the company commander should-

a. Designate platoon positions and general orientations.

b. Ensure the company occupies the position designated in the OPORD. Tanks are moved to hull down positions, and the platoon leader assigns specific sectors of fire.

c. Prepare for a counterattack.

d. Establish security and mutual support between adjacent units.

e. Eliminate any remaining pockets of enemy resistance and secure EPWs.

f. Prepare to continue the mission.

0350. **Reorganization**. Reorganization must also be conducted rapidly. Each platoon leader should report his casualties and ammunition, fuel, and tank status to the commander and XO. The XO or 1SG reports the status to the TF. The commander, XO, and 1SG should ensure that the following activities are accomplished during reorganization:

a. Reassignment of crew members if losses were heavy. Survivors of damaged tanks can replace casualties elsewhere.

b. Cross-level ammunition, where necessary.

c. As many repairs by the maintenance section as time allows.

d. Evacuation of wounded soldiers.

e. Evacuation of EPWs and intelligence information.

f. Evacuation of bodies of soldiers KIA.

Tank Company during Limited Visibility

0351. Limited visibility attacks are preferred by the infantry. They are also the most difficult to execute. They require detailed planning, preparation, and rehearsals. Movement will be slower, confusion more likely, and consolidation and reorganization more difficult.

0352. Control of platoons is more difficult at night in limited visibility. Strict light and noise discipline is required. Control measures are more restrictive. The following are some control considerations that apply during company planning for limited visibility operations:

a. Assembly areas may be smaller and closer to the LD. Standardize platoon/company movement in and out of assembly areas.

b. Guides may be posted to assist movement to the LD and through the points of departure.

c. Allows more time for movement and positioning of key weapon systems.

d. Objectives and routes must be more clearly defined.

e. Intermediate objectives may be necessary to maintain control and direction.

f. Graphic control measures are usually more restrictive. For example, restrictive fire lines (RFLs), no-fire lines, and limits of advance (LOA) may be introduced to further control unit movement and fires.

g. Illumination rounds timed to burn on the ground can mark objectives and keep the company oriented.

h. Navigational aids and positioning systems greatly aid in unit movement.

i. Thermal night vision capability gives the tanks the ability to see farther than the passive sights of the infantry.

0353. One of the greatest dangers in a night attack comes from the risk of friendly fire. Some techniques to help reduce the risk of fratricide include:

a. Improving gunner vehicle identification and thermal image identification skills.

b. Coordinating laterally before and during the battle. Knowing where units on the left and right will be, as well as their formations and their exact routes. During movement, everyone must be informed of relative locations and any route changes or delays.

c. Ensuring that subordinate leaders understand all aspects of the operation. Rehearse the plan over similar terrain, at night if possible. Conduct brief backs with each key leader (including attachments).

d. Tightening control measures, especially when units are at the front or in a known direction. Use weapons hold or weapons tight status (for example, weapons hold left).

Tank Platoon in Offensive Operations

Direct Fire Planning

0354. The platoon offensive fire plan establishes how direct fires will be used to support movement or other actions during an operation (assault or support by fire, for example). The platoon leader considers the following factors when developing his plan:

a. Enemy situation.

b. Friendly situation.

c. Platoon's mission.

d. Commander's scheme of maneuver and plan for FS.

e. Ammunition status and plan for resupply.

f. Special signals and communications to be used.

0355. The following issues should be addressed in the plan:

a. Engagement of known enemy positions.

b. Use of fire or smoke to conceal or cover movement.

c. Any support by fire missions.

d. Graphics used to control fires.

e. Engagement techniques or fire patterns if different from unit SOP.

Techniques of Engagement

0356. The platoon uses one of three techniques to engage enemy units.

a. **Assault.** Assault fire is employed when the platoon is assaulting the enemy. All weapons are used in assault fire.

b. **Support by Fire.** Support by fire is used to kill or suppress an enemy position while other elements move, assault, or withdraw. The platoon leader can direct the intensity of fire in his OPORD, or he may have to determine the intensity during the operation. When delivering supporting fire, the platoon can expect to draw enemy return fire. Hull-down positions are desirable, and firing tank should alternate firing positions.

c. **Surprise and Ambush Fires**. In the offense, targets can appear on the battlefield without warning. These targets must be immediately engaged and destroyed to maintain momentum. Platoon battle drill procedures provide for positive action when such targets appear. The first tank to observe surprises the target to engage them immediately and the remaining platoons provide suppressive fire or move to support the engaged platoon.

Mounted Movement

0357. **Techniques of Mounted Movement.** The following are the primary mounted movement techniques:

a. **Traveling**. Traveling is characterized by continuous movement of all elements and is best suited to situations where enemy contact is unlikely and speed is important. This is the most likely technique when infantry rides inside the tank.

b. **Traveling Overwatch**. Traveling overwatch is an extended form of traveling that provides additional security when contact is possible but speed is desired. The lead element moves continuously. The trail element moves at various speeds and halts periodically to overwatch the movement of the lead element. The trail element maintains a minimum distance of 500 meters, depending on terrain, to permits the movement in case the lead element is engaged.

c. **Bounding Overwatch**. Bounding overwatch is used when contact is expected. It is the most secure and slowest movement technique. There are two methods of bounding: 1) Alternate bounds. The lead element halts and assumes overwatch positions. The rear element advances past the lead element and assumes overwatch positions. The initial lead element then advances past the initial rear element and assumes overwatch positions. Only one element moves at a time. This method of bounding is usually more rapid than successive bounds. 2) Successive bounds. In this method, the lead element, covered by the rear element, advances and assumes overwatch positions. The rear element advances to a position abreast of the lead element, halts, and occupies overwatch positions. The lead element then moves to the next position. Only one element moves at a time, and the rear element does not advance beyond the lead element. Successive bounding is easier to control and more secure than alternating bounds.

0358. **Overwatch**. When tank elements are given the task to overwatch, they should occupy positions that offer cover and concealment, good observation positions, and clear fields of fire. Elements occupying overwatch positions should:

a. Visually check the security of the positions. Occupy hull-down firing positions.

b. Assign sectors of fire.

c. Orient weapons on likely or suspected enemy positions.

d. Search for targets (see Figure 3.17).

e. Use thermal sights to find heat sources not visible to the naked eye. Thermal signatures may reveal tanks or troops in tree lines or wooded areas; towns or villages; depressions; or potential observation points, such as church steeples, silos, or hilltops.

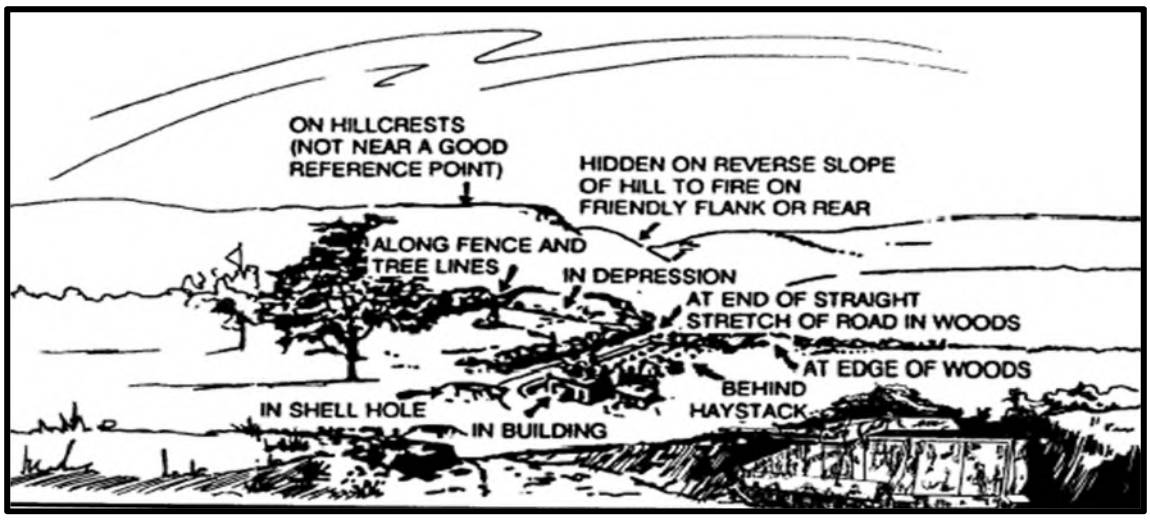


Figure 3.17 – Search for potential target

Methods of Tank/Infantry Movement

0359. When infantry and tank move together in any operation, the infantry moves using one of three methods: dismounted, truck-mounted, or tank-mounted.

0360. **Dismounted.** There are two ways dismounted infantry and tank can move together: tank lead, followed by the dismounted infantry, or infantry leads and tank follows.

0361. Movement with tanks followed by infantry is used when the terrain is relatively open and the infantry has little or no cover and concealment. Moving behind the tank vehicles provides some protection and concealment from enemy small-arms fire.

0362. Dismounted infantry followed by tank is used in restricted terrain when visibility is limited. Infantry provides security for tanks, clearing lanes or zones in front when fields of fire and observation are limited. Considerations for using dismounted movement includes the following:

a. Speed is reduced to dismounted elements. If tanks lead, leaders must guard against leaving infantrymen too far behind.

b. Communications between infantry and tank elements can be accomplished through radio, hand-and-arm signals, or the external telephone mounted on the tank.

c. Prearranged signals (flags, smoke, panels, or hand-and-arm) are coordinated and practiced to ensure close team cooperation.

d. Infantry do not move in front of vehicles unless told to do so.

0363. **Truck Mounted**. In some cases, the infantry/tank team may have trucks and HMMWVs or KM 450s/250s to transport infantry. This method may be used in several situations:

a. When traveling over long distances to a LD.

b. When speed is important and outweighs the risk of exposure to enemy tire. Examples include penetration, breakout from encirclement, exploitation, or pursuit.

c. When tank and truck-mounted infantry move together, the tanks usually move in front to provide protection and firepower. One platoon or section may also be placed on the rear of the infantry formation for protection.

0364. **Tank Mounted**. In the infantry division, some of the infantrymen in the infantry/tank team may be transported on tanks. This occurs mostly during a march, exploitation, or pursuit. This method has the advantage of maintaining speed while keeping the infantry and tank together. It also saves infantrymen from fatiguing dismounted marches. The decision to carry infantry on tank requires careful planning. Tactical unity of the infantry and tank teams must be maintained. Infantry leaders should mount tank leaders' vehicles. Usually, an infantry squad (nine infantrymen) can fit on one tank. During planning, each squad should link up with the vehicle to allow the squad and tank crew to work together and rehearse mounting, dismounting, and action drills. Infantry platoon leaders should ride platoon leaders' tank, and infantry platoon sergeants should mount at the platoon sergeants' tank. The infantry leaders should locate next to the tank commanders (see Figure 3.18).

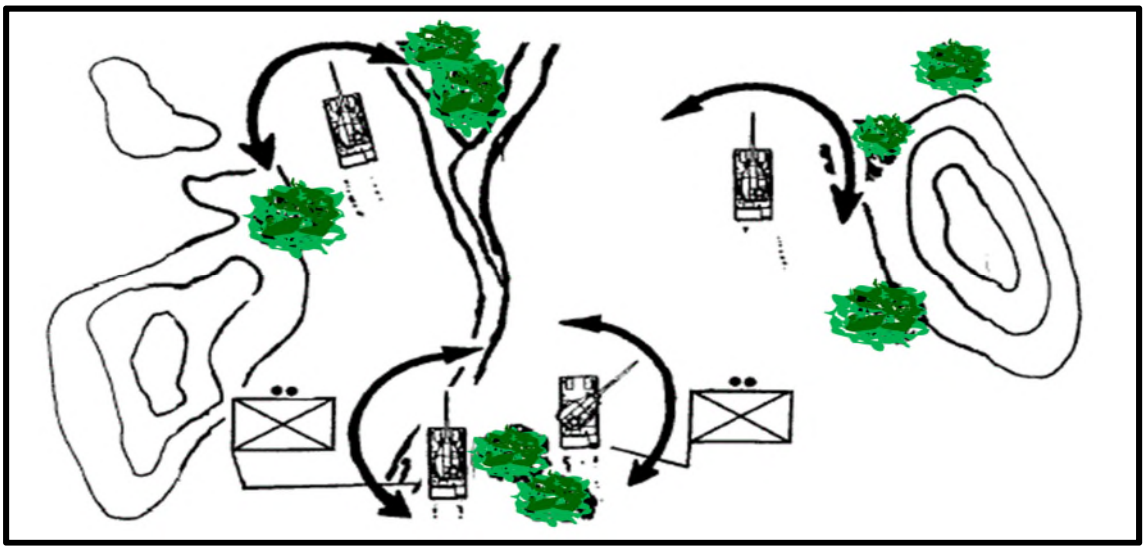


Figure 3.18 - Tank formation with mounted infantry

0365. The following are considerations for infantry and tank leaders when mounting infantrymen on the tank:

a. Always alert the tank commander before mounting or dismounting.

b. Follow the commands of the tank commander.

c. Infantry platoons should be broken down into squad-size, with the infantry platoon leader on the platoon leader's tank and the infantry platoon sergeant on the platoon sergeant's tank.

d. Whenever possible, mount and dismount over the ramps and side doors of the tank. This ensures the driver to see the infantry and does not pass in front of the machine guns.

e. Rider should take the following actions on contact: 1) Wait for the vehicle to stop. 2) Dismount IMMEDIATELY on the vehicle commander's command (one fire team on each side if situation permit). DO NOT move forward of the turret. 3) Move at least 5 meters from the vehicle.

f. Practice mounting, dismounting, and actions on contact so your team will be trained.

g. DO NOT move in front of vehicles unless ordered to do so.

h. DO NOT move off a vehicle unless ordered to do so.

i. DO NOT smoke when mounted on a vehicle.

j. DO NOT stand near vehicles during refueling and rearming.

k. DO NOT stand near a moving or turning vehicle at any time. Tanks have a short turning radius.

l. Stay clear of the vehicle's canister ejection device. Canisters ejected from main gun rounds can cause serious injury or death.

Fire and Maneuver

0366. In fire and maneuver, one element maneuvers while another overwatches or supports by fire to suppress or destroy the enemy. Maneuver elements use movement techniques and covered and concealed routes to maneuver to the enemy's flank, maneuver to dominating terrain, or bypass. When the company commander uses one platoon to support by fire, the platoon occupies dominating positions and suppresses the enemy (see Figure 3.19).

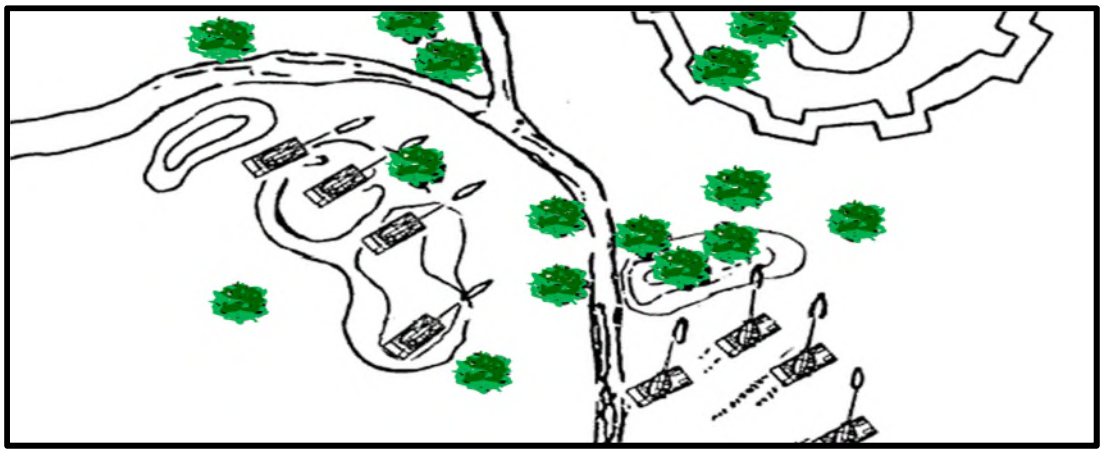


Figure 3.19 – Fire and maneuver

0367. **Support by Fire Element**. The company commander designates element to support by fire while the remainder of the company moves for an assault. The support by fire element's primary mission is to destroy as much of the enemy as possible by long-range fires before the assault. The element uses direct and indirect fires to prevent the enemy from engaging the assault unit or from adjusting positions to counter the assault unit.

0368. The support by fire element monitors movement of the assault unit; it shifts fires as the assault unit begins to move across the objective. To be effective, the element should be positioned on dominating terrain overmatching the enemy position. The distance between the support by fire element and the enemy position will vary.

Several factors should be considered in positioning of the support by fire element:

a. Available terrain. These considerations apply: 1) What terrain dominates the enemy position? 2) What dominant terrain supports the scheme of maneuver of the assault unit?

b. Enemy weapon system capabilities. These considerations apply: 1) Does the enemy have tanks, ATGMs, attack helicopters, or prepared antitank positions? If so, the support by fire position should be 1,500 to 3,000 meters from the enemy to maximize the advantages of the tank in an open terrain. 2) Is the enemy dismounted infantry? If so, the element is positioned to allow effective machine gun fires (900 meters). 3) Does the enemy use mounted infantry? If so, the element is positioned beyond 1,000 meters, then repositioned closer once enemy vehicles are destroyed or enemy dismounts. 4) Does the enemy have a combination of vehicles, mounted infantry, and dismounted infantry? If so, the element is positioned between 1,500 and 3,000 meters to destroy enemy tanks, then moves closer to destroy enemy infantry.

c. Support by fire weapons capabilities. These considerations apply: 1) The tank provides the crew with armor protection. 2) The tank can implement a ballistic solution to 5,000 meters or more. 3) Maximum effective range of the 7.62mm coaxial machine gun is 900 meters. 4) Maximum effective range of the Cal .50 machine gun is 1,800 meters and the maximum effective range of Tank Guns is ranges from 3,500 to 5,000 meters.

d. Time available. These considerations apply: 1) How much time is available to position the overwatch element? 2) Is any platoon currently in contact with the enemy?

0369. **Assault Unit.** The mission of the assault unit is to close with and destroy the enemy. Normally composed of tanks and infantry elements under the control of the company commander, the assault unit moves along covered and concealed routes to the flanks or rear of the enemy. The elements move until they reach their last covered and concealed position (assault position). Once the commander has determined that all observed enemy vehicles and antitank weapons on the objective have been destroyed or suppressed, he orders the assault. The assault elements move rapidly in a line formation, under cover of direct and indirect fires, to the objective. En route, they

engage enemy targets on the move (stabilized) or from a temporary halt (non-stabilized). The assault unit calls for the lifting and shifting of supporting fires (direct and indirect) at the fire support coordination line (FSCL).

0370. **Attack by Fire.** Some objectives lend themselves to an attack by fire. The purpose is to destroy the enemy from a distance. This method can be used when the enemy consists of tanks and the mission does not dictate or support occupation of the objective. The support by fire element suppresses the enemy while the assault unit moves in dominating terrain and, with the support by fire element, completes the destruction of the enemy (see Figures 3.20).

Battle Drills

0371. These drills are a basic guide to infantry/tank combined arms tactics. Tank platoons and the infantry they support, should be able to execute them upon command. Repetition in training, conducted to standard, is the key to the proper execution of drills in combat. When habitual relationships are maintained, the execution of such drills is enhanced. If habitual task organization is not possible, rehearsals are key to proper performance.

0372. This discussion describes the drills that each tank platoon must become familiar with and be able to execute, whether working alone or with infantry. Each drill will show infantry moving dismounted with the platoon, as well as mounted where applicable.

0373. Platoons must practice drills to prepare for the C2 problems inherent in battle. Drills teach platoon members virtually automatic responses to combat situations, outlining actions to be taken immediately upon contact or in response to brief oral commands or visual signals. Crews and units gain proficiency only through practice before the battle. Drills can be carried out from any formation or movement technique.

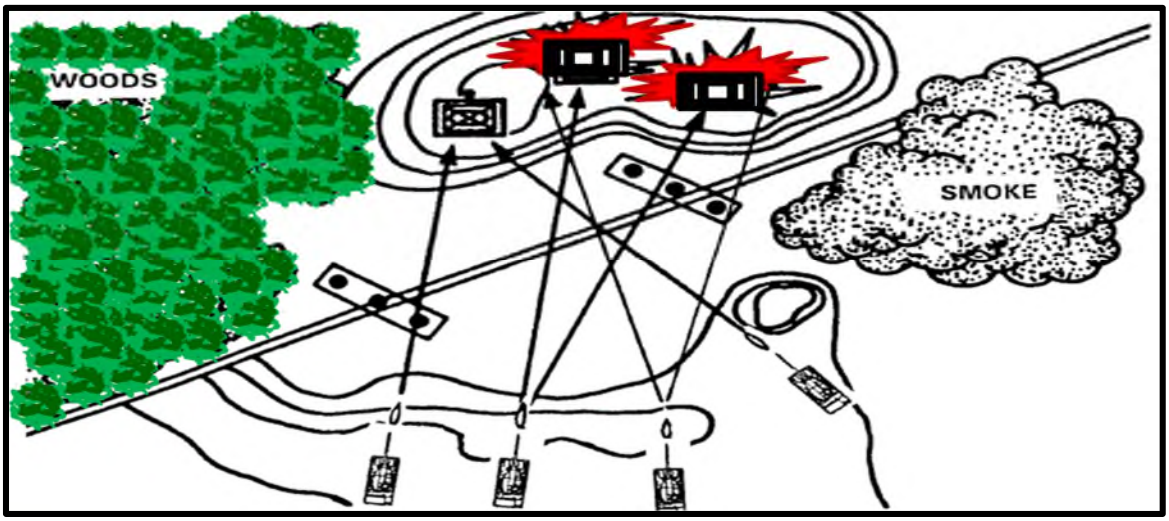


Figure 3.20 - Attack by fire

0374. **Actions on Contact.** When enemy fire is encountered, the platoon leader should execute the following actions on contact:

a. Return fire and alert the rest of the platoon.

b. Initiate a battle drill (action drill or contact drill). If no drill is specified, the platoon should seek cover and concealment.

c. Send a contact report to the company commander.

d. Develop the situation through fire and movement to determine the size, type, and location of enemy units.

e. Send a spot report to the commander. The platoon may destroy the target if initial fire is effective, or it may have to continue fire and movement to fix or destroy the enemy (based on instructions from the company commander).

0375. **Change of Formation Drill**. To accomplish a rapid change of formation, each tank commander must know the following information:

a. The new formation.

b. The relative position of each tank in the new formation.

c. The position of infantrymen if they are moving dismounted within the formation.

0376. The platoon leader can use hand-and-arm signals or the radio to inform the platoon of the new formation. He should always use hand-and arm signals when dismounted infantrymen are in the vicinity, regardless of the method used to communicate to the other vehicles. Each tank commander will know his position in the formation by following an SOP that specifies tank positions when assuming each formation. Figure 3.21 shows the movement of individual tank during the change from column to wedge to line. The key to a successful change is practice.

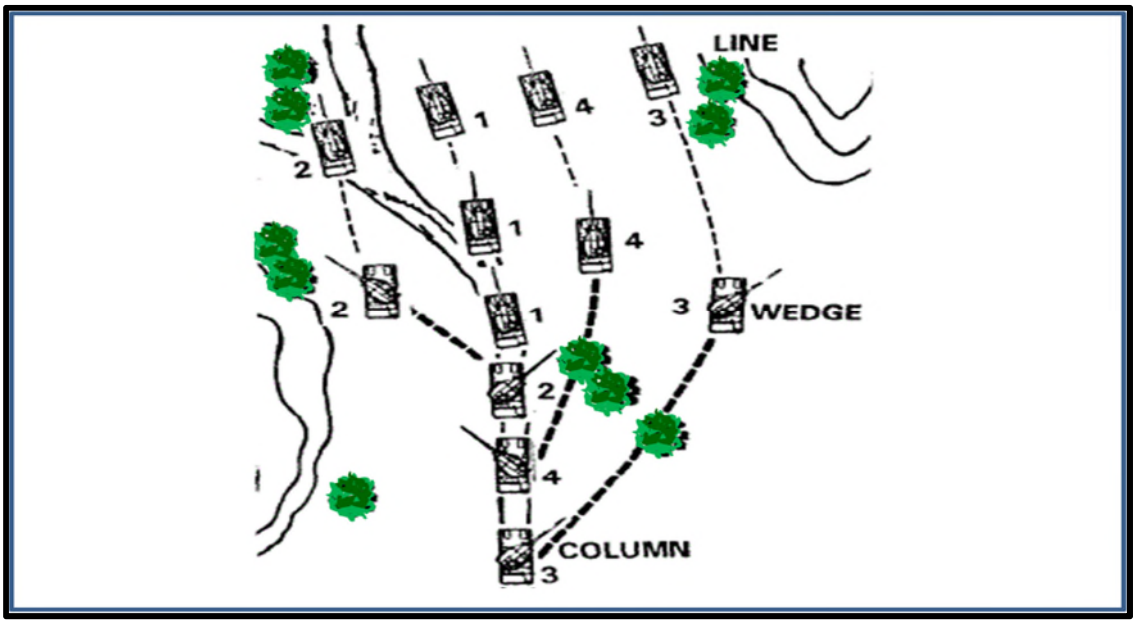


Figure 3.21 – Change of formation drill

0377. **Contact Drill.** Contact drills teach the platoon on how to orient weapon systems and engage an enemy without changing the direction or speed of movement along the axis of advance. This can be used when contact is made with small fire arms or when the platoon sights the enemy without being detected and does not want to stop or slow its momentum. The platoon leader can initiate a contact drill by hand-and-arm signals or radio. Even when using the radio, he should also use hand-and-arm signals when moving with dismounted infantrymen.

0378. **Action Drill.** Action drills permits the platoon to change direction when reacting to changes in terrain or enemy contact.

0379. **Changes in Terrain.** Using the wingman concept, this change occurs automatically when the platoon leader's tank changes direction. To speed up a change in direction, the platoon leader can direct an action drill using hand-and-arm signals or the radio. After executing the change of direction, the platoon automatically comes on line and continues to move. To return in the original formation, a hand-and-arm signal or radio command can be given. Figures 3.22 through 3.24 show the tanks' relative positions during changes of direction.

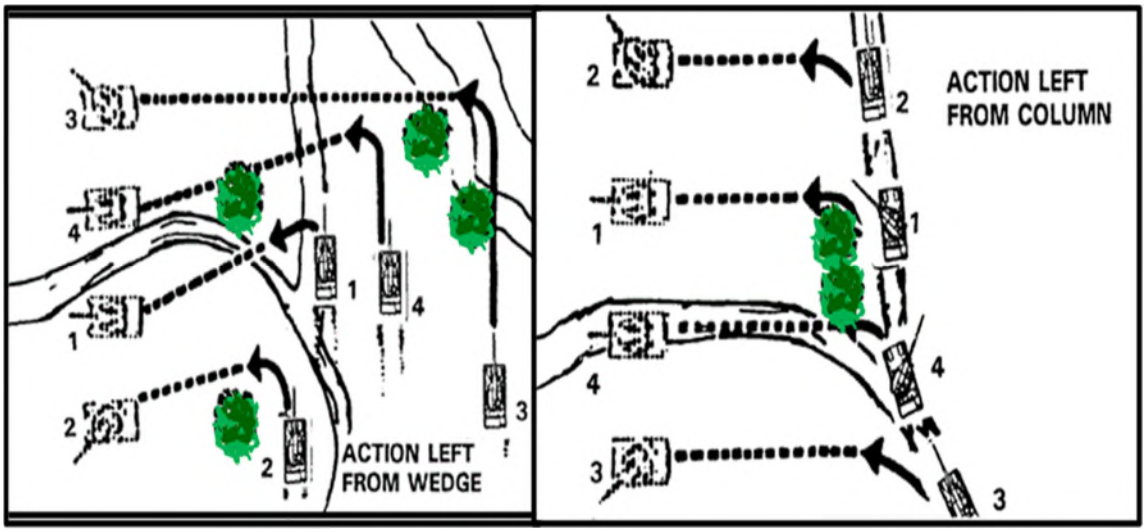


Figure 3.22 – Action left drills.

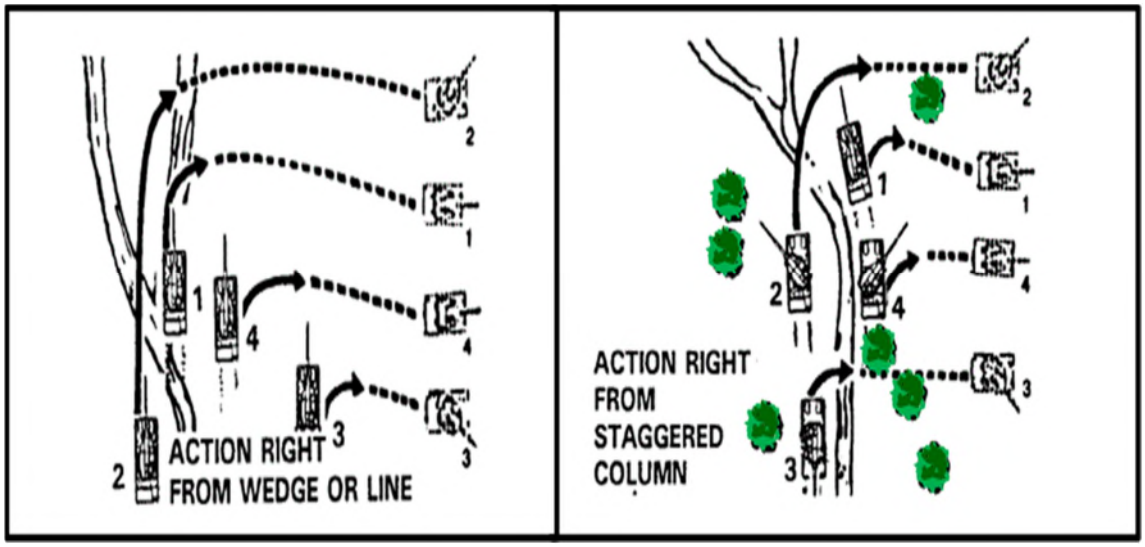


Figure 3.23 – Action right drills.

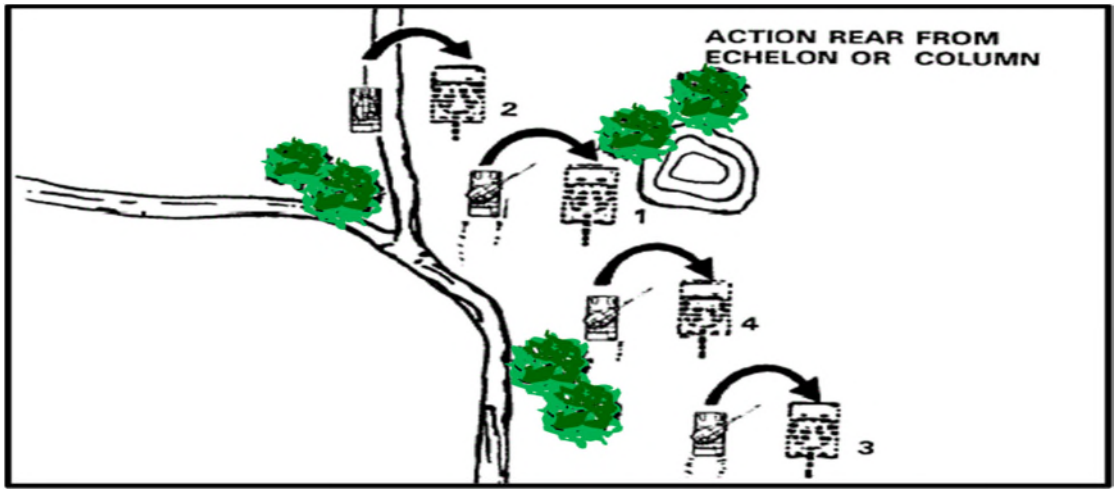


Figure 3.24 – Action rear drills

0380. **Enemy Contact**. Following a contact report involving antitank weapon systems, the platoon leader can direct an action drill to orient the platoon's frontal tank towards the anti-tank fire while moving to cover and concealment. Figures 3.25a through 3.25e show examples of action drills when used to react to enemy contact.

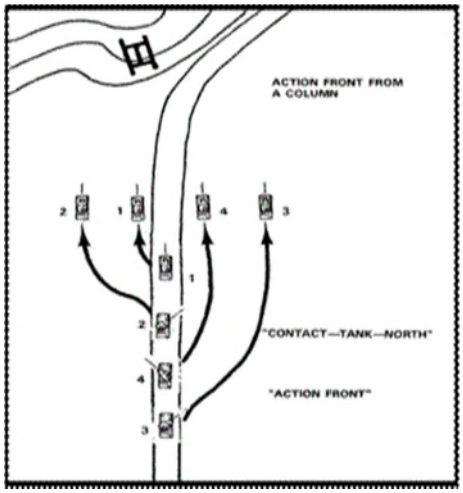


Figure 3.25a – Use of action drill to react to enemy contact (action front from a column)

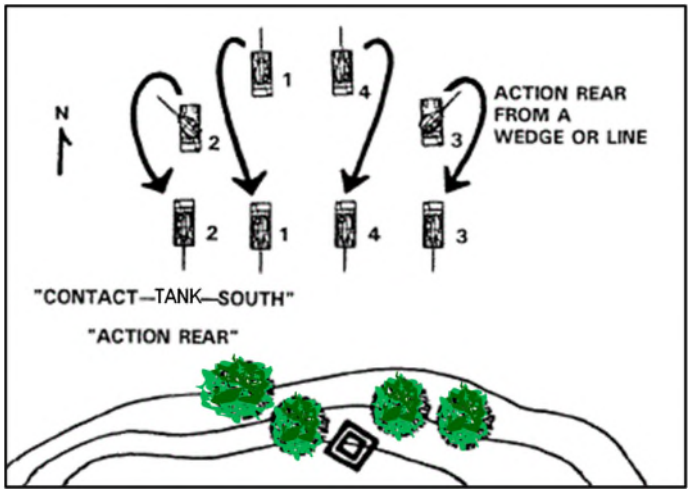


Figure 3.25b – Use of action drill to react to enemy contact (action rear from a wedge or line).

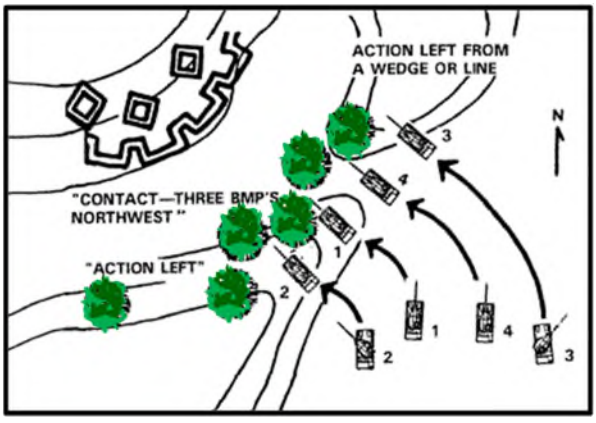


Figure 3.25c – Use of action drill to react to enemy contact (action left from a wedge or line).

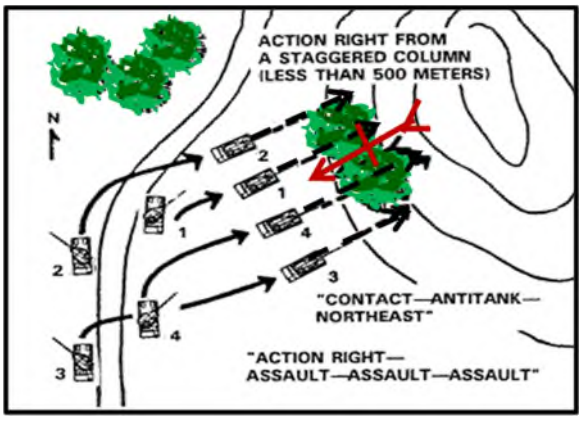


Figure 3.25d – Use of action drill to react to enemy contact (action right from a staggered column).

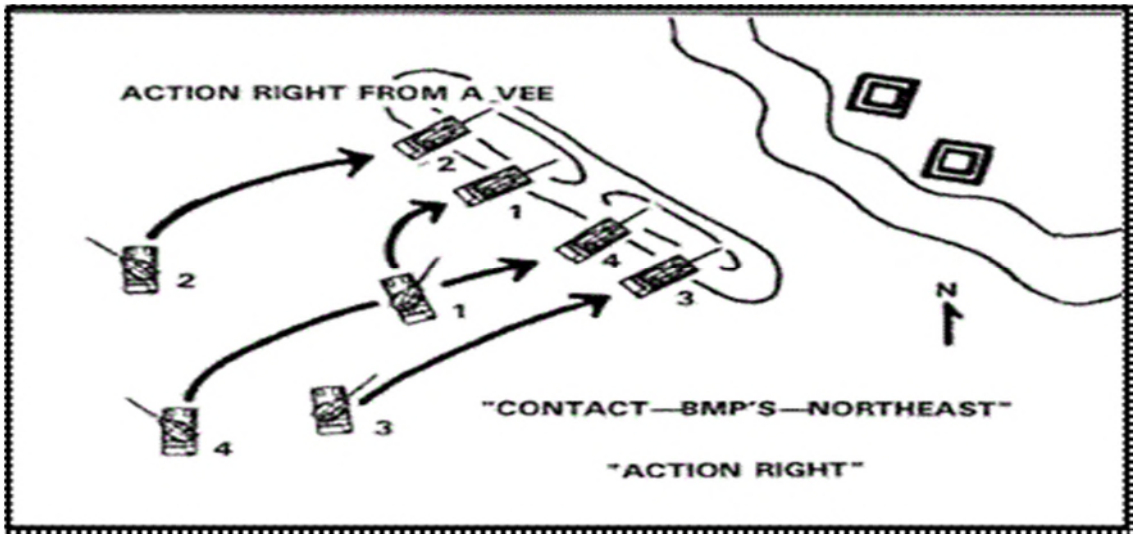


Figure 3.25e – Use of action drill to react to enemy contact (action right from a vee).

0381. **React to Indirect Fire Drill**. When the platoon receives indirect fire, crewmen close their hatches and ballistic doors. If on the move, the platoon maintains their speed and direction. With infantry mounted, the platoon leader may issue the command for an action drill to avoid casualties to the infantry. If maneuvering with infantry, a common rally point is desirable. The platoon leader sends a spot report to the higher headquarters. If the situation permits, a more detailed shelling report (SHELREP) follows the spot report. If the mission requires the platoon to remain stationary, permission must be obtained from the higher headquarters before moving. Once clear of the indirect-fire effects, crews can open hatches and necessary ballistic doors. If masking was required by the OPORD (for example, when the enemy has NBC capability), the platoon leader has the flexibility to modify the mission-oriented protection posture (MOPP) guidance based on immediate test results. See Figure 3.26 for an example of this drill.

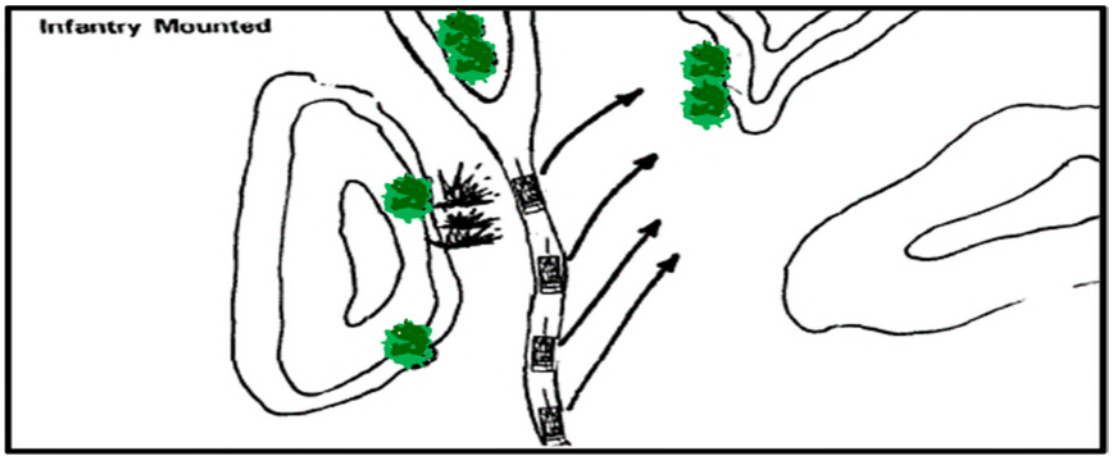


Figure 3.26 – React to indirect fire drill.

0382. **React to Air Attack Drill**. The platoon should practice passive defense against air attack. Use of cover and concealment can frequently prevent high-performance aircraft and helicopters from detecting and attacking the platoon. The air attack drill

involves the four steps discussed in the following paragraphs. See Figure 3.30 for example.

0383. **Alert the Platoon**. Air guards can alert the platoon using one of two methods: announcing "CONTACT—ENEMY AIR—-(direction)" over the radio or using hand-and-arm signals.

0384. **Seek Cover and Concealment**. When moving, tanks seek immediate cover and concealment. If concealment is not available, moving vehicles should stop. A stationary tank is harder to see than a moving one. If enemy aircraft detect the vehicles and initiate an attack, the platoon leader announces "ENEMY AIR" and exposed vehicles to immediately move at a 45-degree angle toward or away from the attacking aircraft. Tanks should maintain a 100-meter interval and avoid presenting a linear target in the direction of attack. See Figure 3.27 for a depiction of this drill.

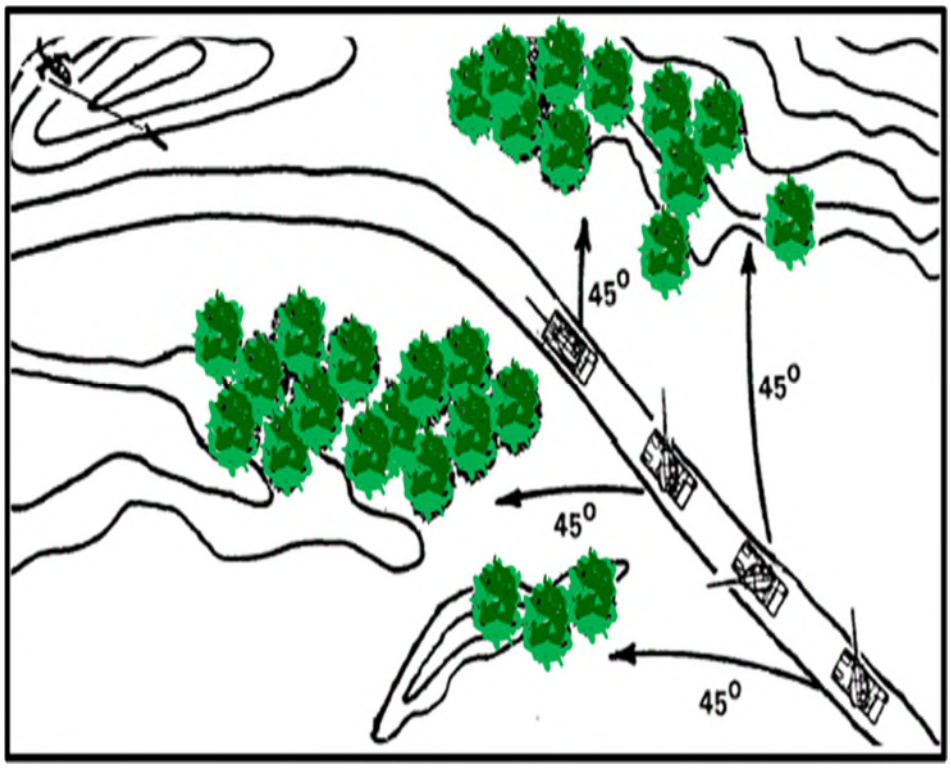


Figure 3.27 – React to air attack drill.

0385. **Prepare to Engage.** Tank commanders should prepare to engage aircraft with a high volume of machine gun fire on order of the platoon leader. The platoon leader must be sure that the aircraft are attacking, since firing machine guns could give away their positions. Volume is the key to effectiveness. The idea is to throw up a wall of fire and let the aircraft fly into it. In some cases, the main gun can be used against hovering helicopters. Figure 3.28 illustrates aiming points for engaging enemy aircraft.

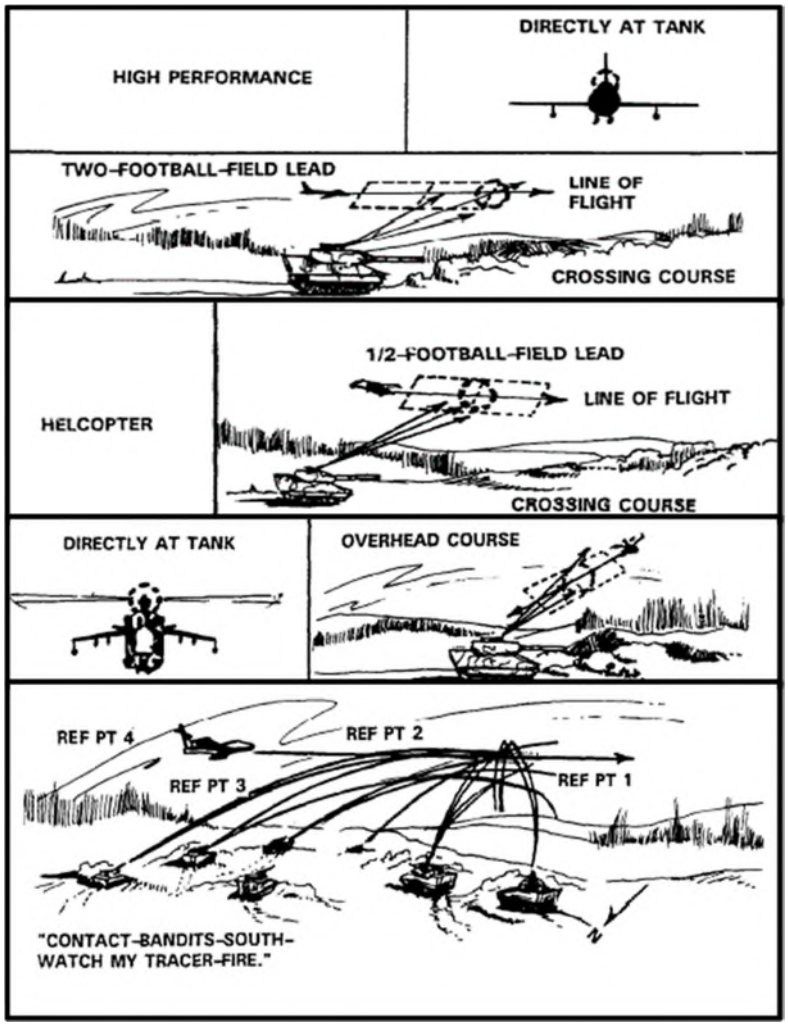


Figure 3.28 – Aiming point for engaging enemy aircraft

0386. **Report.** The platoon leader sends the higher headquarters a contact report. Example: "CONTACT—HELICOPTER—SOUTH." The platoon leader sends a complete spot report as soon as possible. If the platoon is engaged by bombs or spray, the reaction to indirect fire drill is used. Enemy aircraft operates in pair, with two to eight or more aircraft in each flight. After the first aircraft passes overhead, another may follow. Tanks should remain in covered and concealed positions for at least 60 seconds after the first aircraft leaves.

0387. **Break Contact Drill.** When the tank platoon makes contact with a superior unit, the commander may give the order to break contact. The voice command is "BREAK CONTACT," followed by a direction (clock method, or front, rear, left, or right).The visual signal is a red flag raised, then dropped in the direction, the break is to be conducted (see Figure 3.29). The following are steps in successfully executing a break contact fire and movement drill:

a. **Make the initial break**. The commander designates a support by fire element to provide overwatch and supporting fire, if needed. Dismounted infantry moves to a designated covered position in the direction of the break (clock method). If

infantrymen are mounted on the vehicles not in contact (not the support by fire unit), they may remain mounted as the vehicles move to break contact.

b. **Tanks disengage**. The support by fire element uses successive bounds to join infantry. These vehicles may employ smoke as needed. The bounds continue until contact is broken.

c. **Report. The** platoon leader sends a spot report to the higher headquarters and prepares to continue the mission.

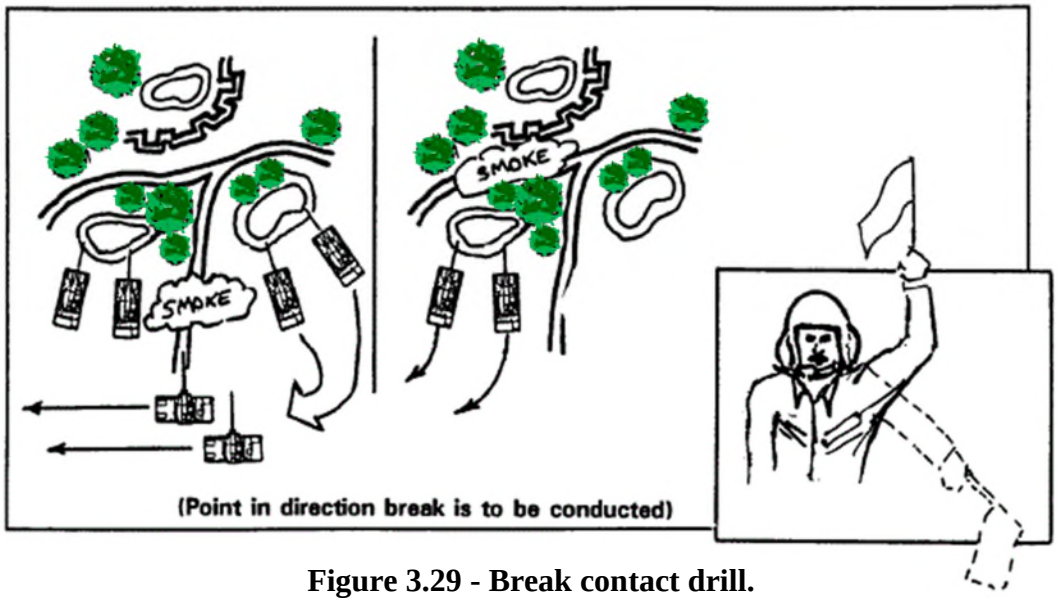


Figure 3.29 - Break contact drill.

0388. **Attack an Objective Drill**. This drill can be used during a hasty attack as a reaction to enemy contact. The voice command is "ASSAULT, ASSAULT, ASSAULT— (followed by a direction)." Steps for execution of this drill, which is illustrated in Figure 3.30, are as follows:

a. The commander or platoon leader must determine and announce how the basic maneuver will be conducted. 1) Tank attacks by fire while infantry assaults. 2) Tank attacks by fire while infantry assaults, then tank joins in the assault. 3) Tank and infantry assault the objective on different axes. 4) Tank and infantry assaulting together.

b. If necessary, infantrymen dismount the vehicles and move to the flanks.

c. The designated element establishes a base of fire and suppresses the objective. Deception smoke is used to confuse the enemy as to the true location of the assault unit. This support by fire position may later serves as the rally point for all elements if needed.

d. The assault element maneuvers to the assault position (nearest covered and concealed position).

e. The support by fire element shifts tires when the prearranged signal is given by the assault element as it reaches the final coordination line.

f. The assault element conducts fire and movement onto the objective, secures it, begins consolidating its position, and reports to the commander or platoon leader.

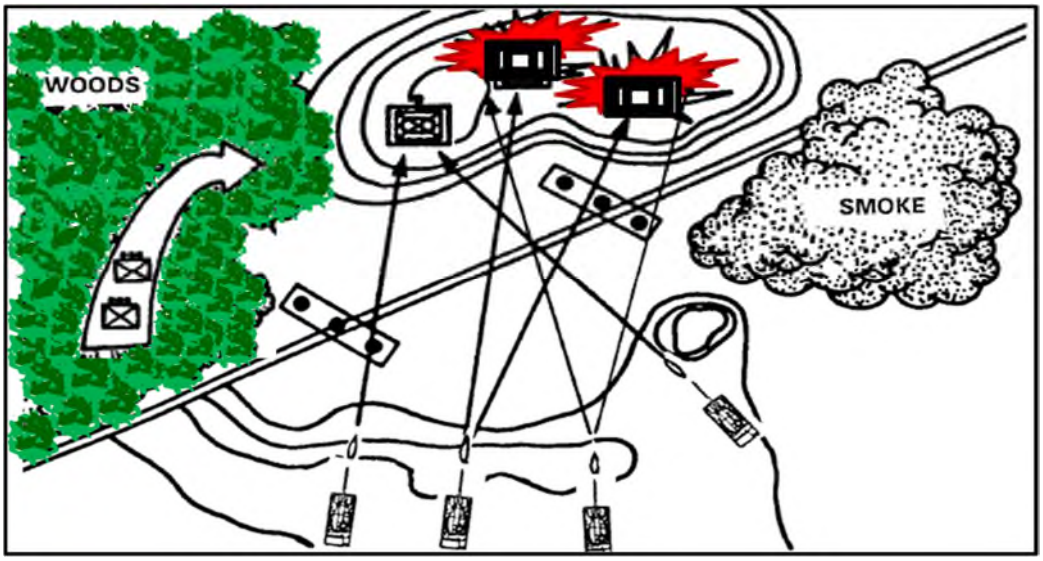


Figure 3.30 – Attack an objective drill.

Actions at Danger Areas

0389. A danger area is any place that the tank platoon, or any unit maneuvering with the tank platoon, might be exposed to enemy observation, fire, or both. If a danger area cannot be avoided, tank crosses it with great caution and as quickly as possible. Tank and infantry maneuvering together uses the same techniques, using tank's overwatch capabilities and infantry's ability to clear areas where hidden enemy infantry may engage friendly units.

Types of Danger Areas

0390. **Open Areas.** The infantry on the near side observes and provides local security, including the near side flanks. Using an infantry guide, tank maneuvers to the flanks to provide overwatch while the infantry moves across in open area to secure the far side (see Figure 3.31). When cleared, the remaining infantry moves across at the shortest exposed distance as quickly as possible. The tank platoon then bounds by section across the open area and reassumes the previous designated formation. After clearing the far side, tank would provide flank security.

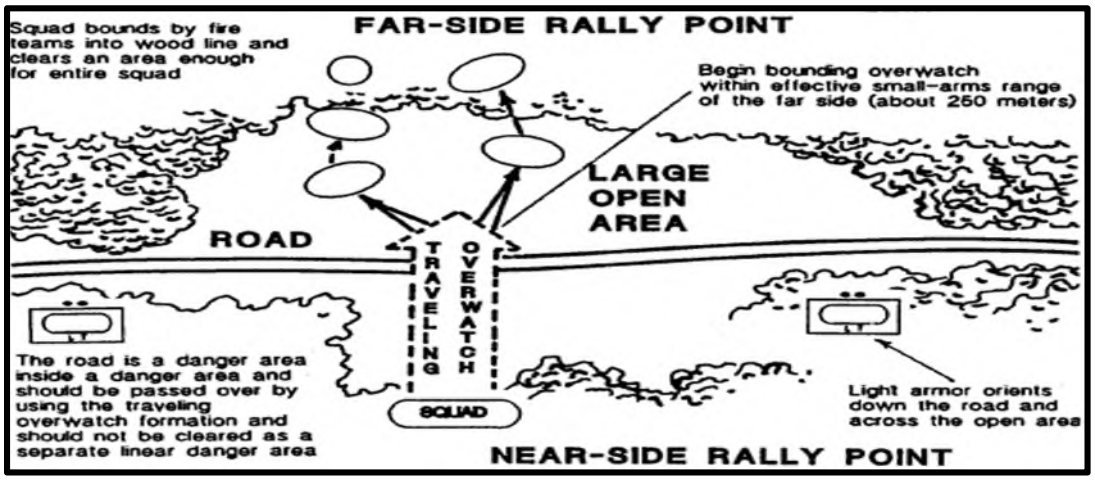


Figure 3.31 – Example of actions at a danger area with tank overwatching

0391. **Linear** (Roads and Trails). Cross roads or trails at or near a bend, a narrow spot, or on low ground. Use the same techniques for linear areas as those used for open areas, except the tank platoon may orient down the road or trail while providing near side flank security.

0392. **Defiles.** Crossing defile requires the infantry to clear the flanks of the defile (often high ground), the far side of the defile, and then the defile itself. If engineers are available they must help to locate and neutralize mines in the defile. Once the defile is cleared, tank moves through quickly and augments the infantry on far side security until the remaining infantry moves through or past the defile.

0393. **Built-up Areas.** Cities, Towns and Villages.

0394. **Trenches, Gullies, Wadis, and Tunnels.** Typically in operations other than war, the enemy will not have the resources to build fortifications and will maximize use of below-ground protection. Tank provides overwatch while infantry fights the below ground battle. Recognition signals such as smoke grenades, flaxes, and flags on antennas can indicate the location of friendly infantry to TANK crews. Tank must be prepared to engage enemy reinforcements or any retreating enemy as they go aboveground. Tank can also advance to engage the enemy with machine guns along the length of a trench, gully, or wadi.
Obstacles.

0395. **Planning Considerations for Crossing Danger Areas.** The following planning considerations apply for tank and infantry when crossing danger areas: Plan rally points, the use of hand-and-arm signals (for example, signals for danger areas or bypasses), and for the use of indirect fire targets around suspected danger areas.

a. Rehearse avoiding danger areas by practicing maneuvering around them.

b. Rehearse actions to be taken at unavoidable danger areas; upon enemy contact (both direct and indirect fires); to control friendly direct and indirect fires; when consolidating; and upon reassuming the formation.

0396. **Breach an Obstacle (In-stride**). The tank/infantry team may not be able to bypass a small or unmanned obstacle, so the commander determines to conduct an in-stride breach. The voice command for this exercise is “BREACH.” The hand-and-arm signal is shown in Figure 3.32. The steps for this drill are as follows:

0397. **Organization**. The commander should organize the team into three elements as described below:

a. **Support unit**. This is usually a tank platoon or section with an infantry weapons section (M60 machine gun and/or 60-mm mortar). It leads in the movement to the obstacle.

b. **Breach unit.** This consists of infantry and engineers if available. It follows the support unit to the obstacle.

c. **Assault unit.** This is usually infantry, but the commander may be able to add a tank section. It follows the breach unit when moving to the obstacle.

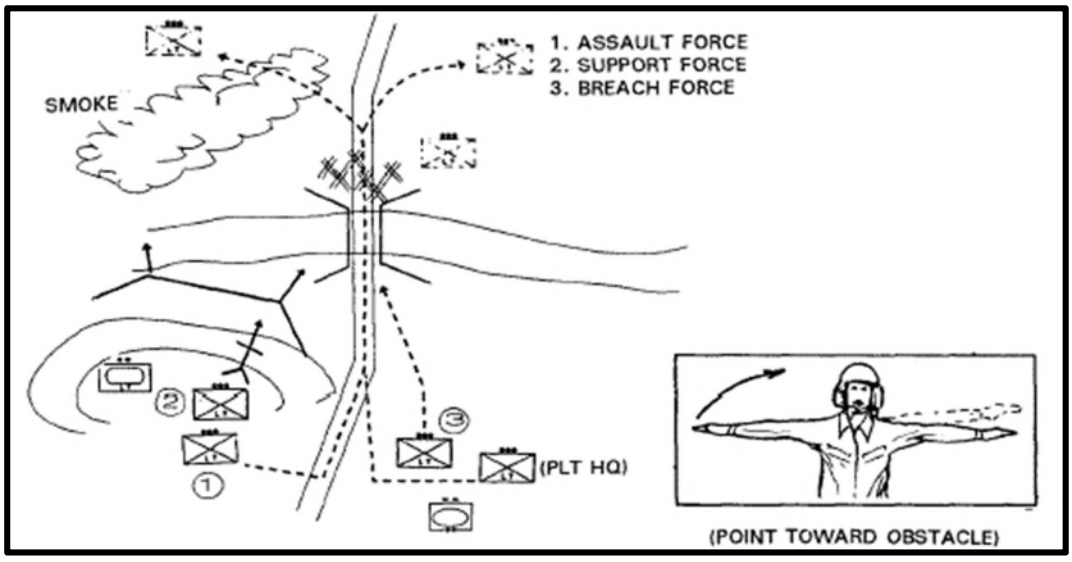


Figure 3.32 – Breaching an obstcle (in-stride)

0398. **Suppression.** The support unit (#2) establishes a base of fire and suppresses enemy direct-fire defenses. It must be able to suppress the enemy by direct and indirect fires, to include providing air defense coverage.

0399. **Obscuration.** The support unit adjust artillery-projected smoke on the far side of the obstacle to prevent enemy observation of the breach and assault units.

03100. **Security.** The breach unit (#3) secures the near side of the obstacle while the assault unit (#1) maneuvers to a better position to prepare for an assault. Ideally, the obstacle is breached after securing the far or enemy side. Friendly units from the support unit can reach the far side of the obstacle by infiltration, bypass, air assault, or minor breach.

03101. **Reduction.** The breach unit clears, marks, and secures the lane. The support unit continues to suppress and shift indirect fires and smoke while the assault unit is

poised to move quickly through the breach once it is cleared and marked. The assault unit moves through the breach, conducts below ground battle and provides far-side security by adding the shoulders against enemy counterattacks. The remainder of the breach unit, followed by the support unit, moves through the breach and continues the designated mission.

Movement to Contact

03102. A movement to contact is conducted to gain, maintain, or re-establish contact with the enemy. Once contact is made, units move quickly to develop the situation. The battalion makes contact with the smallest possible element to maintain flexibility and security. This is important for the infantry because limited mobility and dependence on restrictive terrain makes it quite vulnerable. Since movements to contact are usually characterized by lack of information about the enemy, commanders must plan for continuous and extensive reconnaissance and security. Movement to contact will terminate in a hasty attack or hasty defense. Two techniques are most commonly used by infantry battalions to conduct a movement to contact, an approach march and a search and attack.

03103. **Approach March**. This is an advance of a combat unit when direct contact with the enemy is imminent. Troops are fully or partially deployed. The approach march ends when ground contact with the enemy is made or when an attack position is occupied. Using this technique, infantry battalions normally organize into a security unit, advance guard, main body, flank guards, and rear guard. The guard elements move with and secure the main body. Figures 3.33 through 3.36 show examples of infantry battalion formations (with attached tank platoon) in a movement to contact using the approach march technique.

03104. **Search and Attack**. This technique is a decentralized movement to contact, requiring multiple, coordinated patrols (infantry squad- and platoon-size) to locate the enemy. It is most often used against an enemy operating in dispersed elements. When conducting a search and attack, infantry units spend more time operating in an AO rather than simply sweeping through it. Tank platoons can have great value during search and attack operations. They primarily serves as a reserve to conduct a hasty attack to defeat the enemy by assaulting critical sites (CPs, supply points) once the infantry has found and fixed enemy units. Figure 3.33 shows an example of a tank platoon conducting a hasty attack on a fixed enemy unit. Whether the purpose of the search and attack is destruction of the enemy, area denial, or unit protection, the critical execution factor for tank is always to be capable of rapidly massing combat power. In conducting movement to contact, infantry units protect the unit by:

a. Moving during period of limited visibility and using stealth and surprise.

b. Conducting thorough reconnaissance and being able to attack at the time and place of the commander’s choice.

c. Using all available combined arms assets.

03105. **Scouts** are usually employed well forward of the advance guard to conduct reconnaissance for the battalion movement. Tank elements should be given checkpoints

along the movement route from which they can cover the most likely enemy tank avenues of approach. Commanders and S3s must keep in mind that tanks need additional security units. When overwatching the movement of infantry, the tank platoon can either move from one dominating piece of terrain to another, move with the infantry, or move behind the infantry. The infantry commander must designate specific movement techniques and formations to reduce danger to the unit while moving. Once contact is made, platoons use fire and movement to develop the situation.

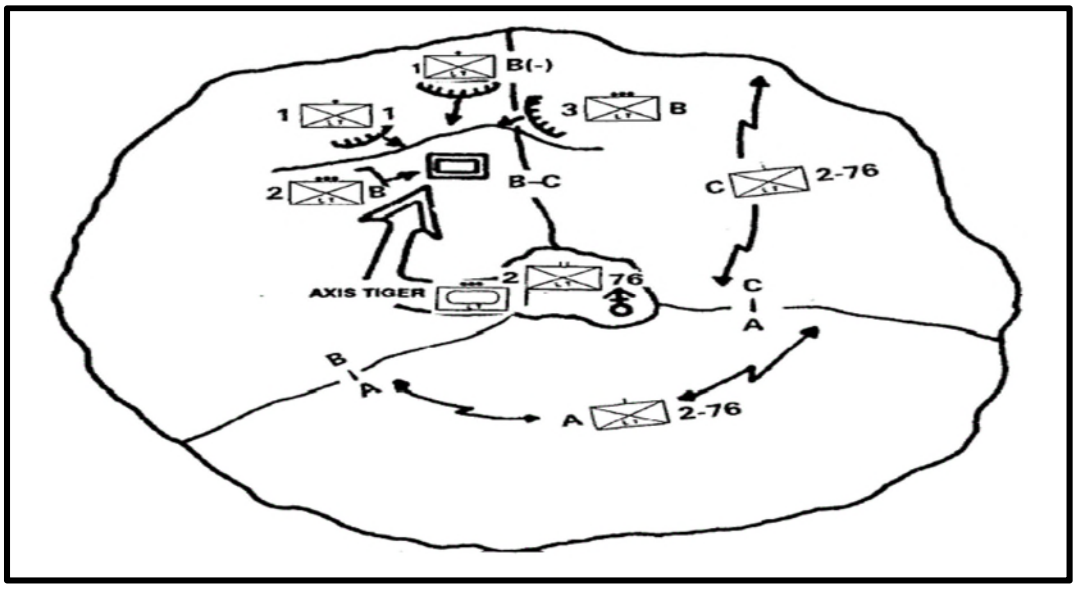


Figure 3.33 – Example of a tank platoon conducting a hasty attack on a fixed enemy force during search and attack operation.

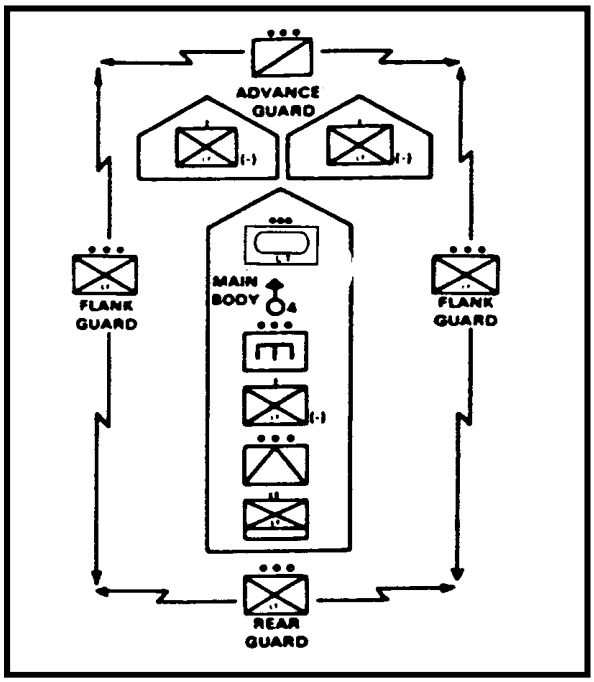


Figure 3.34 – Movement to contact using the approach march technique (battalion vee)

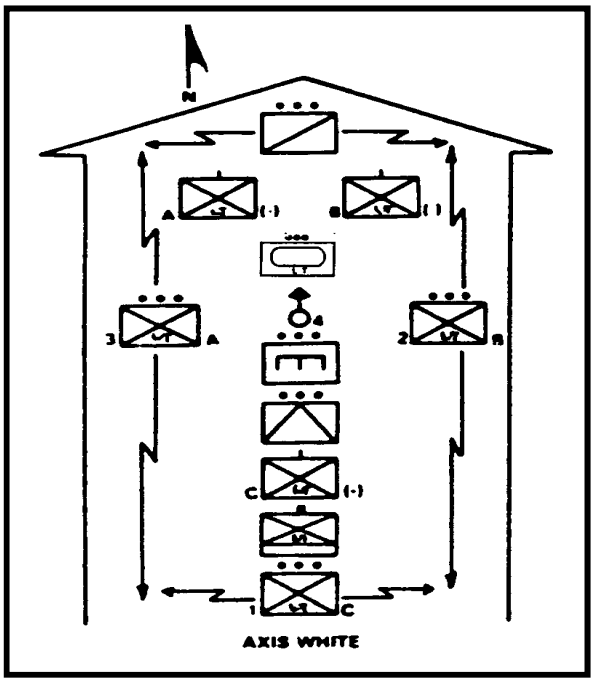


Figure 3.35 – Movement to contact (wedge)

03106. Normally, priority of fires will be given to the advanced guard until contact is made and the main body assumes the attack; at this point, the main body receives priority of fires. The advance guard may disperse or concentrate during movement to contact. This decision is based on METT-T. Units are concentrated when intelligence indicates the enemy is operating in company- or larger-size units or speed is a consideration. Units are dispersed when intelligence indicates the enemy is operating in dispersed, small units or coverage is more important than speed. When the movement to contact culminates in an attack, the battalion may attack from a wedge, vee, single-column, or multiple-column formation along the axis of advance or in a zone of action. The following considerations influence the decision of which formation to use:

a. A wedge is normally used to allow the battalion to mass faster; give greater flexibility, increase the probability of contact, and increase the ease of movement. The primary disadvantage of the wedge is difficulty to control. This formation allows the tank to be forward for quick response.

b. The vee formation is used to increase frontage, speed of reaction, and capability for envelopment. The disadvantages are that it is difficult to control and has fewer uncommitted units and a smaller reserve. In this formation, the tank platoon is farther back in the formation and requires greater response time.

c. An attack using a single column is normally used when time is not critical. The primary disadvantages of using a single column are that it is susceptible to enemy delay tactics and takes longer to get the rear company into action.

d. Multiple columns are normally used when speed is critical and wide deployment is necessary. The primary disadvantage of multiple columns is that C2 become more difficult.

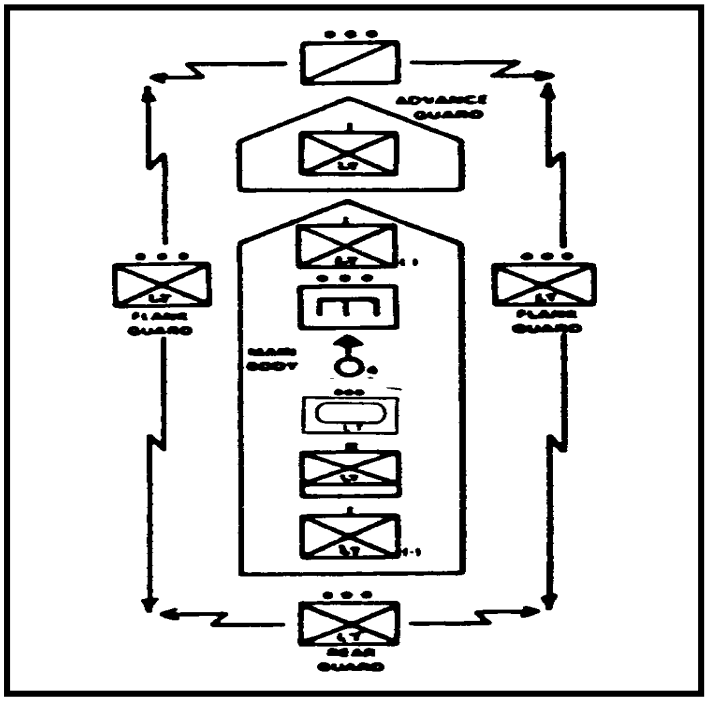


Figure 3.36 – Movement to contact (column)

Methods of Attack

03107. The tank platoon and the infantry can operate attacks in many ways. The infantry battalion commander, supported by the recommendations from his staff and the tank platoon leader will decide on which method to use based on the specific tactical situation. The following discussion examines the four most basic methods for conducting light/heavy attacks.

03108. **The tank platoon attacks by fire while the infantry assaults the objective**. The tank platoon occupies hull-down defilade positions until the infantry masks the tank fires (see Figure 3.37). The tank fires can be timed to divert the enemy's attention and cover the sounds of the infantry's approach or breach. This method is most often used when enemy antitank weapons or obstacles block the only possible tank avenue of approach. Close coordination between the heavy and light units is vital to ensure effective fire control and prevent fratricide.

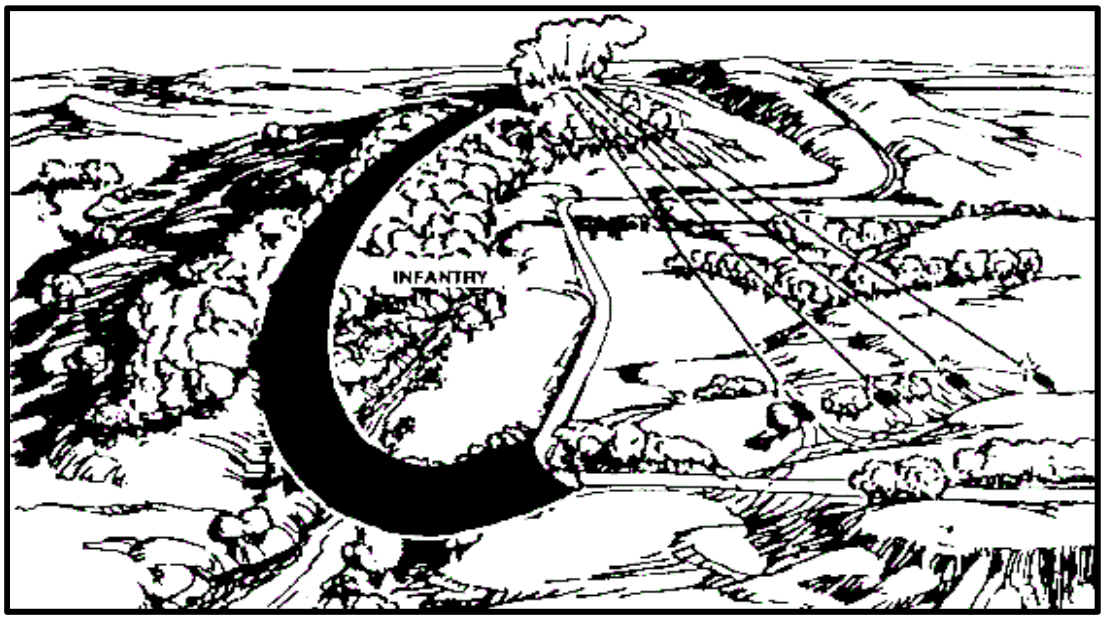


Figure 3.37 – Tank platoon attacking by fire

03109. **The tank platoon initially attacks by fire, then moves forward rapidly and joins the infantry for the assault.** As in the first method, the tank platoon first suppresses the objective from hull-down defilade positions while the infantry moves to an assault position.

03110. When the infantry masks the tank fires, or upon a prearranged signal, the tank platoon moves forward quickly and assaults slightly ahead of the infantry. This method is used when the enemy has prepared obstacles on the mounted avenues of approach. The infantry must first breach the obstacles and clear a lane for the tanks to reach the objective. Careful coordination and preparation of a detailed fire plan by the tank platoon are essential to keep indirect fires on the objective until the final assault.

03111. **The tank platoon and infantry converge on the objective from different directions and assault at the same time.** Open or partially open terrain that is free of mines and other tank obstacles is vital. Effective neutralization of enemy anti-tank

weapons by direct and indirect supporting fires and smoke is also necessary. However, neutralization is needed only during the time required for tanks to move from their LD to the near edge of the objective. The tank platoon must coordinate directly with the assaulting infantry for timing of the assault (the infantry will have to move earlier) and fire control on the objective. Even though this method provides surprise, increases the fire effect, and maximizes shock, actions on the objective are complex. The tank platoon must tightly control its fires while friendly infantry clears restrictive terrain on the objective.

03112. **The tank platoon and infantry advance.** The tank platoon may bound short distances, stop to fire, then bound forward again as the infantry comes abreast. This method is used when the enemy situation is vague; when the objective is very large and consists of both open and restrictive terrain; or when visibility, fields of fire, and MS movement are restricted, such as in fog, in towns, in woods, or at night. The tank platoon provides immediate, close, direct fires, and the infantry protects the tank from individual anti-tank measures. Rather than bounding, the tank platoon may move at the same rate of speed as the infantry. The infantry may follow closely behind the tanks for protection from small arms fire; in turn, the infantry protects the tanks from handheld anti-tank weapons by providing security to the flanks and rear. When this method of attack is used, it is imperative that the infantry and each MS use the external phones to coordinate fires and maneuver. Figure 3.38 depicts an attack with infantry and tank units advancing together.



Figure 3.38 – Tank platoon and infantry advance.

Types of Attack

03113. There are two types of attacks-the hasty attack and the deliberate attack. The main difference between them is in the depth of planning. All attack plans address intent, maneuver, and tires; all seek to strike a weak point, flank, or rear area of the enemy unit.

03114. **Hasty Attack**. Light units must seize every opportunity to destroy enemy with violence, offensive actions. The hasty attack is used when such an opportunity is presented and little time is available for detailed planning. It is also used to gain or maintain the initiative. A hasty attack can develop when—

a. A movement to contact results in contact.

b. A deliberate attack changes after it is under way.

c. Further advance is ordered after securing an objective.

d. A counterattack is ordered in the defense.

03115. **The commander must rapidly assess the situation, formulate a scheme of maneuver** and a supporting fire plan, and communicate the scheme of maneuver to his subordinates using FRAGOs. The unit then conducts the hasty attack using infantry, tank fire and movement. The primary employment of the hasty attack is in conjunction with a movement to contact, during which the unit may be assigned the mission of securing a terrain feature or of destroying an enemy unit. When enemy contact is made en route to securing an objective, the tank platoon may conduct the hasty attack with the infantry by:

a. Fixing and bypassing the enemy, depending on enemy strength and the unit's orders.

b. Attacking by fire to destroy the enemy and then bypassing.

c. Conducting a hasty attack to kill the enemy and continuing the attack to the objective.

0401. Depending on his orders and the size and location of the enemy, the commander develops a plan to conduct a hasty attack when enemy contact is made. He designates an objective, a support by fire element, the support by fire element's overwatch positions, an assault element, and covered and concealed routes into the flanks of the enemy. He then issues a FRAGO to his platoon leaders.

0402. The first unit to make contact with the enemy executes action on contact and assumes the role of the support by fire element. As discussed in the methods of attack, the commander employs the tank as the assault unit or to engage the enemy upon contact while he develops and issues his plan to assault the objective with infantry.

03116. **Deliberate Attack.** If the tank platoon can execute a hasty attack, it can execute a deliberate attack. A deliberate attack is distinguished from a hasty attack by a more detailed knowledge of the enemy; a larger amount of time devoted to planning, coordination, and preparation; and more extensive collection and use of intelligence. Once begun, the deliberate attack is executed with the same speed, violence, and application of concentrated combat power as a hasty attack. Units normally conduct a deliberate attack from defensive positions.

Offensive Operations in Built-Up Areas

03117. Because of the nature in the terrain, offensive operations in BUAs are normally conducted by dismounted infantry. Combat is characterized by house-to-house fighting; restricted observation, fields of fire, and maneuver space for tanked vehicles; and difficulty in C2. Tanks are employed as much as possible in close support of dismounted teams to secure locations and provide direct FS.

03118. **Types of Built-up Areas (BUAs)**. The following are general categories that help separate BUAs by type:

a. Villages, with populations of 3,000 or less.

b. Strip areas, which are BUAs along roads connecting towns or cities.

c. Towns or small cities, with populations of up to 100,000 and not part of a major urban complex.

d. Large cities and associated urban sprawl, with populations of up to 20 million and covering hundreds of square kilometers.

03119. **Basic Building and Street Patterns**. Each BUA or urban area has its own characteristics, making each objective a unique challenge for the infantry/tank team. The following list includes the typical building/street patterns that the army will encounter in urban terrain:

a. Dense, random construction.

b. Closed, orderly blocks.

c. High-rise areas.

d. Industrial/transportation facilities.

e. Dispersed residential areas.

03120. On the outskirts of suburbs, small plots of land, gardens, farms, fields, or vacant lots surround isolated houses or group of houses. The attacking unit usually starts at this part of the BUA. It should treat houses as inferior bunkers or individual emplacements.

03121. In the residential district, streets, gardens, and grassy plots usually flank closely spaced, detached, or semidetached buildings. The arrangement may or may not follow a geometric pattern. The center of the BUA is usually the business section. It will almost always consist of buildings of block-type construction, with little or no space between them, except for an occasional park, street or alley. This will require fighting from building to building and block to block.

03122. **Attacking in Built-up Areas.** A detailed study of the city and the enemy's dispositions in and around it forms the basis for planning the attack and seizure of a

BUA. Planning may include tanks for both maneuver and FS. The attacking unit is normally separated into two units-the enveloping unit (tank heavy) and the direct assault unit (infantry heavy). Follow the procedures and considerations listed below when attacking a BUA:

a. Dissipate the enemy's strength by causing him to react to demonstrations, feints, or ruses.

b. Concentrate overwhelming combat power to unit a quick and violent disruption of the defenses, envelop the BUA, and move rapidly to the enemy's rear.

c. When possible, reduce strongpoints with fires only, secure them with follow-on units, and keep moving.

d. Cut lines of communication and defeat the enemy through isolation.

e. Attack at night to gain surprise or to take objectives whose assault during daylight would be too costly. An attack at night will take advantage of the TANK's thermal sight capability.

f. Once momentum has been gained, attack continuously until defenses have been splintered.

03123. Attack Phases. An attack of a BUA comprises three phases:

a. Isolation of the BUA.

b. Gaining a foothold at the edge of the BUA.

c. Systematic clearance and seizure of objectives.

03124. **Isolation.** The first phase is the isolation of the city and the seizure of terrain features that dominate approaches to it. The attacker has the advantage of maneuver to isolate the city he will seize. The enveloping unit:

a. Prevents the escape of the enemy.

b. Prevents reinforcements from entering the BUA.

c. Provides direct fire support for the direct assault unit.

d. Protects the direct assault unit from counterattack.

03125. Once he has isolated the city, the attacker can either press the attack or contain the defender and unit him to capitulate. If necessary, the unit then secures positions outside the BUA from which to support entrance into the city itself. The tactics and techniques for this phase of operations are similar to those used in an attack against an enemy strongpoint (see Figure 3.39).

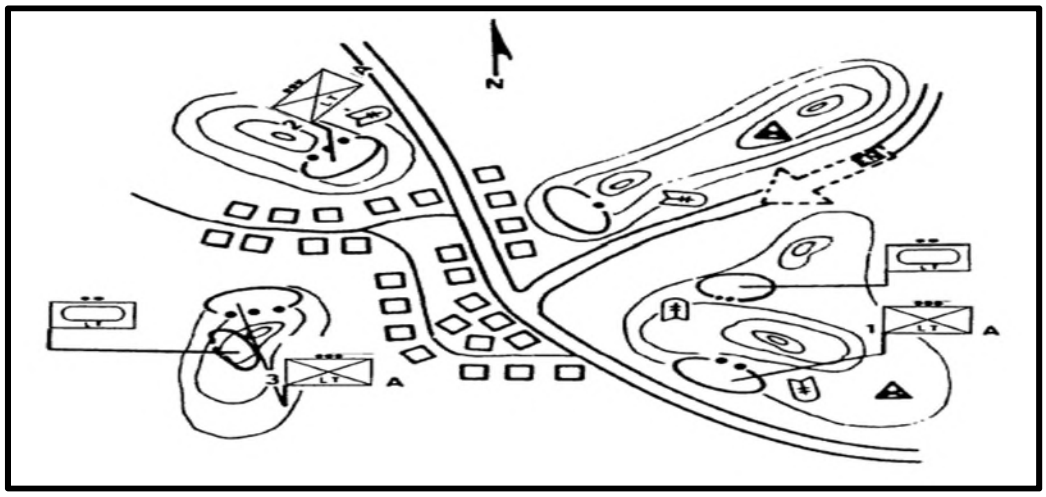


Figure 3.39 – Isolating the built-up area.

03126. **Gaining a Foothold.** In the second phase, the attacker selects his best point of entry into the city. He can attack from any direction; he can bypass strongly defended buildings by going under, over, and around them and by using cellars, sewers, subways, or other underground passages. The unit advances to the edge of the BUA to gain a foothold and eliminate the defender's observation of (and direct fires on) approaches into the area. The assault unit uses the foothold area to reorganize, decentralize control, and displace units in supporting positions. The attacking unit penetrates the area on a narrow front; tanks with infantry lead the way.

03127. The commander concentrates all available supporting fires on the entry point (see Figure 3.40).

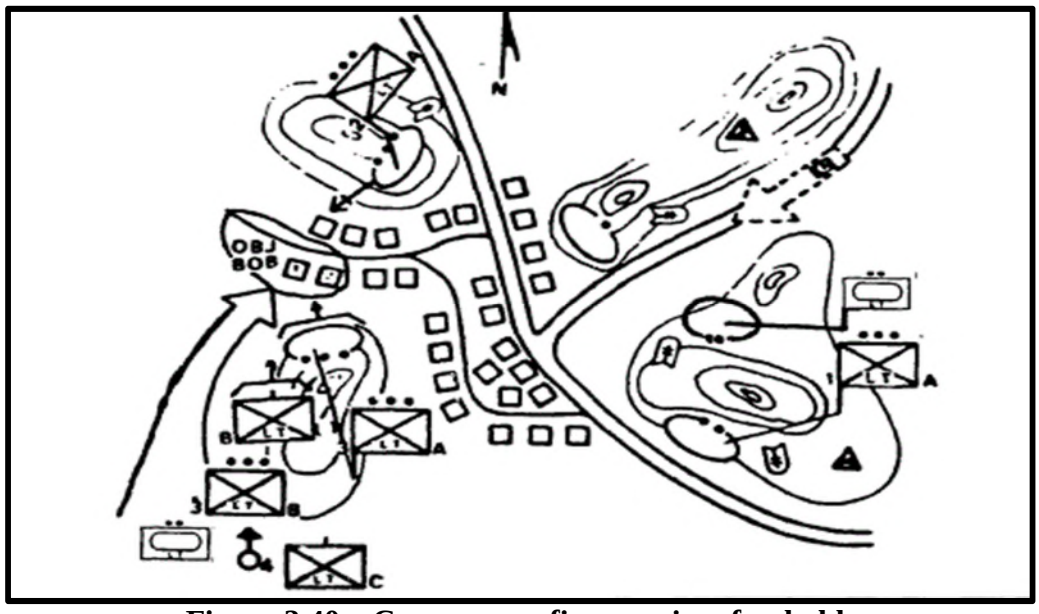


Figure 3.40 – Concentrate fire to gain a foothold.

03128. The probability of success increases when the commander launches his assault from an unexpected direction at early morning just before light or under the cover of smoke. The commander normally employs a column formation in the initial assault. Assaulting units can expect to encounter barricades, antitank mine obstacles, and antitank fire.

03129. The commander may employ variations of the column formation. For example, a battalion TF may use a column, with each of its company teams in a line, wedge, or echelon. These formations tend to shorten the length of the column, reducing the time necessary to move into the BUA. The leading elements of the unit should use a formation that speeds the delivery of maximum fire in the point of penetration. The commander should place artillery air bursts over the entry point to prevent the enemy from manning crew-served or individual antitank weapons. The infantry moves as close to the objective as possible. When the infantry attacks a strongly defended area, it provides close-in protection for the tanks. Unit leaders may assign fire teams or squads to work with each tank. Visual signals help maintain direct communication between the rifle squad or fire team leader and the tank commander. The infantry maneuvers engage or destroy the resistance. Tanks move forward as soon as possible to support them. When possible, the tank fires augment the assault or cover critical areas on the unit's flanks. When buildings on the periphery of a town that heavily fortified, the commander may have to employ techniques for the attack of a fortified area.

03130. **Clearance and Seizure**. Phase three can vary from a systematic block-by-block, house-to-house reduction of the BUA to a rapid advance with clearance of critical areas and buildings. It begins without pause after the completion of phase two. Clearance and seizure techniques depend on the mission, the size of the town, construction and building arrangement, and the enemy's disposition, strength, and objective. The direct assault unit clears the city of enemy resistance and links up with the enveloping unit when the unit must continue to move (see Figure 3.41).

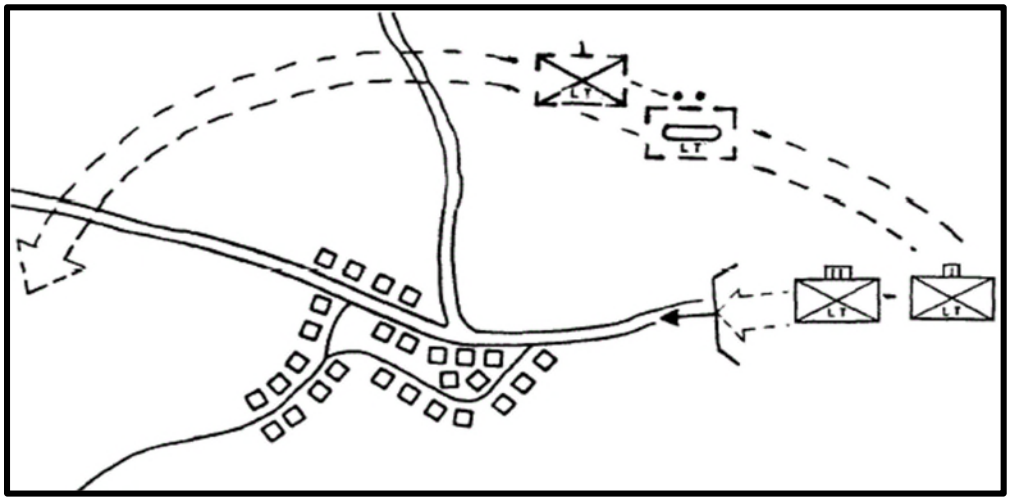


Figure 3.41 – Enveloping and assult forces.

03131. When the BUA is large and heavily fortified or when the mission requires it, units may have to perform a methodical house-by-house, block-by-block clearance operation. The commander should divide the area into zones of responsibility. Each subordinate unit must clear its zone completely, leaving no enemy to its rear.

03132. When the BUA is small or lightly defended, the attacking unit should drive through or into it as rapidly as possible. Tank should lead the column in this instance, closely followed and supported by infantry. It will rarely be possible to employ more than two tanks at the head of the column except when advancing on a wide street. Tanks continuously concentrate main gun and automatic weapons fire on windows and the rooftops of buildings (see Figure 3.42). The infantry protects the tanks from close-in

enemy fire. When required to protect tanks from fire to nearby buildings, an infantry squad moves along each side of the street, keeping abreast of the lead vehicles. Depending on the resistance, the squad may challenge every doorway or ground floor window by throwing in hand grenades and spraying the interior with small arms fire. Unit leaders will usually assign soldiers in each squad to locate and engage targets on the upper floors and rooftops of the buildings. The infantry may also assist in the removal of obstacles or barriers halting the advance.

03133. When seizing buildings, the tanks support the assault by isolating the building and providing suppression during entry (see Figure 3.43). The tank can also create a hole in a wall of a building with main gun fire to allow the infantry to enter the building through an unexpected entrance.

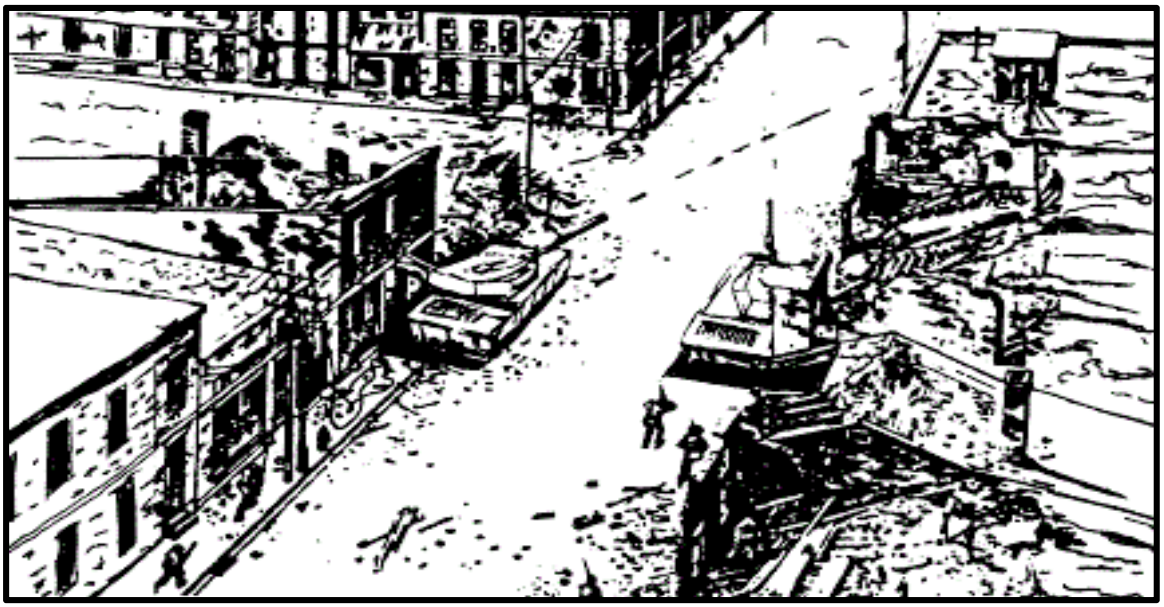


Figure 3.42 – Tanks advancing with infantry

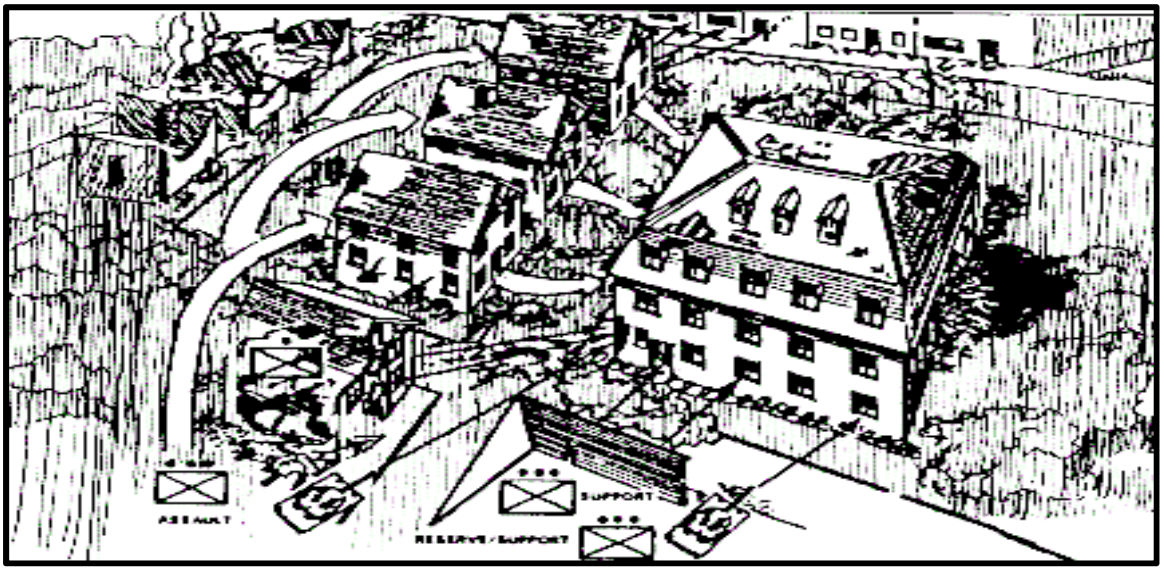


Figure 3.43 – Isolating and seizing a building

03134. The following procedures apply when attacking a BUA:

a. Dissipate the enemy's strength by causing him to react to demonstrations, feints, and ruses.

b. Concentrate overwhelming combat power to unit a quick and violent disruption of enemy defenses, envelop the BUA, and move rapidly to the enemy's rear.

c. When possible, reduce strongpoints with fires only, secure them with follow-on units, and keep moving.

d. Cut lines of communication (LOC) and defeat the enemy by isolating him.

e. Attack at night to gain surprise or to take objectives that would be too costly to assault during daylight. A night attack will take full advantage of the tank's thermal sight capability.

f. Once momentum is gained, keep the attack continuous until enemy defenses have been splintered.

03135. **Tank Tasks**. When the tank platoon is included as part of the direct assault unit, it should be employed as a platoon. In BUAs that are very restricted, however, sections may have to operate separately, each under the control of an infantry commander. Tank supports infantry in BUAs by:

a. Providing shock action and firepower.

b. Isolating objectives with direct fire to prevent enemy withdrawal, reinforcement, or counterattack.

c. Neutralizing or suppressing enemy positions with smoke, high-explosive (HE), and automatic weapons fire as infantry closes with and destroys the enemy.

d. Assisting opposed entry of infantry into buildings when doorways are blocked by debris, obstacles, or enemy fire. Smashing through street barricades or reducing barricades by fires.

e. Using fires to reduce enemy strongpoints in buildings.

f. Obscuring enemy observation using on-board smoke generators.

g. Holding cleared portions of the objective by covering avenues of approach.

h. Attacking by fire any other targets designated by the infantry.

i. Establishing roadblocks.

j. Suppressing identified sniper positions.

03136. **Infantry Tasks**. Infantry supports tank in urban terrain by:

a. Locating targets for engagement by tank weapons.

b. Suppressing and destroying antitank weapons with mortars, automatic weapons, and grenades.

c. Assaulting positions and clearing buildings.

d. Providing local security for tanks at night or during other periods of reduced visibility.

03137. **Tank Employment Considerations**. The following are some techniques and concerns that the infantry and/or tank leader should consider when employing tanks in urban terrain:

a. Tanks main gun fire is an effective method for eliminating a sniper in a building or creating a psychological effect that destroys his will to continue.

b. Streets and alleys constitute ready-made fire lanes and firing zones. They can greatly restrict and canalize vehicular traffic, making it vulnerable.

c. In urban terrain, tank works best in platoons or in sections of two tanks. In extreme cases, tanks can work individually, but this is not recommended.

d. At least one infantry squad should remain with each tank to furnish local security.

e. The external phone is an excellent means for communication between the infantryman and the tank commander.

f. The tank should use HE ammunition to create holes in the walls of buildings so the infantry can enter.

g. The tanks should use HE ammunition against barricades. HE will demolish steeples, chimneys, and other tall structures likely to contain enemy artillery observers. This technique is dependent on the established rules of engagement. In operations other than war, minimizing collateral damage may be a priority. However, crews can fire on observed or known hostile enemy positions at all times.

h. Crew members must be on alert for bunkers or pillboxes in houses along the street.

i. Tanks should avoid stopping or moving slowly near non secure buildings.

j. Units should check all bridges and overpasses for mines and should determine their weight-carrying capacity.

k. Tanks should stay near the buildings held by friendly troops. Crew members should watch for signals from the infantry inside the buildings on their flanks.

l. Tank crew members should keep their personal weapons ready for close-in combat.

m. When possible, tanks should destroy enemy strongpoints with main gun fire. One technique is to fire tank-piercing ammunition to penetrate the reinforced wall of a building followed by high-explosive antitank (HEAT) rounds to kill or neutralize the enemy. Tanks should fire first into the ground floor to drive the enemy into the basement, where infantry can attack them, or to higher floors, where tank fire can destroy them.

n. Tanks are at a disadvantage because their main guns cannot depress or elevate sufficiently to fire into basements and upper floors at close range (see Figure 3.44).

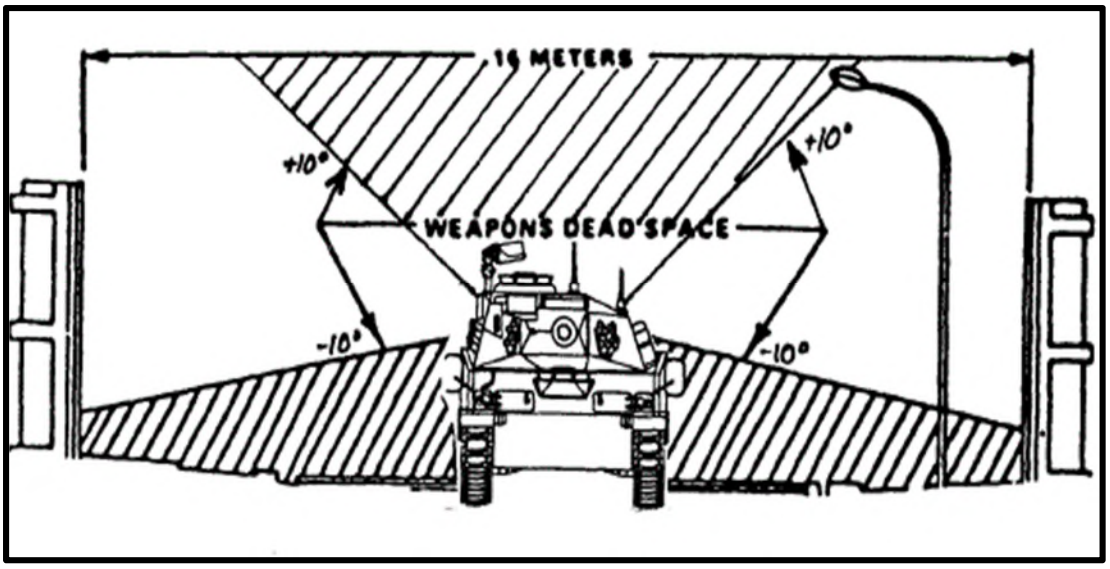


Figure 3.44 – Example of Tank dead space on narrow streets.

03138. **Control Measures**. Combat in BUAs requires control measures in which all troops must be familiar. These includes the following:

a. **Boundaries.** In dense urban areas, units should place boundaries along one side of the street to provide easy and definite identification. In areas where observation and movement are less restricted, they may place boundaries in alleys or within blocks so the one unit's zone includes both sides of the street.

b. **Objectives**. Objectives are specific and limited. Choosing major intersections, principal buildings, or other readily identifiable physical features improves control. Numbering buildings along the route of attack simplifies the assignment. As the unit moves forward through an area, unit leaders should designate the near side of the street as the objective. If they choose the far side of the street, the unit will have to secure buildings on both sides of the street to take the objective. Units must promptly report seizure of objectives.

c. **Frontages, formations, and zones of action.** Attacking battalions normally operate within relatively narrow zones of action. The frontages depend on the enemy's strength, the size of the buildings, and the anticipated resistance. Normally, an infantry battalion TF has a frontage of three to six blocks and attacking companies of one to two blocks. Frontages and zones of action influence tank employment. The tanks should be well forward to add momentum to the attack, exploit success, repel counterattacks, and protect the flanks and rear against enemy action.

d. **Phase lines (PLs)**. PLs increase control by regulating the advance of attacking units. They also indicate where the command passes from one phase of the assault to another. PLs are less restrictive than objectives. They encourage the rapid exploitation of success without halting. Principal streets, rivers, and trolley or railroad lines make good PLs.

e. **Checkpoints and contact points**. Street corners, buildings, railway crossings, bridges, and easily identifiable features can be checkpoints or contact points. They improve the reporting of locations. The commander can use them as specific points where he desires units to make physical contact.

Raid

03139. A raid is a deliberate attack into enemy-held territory for a specific purpose. Infantry units are ideal for conducting a raid. Raids are conducted with swift, violent action to destroy or capture enemy personnel or equipment, rescue friendly personnel, gain intelligence, or gain the initiative. Stealth during movement, indirect approach, violent execution, and precision are all characteristics of successful raids. The same considerations apply as for a deliberate attack, except for the following points:

a. There is always a planned withdrawal from the objective.

b. Planned fire and security elements isolate the objective from enemy reinforcement or retreat.

c. Raids can be done by any size unit.

d. Raids require detailed planning and extensive rehearsals.

03140. The keys to the raid are information, surprise, and timing. Surprise is obtained by using deception, stealth, and speed of execution when moving to the objective area. It is essential that the raiding unit arrive in the objective area without being compromised. Timing is also essential to the execution of the raid. It is difficult to use tank in the conduct of small unit infiltration raids. In some cases, however, the tank platoon can be used as the security element to:

a. Block enemy reinforcement along the mounted avenues of approach.

b. Assist the raid party in breaking contact.

c. Assist the raid party in the withdrawal from the objective.

03141. The raid plan includes a signal to withdraw; well-planned routes to an ORP or rendezvous point; routes covered by preplanned fire; units assigned to cover the withdrawal and assist in breaking contact; a way to evacuate casualties, enemy prisoners of war (EPWs), and captured equipment from the objective; and an order of withdrawal from the objective.

03142. The withdrawal from the objective and the security unit blocking positions must be planned with the same detail as the rest of the mission. Use of preplanned fire is essential to keep enemy units from pursuing. The enemy's anticipated reactions must be considered when planning withdrawal routes. Once the main body has withdrawn from the area, the security unit withdraws. In some cases, the security unit may have to withdraw using delay tactics.

Bypass

03143. The company commander uses fire and movement when he decides to bypass enemy units. The enemy is suppressed by both indirect and direct fires while the unit moves, using covered and concealed routes, past the enemy positions. Where covered routes are not available, the tanks should remain out of antitank weapon range or use smoke to conceal movement.

03144. While bypassing enemy units, moving platoons orient gun tubes on the enemy. The tank platoon (as the support by fire element) suppresses the enemy positions, thus preventing the enemy from firing on or maneuvering against the moving infantry platoons. The infantry unit may halt and provide suppressive fire so the support by fire platoon can move and complete the bypass (see Figure 3.45).

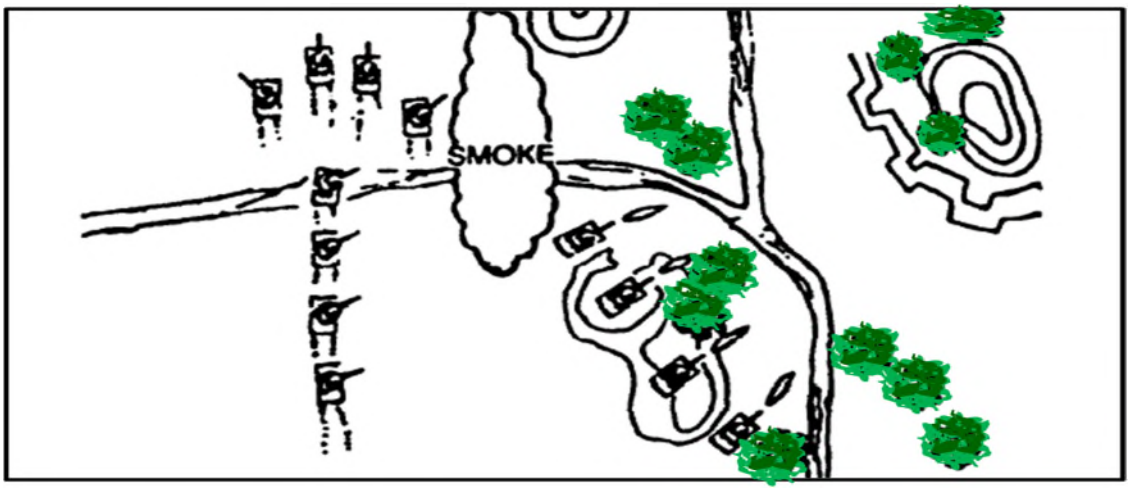


Figure 3.45 – Bypass of an enemy position.

Consolidation and Reorganization

03145. The platoon should consolidate and reorganize as soon as it takes an objective. This is done so that the platoon is prepared to destroy an enemy counterattack or to continue with the attack.

03146. **Consolidation** consists of actions taken to secure an objective and to defend against an enemy counterattack. The commander designates platoon positions and weapon orientations. The platoon consolidates an objective by:

a. Occupying the position designated in the OPORD. Tanks are moved to hull-down positions, and the platoon leader assigns sectors of fire based on the commander's intent.

b. Preparing for an enemy counterattack.

c. Establishing security and mutual support between adjacent units.

d. Eliminating any remaining pockets of enemy resistance and securing EPWs.

e. Preparing to continue the mission.

03147. **Reorganization,** the process of preparing for continued fighting, should be accomplished by SOP. The platoon leader, platoon sergeant, and vehicle commanders are responsible for the following actions:

a. The **platoon leader** should redistribute personnel, oversees consolidation of personnel killed in action (KIA), informs the commander of the platoon's status, Supervises essential maintenance, and establishes communications with units that are out of contact.

b. The **platoon sergeant** compiles status reports on personnel, equipment, and supplies from each vehicle and submits a consolidated report to the platoon leader and 1SG. He then directs cross-leveling within the platoon.

c. The **Tank commander r**eloads machine guns and redistributes main gun ammunition to ready areas, moves wounded crewmen to a covered position and provides first aid, reports his situation, casualties, and status of equipment, ammunition, and fuel to the platoon sergeant, and conducts essential maintenance.

Tank Platoon during Limited Visibility

03148. Limited Visibility. Terrain and weather will not always be ideal for offensive operations. A platoon's ability to move, acquire the enemy, and control or request fires will be affected during periods of limited visibility. Equipment and training can help offset the effects. The platoon leader will have to adapt certain techniques or modify his tactics when operating under adverse conditions. This discussion covers the effects of limited visibility caused by adverse terrain or weather conditions and describes techniques for coping with these limitations.

03149. The platoon often participates in offensive operations during periods of limited visibility. The concealment provided by adverse weather, nightfall, or smoke can favor the attacker by permitting him to mass units and strike an unwarned enemy. The ability to surprise the enemy, plus the increased shock effect caused by an unseen attacker, may permit a smaller unit to successfully attack and destroy a larger unit.

During periods of limited visibility, the radio may become the primary means of communication. The commander decides whether visual signals aided by the use of chemical lights, flashlights, or pyrotechnics can provide secure, adequate communication or whether the radio must be employed.

03150. When planning a night attack, commanders and leaders should make a detailed day and night reconnaissance of the route of march to the platoon points of departure. When a passage of lines is required, the reconnaissance should be conducted jointly with the passed unit at the passage point. Additional control measures such as direction of attack, probable line of deployment, and limit of advance may be used as an aid in the C2 of an operation. Night operations using traditional illumination such as flares should be avoided. Flares may expose silhouette of attacking units or alert the enemy to the pending attack.

03151. The following are some useful ways to offset the effects of limited visibility:

a **Movement.** When navigation is degraded because of inability to see terrain features, a compass and the vehicle odometer are used to navigate. The distance between vehicles should be decreased so that vehicle commanders can see each other to orient and provide overwatch. Crews can use white or luminous tape to outline unit marking panels. Flashlights, chemical lights, or dome lights can be fixed to the turret side or rear, the bustle rack, or the vehicle antenna. The column formation can also alleviate orientation problems. Platoons should employ traveling overwatch and bounding overwatch techniques internally when overwatch by another platoon is not possible. When natural obstacles (cliffs, holes, or cuts) are difficult to see, a detailed map study should be conducted before the operation to determine the best route. When moving with infantry, the infantry can be used to move forward of the vehicles to reconnoiter. The column formation can minimize the chance of encountering unseen natural obstacles.

b **Enemy acquisition.** In limited visibility conditions, the enemy may not be observed until he fires. Therefore, every movement must be overmatched. Elements should use the avenue of approach least likely to be watched by the enemy; this may cause him to shift his positions and give away his location. Every thermal or visual enemy signature, no matter how small, should be engaged. For example, the thermal signature of a tank commander’s face may be the only detectable signature of a hull-down enemy tank. Once a vehicle or section has located the enemy, it may have difficulty in alerting the rest of the platoon. A steady burst of machine gun fire in the direction of the enemy can be used to orient the rest of the platoon to the direction and general position.

c **Control of direct and indirect fires.** Limited visibility conditions or terrain can make it impossible for all of the vehicles in the platoon to see the enemy target. The platoon leader may not be able to determine which vehicles can engage before he issues a platoon fire command. After receiving the fire command, vehicle commanders who cannot see the target should report “CANNOT IDENTIFY” to the platoon leader. When the platoon leader cannot determine a six-digit coordinate for the enemy position (for example, if he only

sees muzzle flashes), he can use the shift from a known point method or polar method.

03152. **Extremes of Terrain**. The following considerations apply to tank employment in several types of difficult terrain:

a **Forest and jungle.** Heavy woods restrict the effectiveness of long-range fires and reduce Tank movement to trails, roads, and cleared areas. Since enemy infantry can operate freely in wooded areas and can surprise Tank positions, tanked units must operate closely with infantry.

b **Mountains.** In rugged mountainous terrain, tank units are restricted to valleys and **roads**. From high ground, the enemy can engage the relatively thin tank on the top of the turret. Tank units should operate with infantry when conducting operations in mountainous terrain.

c **Arctic terrain and extreme cold**. Extreme cold and its effect on men and equipment make operations in arctic terrain difficult. Thin ice, ice ridges, deep snow, and ravines hinder tank movement. The absence of terrain features makes navigation difficult, and the intense white environment makes a camouflaged enemy difficult to see.

d **Desert.** The desert may offer few covered and concealed positions. Once contact is made, units should return fire, continue to move, and use smoke to obscure themselves from enemy gunners. Moving units are exposed for longer periods and must use artillery, careful selection of routes, and high speeds to reduce exposure. Perceived ranges are distorted by the absence of terrain features. Targets that appear 2 or 3 kilometers away may actually be 5 kilometers or more. Shapes are distorted by the effects of heat. The lack of identifiable terrain features also makes navigation difficult, especially at night.

Chapter IV
Defensive Operations

0401. While the offensive element of combat operations is more decisive, the defense is the stronger element. However, the conduct of defensive tasks alone normally cannot achieve a decision.

Purposes of Defense

0402. Commanders choose to defend to create conditions for a counter offensive that allows Army forces to regain the initiative. Other reasons for conducting a defense includes: to retain decisive terrain or deny a vital area to the enemy, to attrite or fix the enemy as a prelude to the offense, in response to surprise action by the enemy, or to increase the enemy's vulnerability by forcing the enemy to concentrate forces.

0403. A defensive task is a task conducted to defeat an enemy attack, gain time while waiting for more favorable conditions to conduct offensive operations, economize forces in one area so superior forces can concentrate for decisive offensive operations elsewhere, and develop conditions favorable for offensive or stability tasks.

0404. The tank unit is not ideally suited for conducting independent defensive operations. It normally operates as part of a larger force and should be assigned missions that capitalize on its capabilities. The tank enhances the overall defense by providing the infantry force with a high degree of mobility and firepower. Tank unit leaders and their troops must understand and employ the following fundamentals of defensive operations:

a. **Understand the enemy**. The tank unit leader must learn the following information about the enemy: 1) Reconnaissance capabilities. 2) Weapon systems available to the enemy. NBC, Artillery, and ATGM capabilities are most critical. 3) The enemy's ability to conduct dismounted attacks and raids under limited visibility conditions.

b. **See the battlefield.** The tank unit leader position is where he can control his forces. He must assertively use a variety of assets to understand the situation to his front, flank, and rear. Thermal sights will **augment** tanks ability to see the battlefield under all visibility conditions. The leader must also attempt to prevent the enemy from seeing the battlefield by using concealment and operations security (OPSEC).

c. **Concentrate fires**. Tank unit leaders must plan to concentrate fires to achieve decisive results. Direct and indirect fires must be brought to bear on the enemy before it can bring effective fires on friendly positions. Fires must cover protective obstacles. Leaders must take the initiative by counter-attacking to exploit enemy weaknesses as they arise.

d. **Use the advantages of the defender.** Maximize combat power by knowing the terrain better than the enemy. Makes the best use of terrain by—using it as a shield against enemy observation and fires. Fighting from covered and prepared

positions when possible. Using obstacles to maximize the effects of direct and indirect fire. Engaging the enemy first at the most opportune time and place, using available weapon systems to best advantage. Rehearsing the plan to ensure every man knows what to do. Using mobility to best advantage.

e. **Fight as a combined arms team.** Tanks will always fight as part of a combined arm team. As noted previously, the capabilities and limitations of tanks and infantry make them complementary when employed as a team.

Defensive Tasks

0405. **Area Defense.** The area defense concentrates on denying enemy forces access to designated terrain for a specific time rather than destroying the enemy outright. The focus of the area defense is on retaining terrain where the bulk of the defending force positions itself in mutually supporting positions. Units at all echelons can conduct an area defense.

0406. **Mobile Defense.** The mobile defense concentrates on the destruction or defeat of the enemy through a decisive attack by a striking force. The mobile defense focuses on defeating or destroying the enemy by allowing it to advance in a point where he is exposed to a decisive counterattack by the striking force. A fixing force supplements the striking forces by holding attacking forces in position, to help channel attacking forces into ambush areas, and to retain areas from which to launch the striking forces. A mobile defense requires an AO of considerable depth. Division and larger formations normally execute mobile defense. However, subordinate echelons may participate as part of the fixing force or the striking forces.

0407. **Retrograde Operations.** The retrograde involves organized movement away from the enemy. The enemy may force these tasks, or the PL may execute them voluntarily. The higher commander of the force executing the retrograde must approve the retrograde task before initiation in either case. The retrograde is a transitional task; it is not conducted in isolation. It is part of a larger scheme of maneuver designed to regain the initiative and defeat the enemy.

The three types of retrograde operations are delay, withdrawal, and retirement.

0408. **Delay** is one of the most demanding missions any unit can undertake. It is also the most common retrograde mission for tank units, which uses it to trade space for time. The ability to delay is essential to success on the battlefield when an enemy force outnumbers the contingency force or has superior tank units. Tank units are the only force in the division structure that can conduct a high-risk delay operation against a mounted enemy. Success depends heavily upon firepower and mobility. The contingency force commander can use tanks to delay when the force’s strength is insufficient to attack or defend. In a delay, units give ground to gain time. They inflict the greatest possible damage on the enemy while maintaining freedom of action. Enemy forces are delayed by the effective use of obstacles, firepower, and terrain. Delaying forces avoid decisive engagement. The tank company can conduct a delay alone or as part of a larger unit. The considerations of planning and executing a delay at company level are the same.

0409. There are two types of delays:

a **Delay in sector**. This mission allows the most freedom to use the terrain as you see fit. There is usually no requirement to hold key terrain when conducting this type of delay.

b **Delay forward of a line**. This is a high-risk mission that requires preventing enemy forces from reaching a specified area earlier than the specified time or event, regardless of the cost. The commander will normally limit the maneuver from BP to BP or will restrict crossing a particular PL based on a specific time or event.

0410. The two basic methods of conducting delays at the company level are described in the following paragraphs:

a From **Successive Positions or PLs**. This method is used when the mission requires covering a wide sector or avenue that is open and difficult to cover with infantry. All or most of the tank company must be deployed forward to cover the area. This method is also used when terrain does not allow for depth. As the fight progresses, the company fights from PL to PL. The platoons disengage separately from one PL or BP to the next while the remaining platoons provide overwatch (see Figure 4.1). In some cases, the enemy may force the company to disengage simultaneously. Bounding within the platoons may be necessary when the terrain restricts the platoons' ability to provide over-watch for one another.

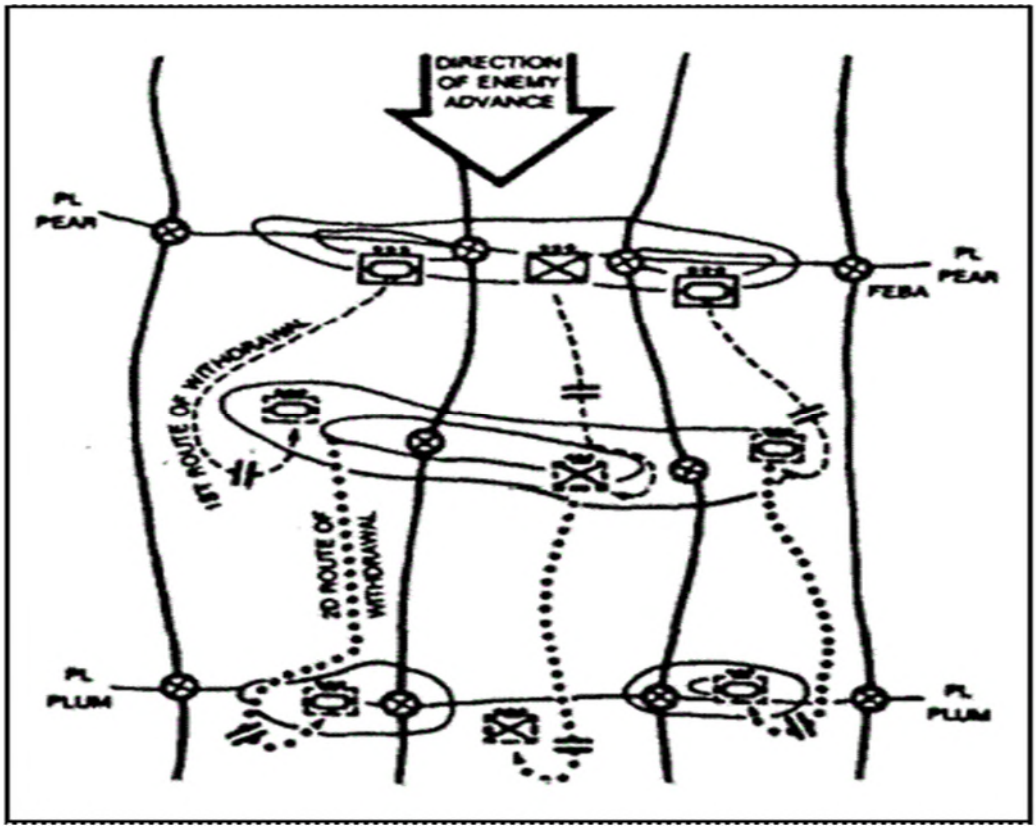


Figure 4.1 – Delay from successive position

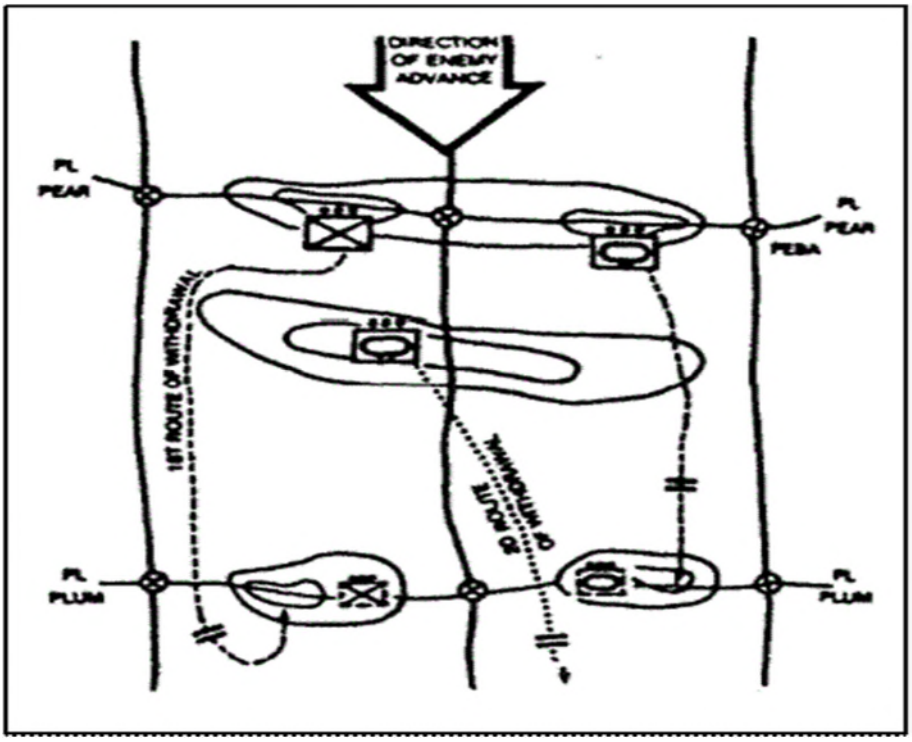


Figure 4.2 – Delay from alternate position.

b **From alternate positions.** When the AO is deep and narrow enough to be covered by one or two platoons, more depth and security can be achieved by delaying from alternate positions (see Figure 4.2). This maneuver does not normally allow all of the platoons to place fire on the enemy at one time. It is more difficult to control because platoons are moving and fighting simultaneously. The platoons have more time to establish their next positions, however, because other platoons are fighting and providing security.

0411. In a **withdrawal,** all or part of a committed force disengages from the enemy voluntarily to preserve the force or free it for a new mission. The purpose of a withdrawal is to disengage from the enemy. To successfully conduct a withdrawal, forces need mobility equal to or greater than the enemy. There are two types of withdrawals:

a **Withdrawal under enemy pressure** requires the unit to maneuver to break contact when it is under attack from the enemy. A unit conducting a withdrawal under enemy pressure is organized into a security force and a main body. The withdrawal should always be conducted with the goal of avoiding discovery. Timing is critical. The unit must disengage by using massed fires and redeploy before the enemy can react to its movement.

b **Withdrawal not under enemy pressure** requires deception and speed. The unit is not under attack and does not expect to be attacked during the withdrawal. During the withdrawal, deception and OPSEC are stressed. A unit conducting this kind of withdrawal from a defensive position is organized into a main body and a DLIC.

0412. The withdrawal plan must be modified to fit the technique used to defend or delay. Defense or delay techniques that are fluid and use a series of ambushes and raids

to accomplish the mission can be used with withdrawal techniques associated with those operations. Defenses or delays that are more static require different withdrawal techniques. The techniques used for a tank company to support the infantry brigade withdrawal are similar to the way the tank platoon supports the infantry battalion withdrawal as described in Chapter 4. The withdrawal requires the force to designate a DLIC, security force, and quartering parties, which are described below:

a **Detachment left in contact.** The size, makeup, and mission of the DLIC are directed by the brigade commander. He will also name the DLIC commander. This is normally the XO. DLIC composition is dependent on the terrain and enemy situation. Although one could be the DLIC, each battalion normally will leave a company as its part of the brigade DLIC. This DLIC must be a mobile force. The tank company can provide the mobility needed to cover the withdrawal of the brigade main body.

b **Security force**. The security force conceals the withdrawal of the main body and deceives the enemy by continuing the normal operational patterns of the battalion. If the enemy attacks during the withdrawal, the security force covers the withdrawal with fires. Priority of artillery and mortar fires is given to the security force. Once the battalion has reached its next position or a designated distance from the old position, the commander withdraws the security force. If it is under attack, the security force may have to maneuver to the rear until contact is broken.

c **Quartering party.** Each battalion sends a quartering party to the next position before the withdrawal starts. As their units arrive at the new location, members of the quartering party act as guides to lead elements into their new positions.

0413. **Retirement.** Retirement is an operation where a force is not in contact, moves away in the rear, normally as a tactical road march from the enemy to avoid combat under unfavorable conditions. A withdrawal from action becomes a retirement after the main force has disengaged from the enemy and march columns have been formed. A retirement may be made to increase the distance between the defender and the enemy, to occupy more favorable terrain, to reduce the distance between maneuver and CSS elements, to conform to the disposition of a higher command, or to permit employment of a unit in another sector.

0414. Planning considerations for a retirement are similar to those for delay and withdrawal. Movement during reduced visibility is preferred. The tank company may be required to move on one or more routes to support the brigade.

0415. Tanks can provide the appropriate forward, flank, and rear security. When contact with the enemy is possible, such as when a withdrawal has preceded retirement, a strong rear security is normally employed. If the enemy attacks the rear, delay tactics are used by the rear guard to extend the distance between the main body and the enemy. The keys to a successful retirement are dispersion, speed, and security.

Forms of the Defense

0416. Subordinate forms of the defense have special purposes and have their own unique planning considerations. The Army recognizes three forms of the defense—

a Defense of a linear obstacle.

b Perimeter defense.

c Reverse slope defense.

Tank Battalion in Defensive Operations

0417. The Tank battalion conducts defensive operations as part of a division defense. It can be integrated into the defense in the following ways:

a As the division's reserve or counterattack force.

b Defending in sector from BPs.

c Task organized with companies supporting each infantry brigade, and a reserve force under the control of the Tank battalion headquarters.

d As a security force.

0418. The Battalion defends in sectors or BPs or using a combination of both. (see Table 4.1)

Table 4.1 – Factors in battle position and sector defense

FACTOR	BATTLE POSITION	SECTOR
Avenues of approach	Well defined; enemy can be canalized	Multiple avenues prohibit concentration
Terrain	Dominates avenues of approach	Dominating terrain not available
Area of operation	Narrow	Wide
Mutual support between companies	Achievable	Cannot be achieved
Higher commanders ability to control	Good	Degraded

Defense in Sector

0419. METT-TC may cause the division commander to deploy the battalion in a sector in which he expects enemy mechanized or armored forces to attack. Tank company teams construct battle positions and defend in depth. Infantry deploys in surrounding terrain that favors dismounted attacks and facilitates supporting fires. Figure 4.3. Shows a defense in sector.

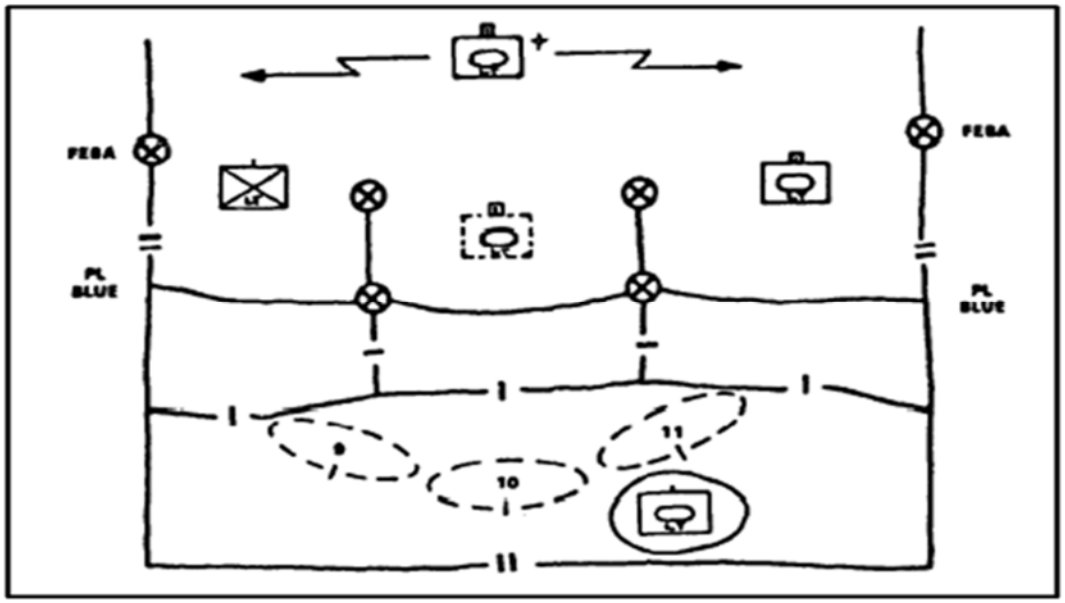


Figure 4.3 – Battalion defending in sector

Battle Positions

0420. A BP is a general location and orientation of forces on the ground from which units defend. The BP can be used for units from battalion- to platoon-size. Tank units defend BPs in the same way armored units do. The difference is that the Tank battalion will not normally operate in terrain that allows it to defend a battalion size BP. The BP defense for Tank units will usually consist of company BPs controlled by the BCT/TF/BDE. Security forces may operate forward and to the flanks of BPs for early detection of the enemy. Figure 4.4 shows a BP defense.

0421. Units can maneuver in and outside of the BP as necessary to adjust fires or to seize opportunities for offensive action in compliance with the commander's intent. The commander may freely move his force within the assigned BP. The commander must notify the higher commander and coordinate with adjacent commanders when maneuvering his force outside the BP.

0422. The BCT/TF/Bde commander allocates space to the subordinate elements within the general area. The BCT/TF/Bde commander-

a. Selects subordinate BPs by considering space two levels down (platoon level). He provides sufficient space in each BP to allow for dispersed primary and alternate positions. Room for operations in limited visibility, supplementary hide positions, and locations for combat trains are also considered.

b. Varies the degree of maneuver of teams by allocating larger company BPs. BPs may also be placed in depth.

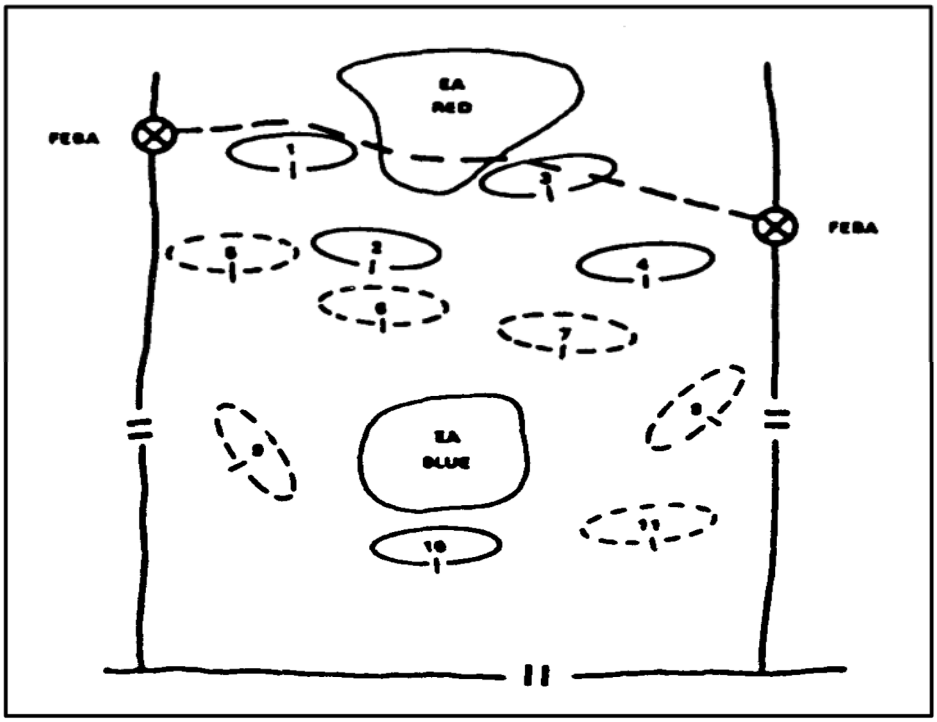


Figure 4.4 – Battle position defense

Other Defensive Tasks

0423. **Reserve.** In some cases, each infantry brigade may be task organized with a Tank company to strengthen the brigade defense. Platoons within the companies may be further attached to infantry battalions. In these cases, the battalion headquarters may become the controlling headquarters for the division reserve consisting of at least one tank company and other division combat forces. This option provides the division commander with a responsive, flexible, potent maneuver force that he can use anywhere in the area of operations. The division can commit the reserve during the battle to spoil the enemy's attack, destroy him, or reinforce success and gain the initiative.

0424. **Rear Operations**. The battalion can provide rear area security for the division. Force buildup can mature to a point in which heavy armor or mechanized follow-on forces enter the theater and assume responsibility for the area of operations once covered by the infantry division. The division may then assume a reserve role, or prepare for future offensive operations. Within the division, the Tank battalion may be tasked to react in rear area threats. The battalion executes this mission similar to a counterattack mission in reaction to forces inserted into the rear area.

0425. **Attacks from a Defensive Posture**. Attacks from a defensive posture include counterattack and spoiling attacks as part of either hasty or deliberate operations. The Tank battalion is an ideal force for the division commander to use for this mission. The following considerations apply:

a **Counterattack.** Infantry defensive operations against a mechanized or armored threat employ defensive positions in depth, coupled with EAs to reduce and then destroy the enemy force as it moves through the AO. The infantry is basically fixed in position once deployed into a defense. The Tank battalion provides a mobile force that can move throughout the defensive sector and counterattack

to defeat the enemy force at the decisive place and time. The counterattack is usually a key element in a coordinated effort to mass fires-CAS, air attack, artillery, and other fires-into a designated EA. BCT/TF/BDEs attack by fire or by fire and movement.

b **Spoiling attack**. The spoiling attack strikes the enemy when he is most vulnerable during preparations for attack in AAs or attack positions or on the move before crossing his LD. The objective of the spoiling attack is the enemy force, not terrain. The reserve is often used in a spoiling attack to allow forward units to concentrate on defensive preparations.

Tank Company in Defensive Operations

0426. This section describes the most common defensive missions conducted by the tank company while operating with an infantry brigade. Organized as a company team, the tank company can conduct defensive combat operations:

a. With its parent tank battalion.

b. Task organized to an infantry brigade.

c. Task organized to an infantry battalion.

0427. The tank company can expect to perform one of many different missions as part of the overall brigade defensive plan. These missions can be as part of the security force or as main battle area (MBA), reserve force, or rear area actions. When planning for the use of tank in the defense, the tank commander should consider the following:

a. **Mission**.

b. **Enemy**. Tank should be placed in defensive positions that capitalize on its long-range firepower and mobility advantages. Tank must concentrate when an enemy mounted threat exists. The company may have platoon(s) task organized to infantry battalions when a dismounted enemy threat exists.

c. **Terrain**. Terrain affects three aspects of tank positioning: 1) Protection (cover and concealment). 2) Firepower (observations and fields of fire). 3) Mobility (obstacles and avenues of approach). Tank should be massed when the terrain allows. In rolling, open terrain, the tank company can defend in platoon BPs to overwatch mounted avenues of approach. In close terrain, further task organization of platoons to infantry battalions may be necessary.

d. **Troops available**. Employment of the tank company may depend on the ability to provide infantry for security. When positioned in or moving through restrictive terrain, infantry is needed to assist in close security. Engineers are needed to prepare tank fighting positions; employment and task organization may be dependent on the availability of engineer support for preparation of positions.

e. **Time available**. Static defensive preparations for tank are time-consuming. The time available may drive the decision to place tank in BPs or in reserve.

0428. Other planning considerations include the following:

a. Determine where to kill the enemy.

b. Determine positions for the Tanks.

c. Plan for security and limited visibility.

d. Assign missions and graphic control measures.

e. Determine the sequence and timing of fire and maneuver.

f. Plan improvement of positions and routes.

g. Incorporate a deception plan.

h. Conduct lateral and higher headquarters coordination.

Company Defensive Fire Planning

0429. The company defensive fire plan enables the company commander to distribute and control fires in support of the brigade commander's defensive concept. In turn, it assists the brigade commander in preparing his fire plan. If part of the brigade area is threatened, the commander can use the fire plan to determine which weapons can cover the threatened area. Using radio or SOP signals, he can direct fires to destroy the threat. To develop a defensive fire plan, the commander must:

a Assign a location for vehicles and sectors of fire. A sector of fire is an area where a tank has a primary responsibility for acquiring and engaging the enemy. The sectors of fire should overlap between individual tanks and with adjacent elements on the companies' flanks, based on where he wants to kill the enemy.

b Designate limited visibility capable TRPs and indirect fire targets in the company sector. The FSO assigns number to indirect fire targets.

c Coordinate with adjacent units.

d Evaluate information from his vehicle commanders to determine if they can effectively observe and engage targets and TRPs within their sectors. Each tank commander prepares a sketch card for each position. The platoon leader consolidates them, prepares his platoon sector sketch, and gives a copy to the commander.

e Develop a sketch of the company's sector, with a list of direct fire engagements and a legend, for all primary, alternate-and supplementary firing positions (see figure 4.5).

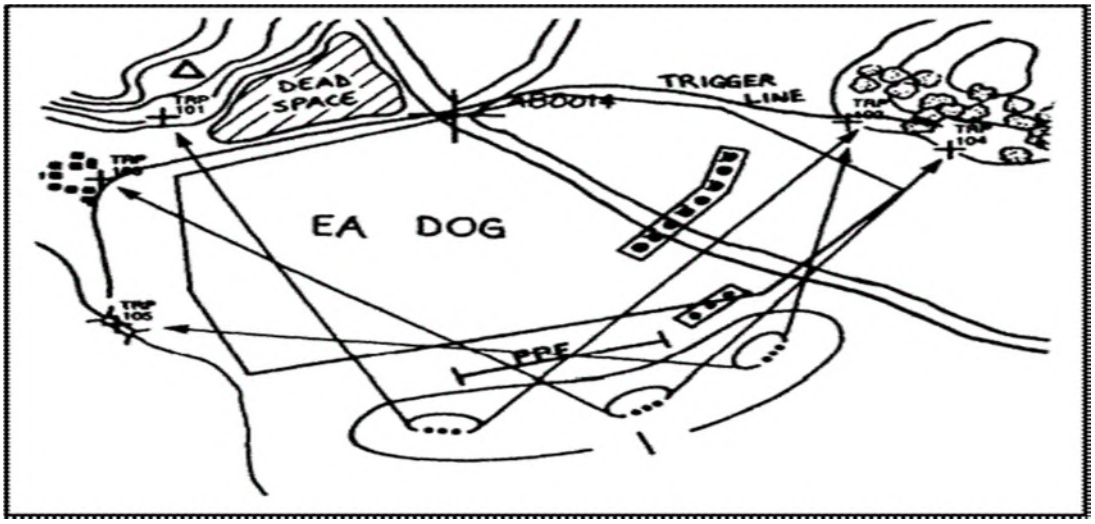


Figure 4.5 – Company direct fire plan.

0430. The sketch should include:

a The company sector.

b Individual tank positions and platoon sectors.

c OPs.

d TRPs and EAs.

e Obstacles.

f Indirect fire targets.

g Dead space.

0431. The legend lists targets in the company's sector and tanks that can fire on those targets. It should also explain the direct fire and indirect fire graphics represented, as well as obstacles and barriers within the sector that can assist in the defense by canalizing the enemy into an EA. Standard military symbols are used to depict the obstacles and barriers, which should be covered by direct or indirect fire.

0432. Give a copy of the company fire plan to the infantry company or BCT/TF/Bde commander as well as each of his tank commanders.

Defense of a Battle Position

0433. The infantry brigade uses the tank company in a BP to concentrate its fires, limit its maneuver, or place it in an advantageous position to counterattack. The tank company defends from a BP in the same way as the tank company. This type of defense is used to control fires and movement. It is designed to concentrate direct fires at critical places and time to take advantage of available terrain. The infantry brigade commander will assign the tank company a BP that dominates a main battalion-size enemy armor (mounted) avenue of approach.

0434. The brigade commander specifies critical tasks for companies defending BPs. A minimum level of preparation is assigned at each battle position (occupy, prepare, or reconnoiter) to enable the company to accomplish its mission. The company is required to orient its weapon systems on an enemy avenue of approach using TRPs or EAs. The tank company defends in a BP to accomplish one or more of the following:

a Destroy an enemy force in an EA.

b Control key terrain by holding the BP.

c Block an avenue of approach.

d Fix the enemy force to allow another unit to maneuver.

0435. When defending a BP, the company positions its element and maneuvers freely within the limits of the position to accomplish the commander's intent. If the company commander needs to position elements outside the BP to make a better use of terrain, increase dispersion, or maximize his firepower, he should coordinate these locations with his senior commander.

Defense in Sector

0436. The infantry brigade uses a defense in sector to prevent an enemy force from passing the rear boundary of the sector. A defense in sector is used when—

a The enemy situation is vague.

b Multiple avenues of approach exist, precluding concentration of fires.

c Retention of terrain is not critical.

d Flexibility is desired.

0437. When defending in sector, the brigade uses the depth of the sector to defeat the enemy within his boundaries while maintaining flank security and ensuring unity of effort. Sectors give the subordinate battalion BCT/TF/Bde commanders the freedom to decentralize fire planning. It allows the commanders to allocate his forces to suit the terrain and to plan in depth.

0438. The tank company will usually have platoons task organized in defending infantry battalions, with at least one platoon (with infantry, anti-tank and/or other appropriate forces) retained as the brigade reserve under the command of the tank company commander. Figure 4.6 shows an example of a brigade defense in sector with a supporting tank company. This example shows three brigades (each with a tank company team) defending in sector. The brigade on the left has one tank platoon defending forward in sector with an infantry battalion and the remainder of the company in reserve. The middle brigade has all three platoons task organized to the battalions and the third brigade (on the right) employs the entire tank company in a blocking position on a flank enemy mounted avenue of approach.

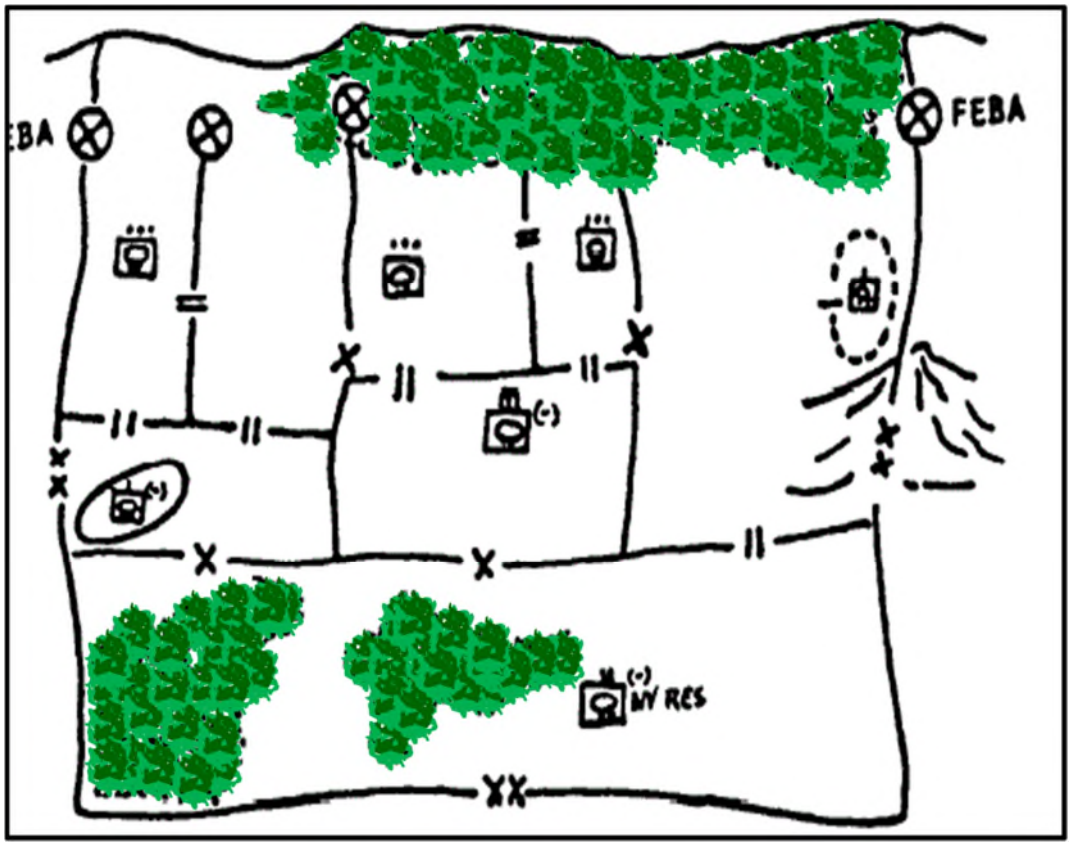


Figure 4.6 – Infantry brigade sectors showing example of tank company employment.

Defense of a Strongpoint

0439. A strongpoint is a defensive position that is fortified as extensively as time and materials allow. It is used to hold key terrain critical for the defense, to provide a pivot for the maneuver of friendly forces, and to canalize the enemy into friendly EAs. A strongpoint is attacked at the risk of high casualties. It cannot easily be overrun or bypassed. It is tied in with existing obstacles, forcing the enemy to reduce it by dismounted assaults and massive artillery and tactical air concentrations.

0440. While the size and type of force selected to execute a strongpoint defense will vary according to the situation, an infantry-heavy unit is normally used, with tank retained for mobility. The tank company can counterattack an enemy force that is halted by the strongpoint defense. Figure 4.7 shows an example of how the tank company can be used in a brigade defense when one of the battalions is defending a strongpoint.

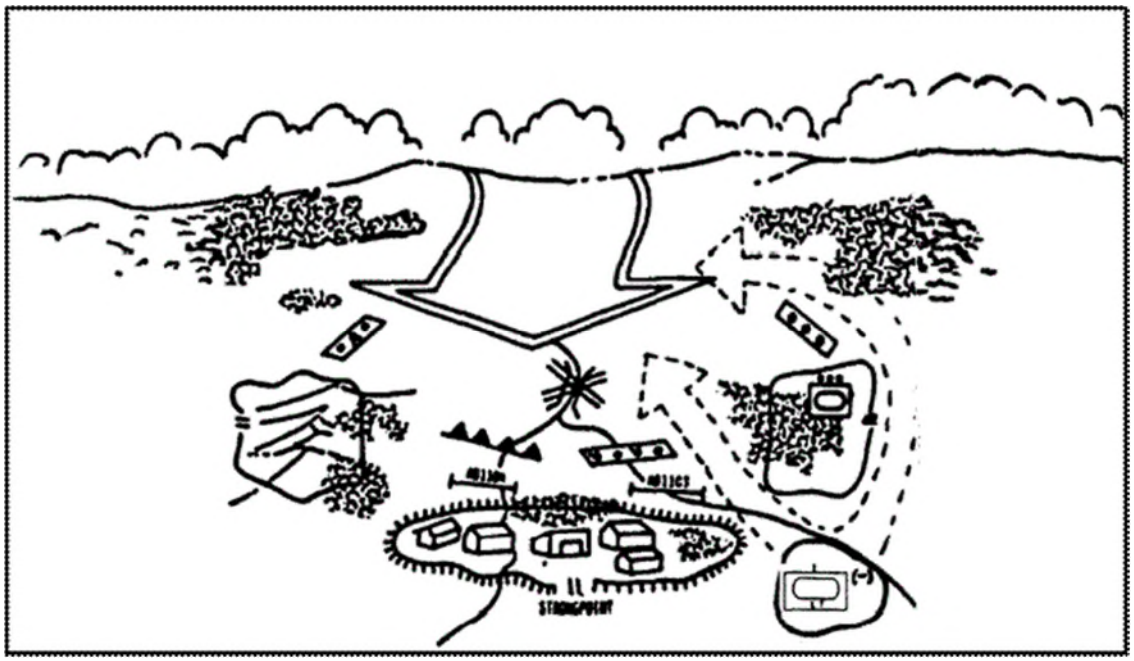


Figure 4.7 – Tank Company supporting a strongpoint defense.

Perimeter Defense

0441. A defense of a perimeter is conducted in the same manner as the battle position except the perimeter defense orients on 360 degrees. Perimeter defenses are used to protect the force, hold specific terrain, or protect a key installation from destruction. The brigade uses this defensive technique when it must hold critical terrain and is not tied in with adjacent units, often employing infantry. Common situations for the use of perimeter defense includes:

a Defense of assembly areas.

b Defense of specific positions or key terrain.

c When the unit has been bypassed or isolated by the enemy.

d When the unit is in an airhead or lodgement.

0442. While a BP can allow some penetration, a perimeter cannot. Subordinate battalions are assigned to defend specific portions of the perimeter. Flanks of all units are tied in to provide mutual support. If the perimeter is penetrated, the reserve blocks the penetration or counterattacks to restore the perimeter. The brigade plans positions in depth. Tank supports the brigade by providing platoons to the battalions that have enemy mounted avenues of approach into their portion of the perimeter. The remainder of the tank company can be retained as a reserve. Figure 4.8 shows how a brigade can use a supporting tank company in a perimeter defense.

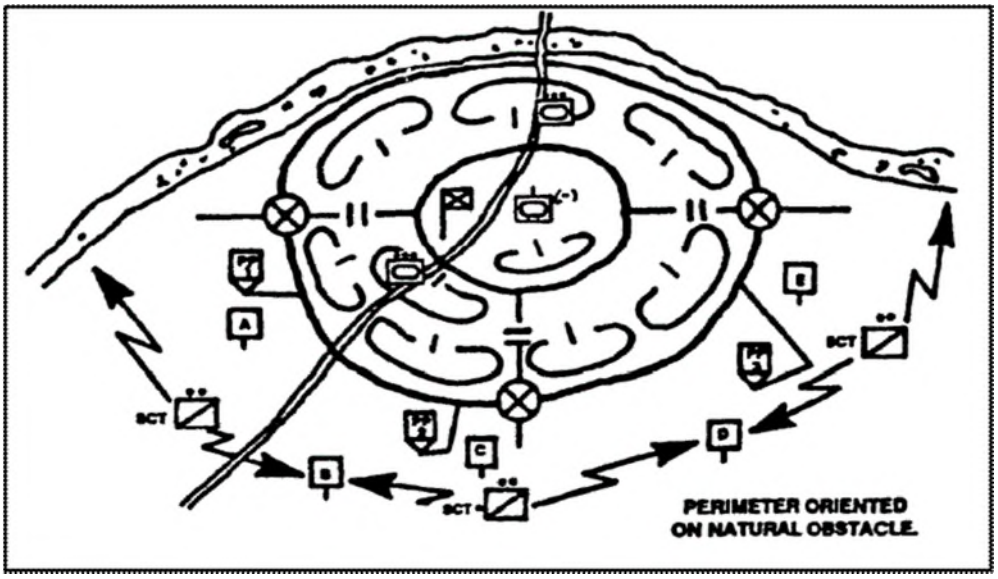


Figure 4.8 – Tank Company supporting a perimeter defense.

Infantry Anti-armor Defense

0443. This defensive technique is normally used against an enemy tank threat in restrictive terrain. It is characterized by decentralized execution of detailed anti-armor ambushes and obstacles. This technique prevents the attacker from focusing his full combat power at one point. Its purpose is to destroy enemy forces through a series of anti-armor ambushes. It can be used to:

a. Deny the enemy that uses of a trail or road network in an area of restrictive terrain.

b. Deny a choke point to the enemy. Based on the situation, this may be a mountain pass, a bridge crossing, or a highway through wooded terrain.

c. Defend against a motorized enemy force that is attempting a move through restrictive terrain.

d. Force the enemy into an EA allowing the unit to mass fires.

0444. This defense allows for planned penetration, ambushes, and counterattacks throughout the enemy formation. It is the most offensively oriented defensive technique that infantry can employ. The brigade assigns sectors to the battalions. The commanders analyze their sectors and organize to make maximum use of dispersed small-unit tactics (down to squad level) to attack the enemy throughout the depth of his formations. The primary focus of this technique is on the enemy force. Indirect fires and obstacles are tied into the tactical plan to slow and/or stop the enemy in the EA. An aggressive patrol plan is used to provide security, report information, and harass the enemy in an effort to confuse him as to the location of the main defense.

0445. An example of a tank company supporting a brigade anti-armor defense is in Figure 4.9. The tank company can be task organized to support the battalion’s sectors

or retained as a reserve and counterattack force to react to penetrations or exploit success. Tank is used:

a. When a motorized enemy is going to attempt a move through restrictive terrain.

b. To maintain a mobile reserve to exploit success.

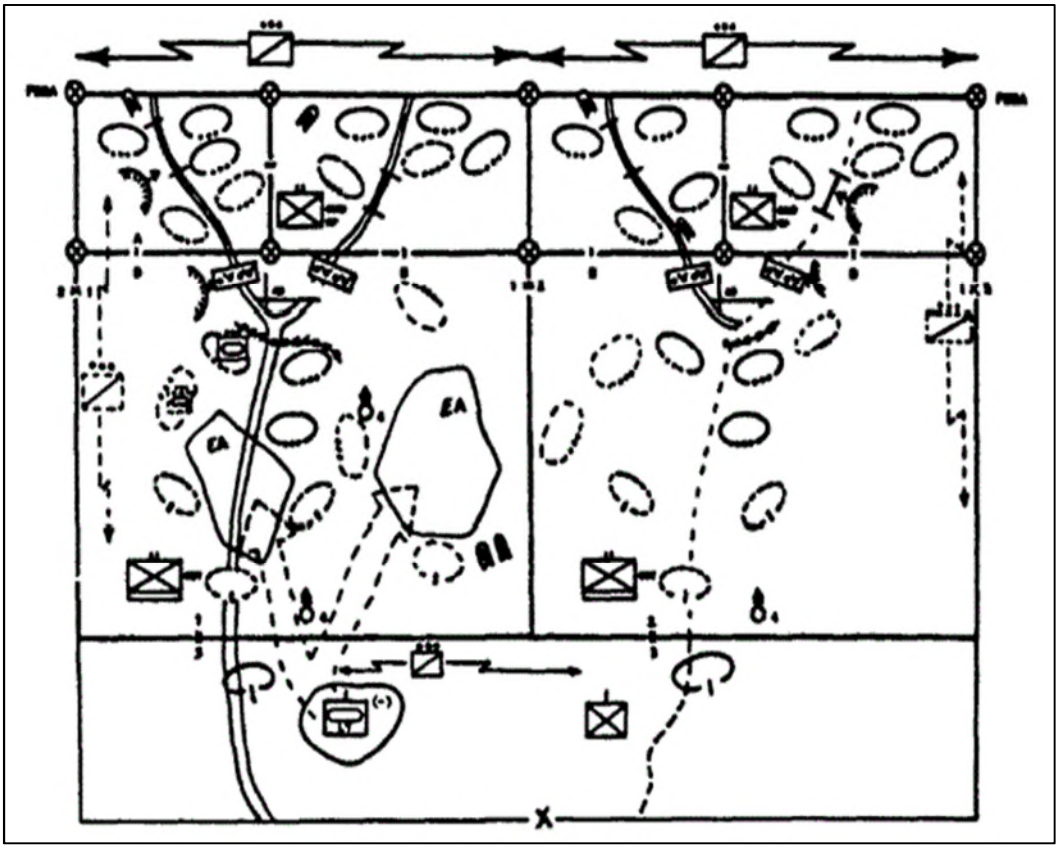


Figure 4.9 – Infantry brigade anti-armor defense.

Defending Built-up Areas

0446. Infantry commanders should consider the following when integrating tanks into the urban defense plan:

a. Avoid combat within BUAs when feasible.

b. Use restrictive missions and detailed control measures to facilitate decentralized execution and prevent fratricide.

c. Provide priorities and deadlines for the accomplishment of assigned tasks.

d. Attach CS and CSS units to the lowest level possible.

e. Employ the combined arm team to maximize individual unit capabilities.

f. Include tank fires in the obstacle plan and counterattack plan.

g. Tank unit leaders must be informed of patrol plans to prevent firing on friendly units. Extensive infantry patrolling is conducted to prevent enemy infiltration. Long-and short-range recognition signals should be incorporated into the security plan.

h. Some Tanks may be held in reserve to counterattack enemy attempts to envelop the town while others provide direct FS for the infantry defense.

i. Use small BUAs as obstacles when they add significant advantage to the direct fire plan.

j. If terrain permits, establish tank defenses as far forward of a BUA as possible.

k. Integrate adjacent terrain into the defense.

l. Use Tanks to assist security forces in limiting enemy ground reconnaissance and infiltration.

m. Tank unit leaders must know what passive resistance measures have been taken along the enemy's avenue of approach, such as removing route indicators and minefield markers, weakening of bridges and culverts. Tank commanders must also know these plans. Failure to know what passive measures have been taken may cause unnecessary mistakes and/or injury to tank crewmen.

0447. The utility of Tanks in defensive street fighting is limited. Streets and alleys provide restricted fields of fire. Restricted observation and the proximity of friendly troops to enemy target the limit use of tank fires. Normally, only security elements are left in the town proper. The defenses are usually concentrated on favorable ground around it. When the town occupies the dominating terrain in the vicinity, however, it may be organized as a key part of the BP or strongpoint. When the town is organized as a defensive position, the tank commanders select primary, alternate, and supplementary positions. Because observation in any BUA is greatly restricted, OPs should be set up and communications improvised between them and the tanks. The OPs should not be placed in steeples, prominent towers, or other obvious locations that the enemy is likely to suspect and take under fire.

0448. Tank units are employed as the defense of an organized BP. A small, well-organized, determined force, defending in a BUA can hold off a much larger attack for longer periods of time. Strongly constructed cities give the defender a decided advantage. Each building or group of buildings is a potential strongpoint.
When reconnoitering for covered routes of advance and withdrawal, tank unit commanders should not overlook the possibility of moving through ground floor lobbies and corridors of the larger buildings. This type of route requires careful marking, but has the advantage of being largely concealed from aerial observation. A careful reconnaissance, made with engineer assistance if possible, is necessary to determine whether the floors will support the tanks.

Other Defensive Tasks

0449. **Reserve Force.** Early in the planning stage, the commander should make important decisions concerning the size, composition, and mission of the reserve. The primary purpose of the reserve is to retain flexibility, reinforce success, or regain the initiative. The tank company must be able to accomplish one or more of the following subsequent missions:

a Counterattack.

b Reinforce a BP, sector, or strongpoint.

c Assume the mission of another unit.

0450. When employed in a positional defense, such as perimeter defense or a BP, reserves can be used to conduct attacks against enemy penetrations by striking a decisive blow against an uncovered enemy flank. Additionally, the enemy's attack fail, reserves could be used to reinforce success. When assigned the reserve mission, the tank company:

a Occupies the reserve position (BP or assembly area).

b Reconnoiters the sector or BPs, concentrating on areas being considered for subsequent missions.

c Rehearses likely subsequent missions.

d Maintains the appropriate readiness condition to react promptly to on-order missions.

e Uses deception to reduce the enemy's ability to locate the tanks.

0451. **Counterattack Force.** The company may participate in a counterattack to exploit an existing enemy weakness in the company's AO. An element counterattacks to:

a. Destroy enemy units.

b. Regain freedom of maneuver.

c. Regain the initiative.

d. Regain key terrain.

e. Relieve pressure on an engaged unit.

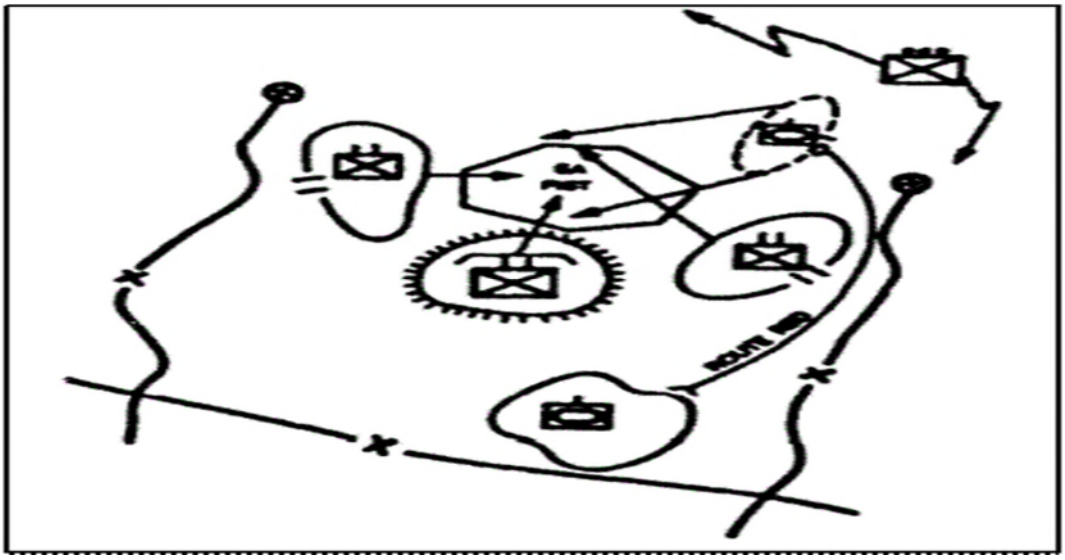


Figure 4.10 – Counter-attack by fire.

0452. **Counterattack by fire.** A counterattack by fire is executed to complete the destruction exposed of enemy elements, to free decisively engaged elements, and to regain the initiative. The company executes a counterattack by fire by moving on a concealed route to a predetermined BP from which it can engage the enemy in the flank and/or rear while the remaining units hold their positions and continue to engage and maintain contact with the enemy (see Figure 4.10). When necessary, the company commander requests permission to maneuver outside the boundaries of the predetermined BP; he accomplishes this through prior planning, coordination with the commander, or an immediate request. If this maneuver influences another unit's mission, the company commander is responsible for coordination within the unit.

0453. **Counterattack by fire and maneuver**. A company normally conducts a counterattack by fire and maneuver to destroy the remaining enemy completely, to relieve pressure on a friendly unit, or to regain key terrain. The counterattack force attacks the enemy from the flank whenever possible, using fire and movement to overwhelm and destroy him.
The company conducts the counterattack by fire and movement in a manner similar to a hasty attack (see Figure 4.11).

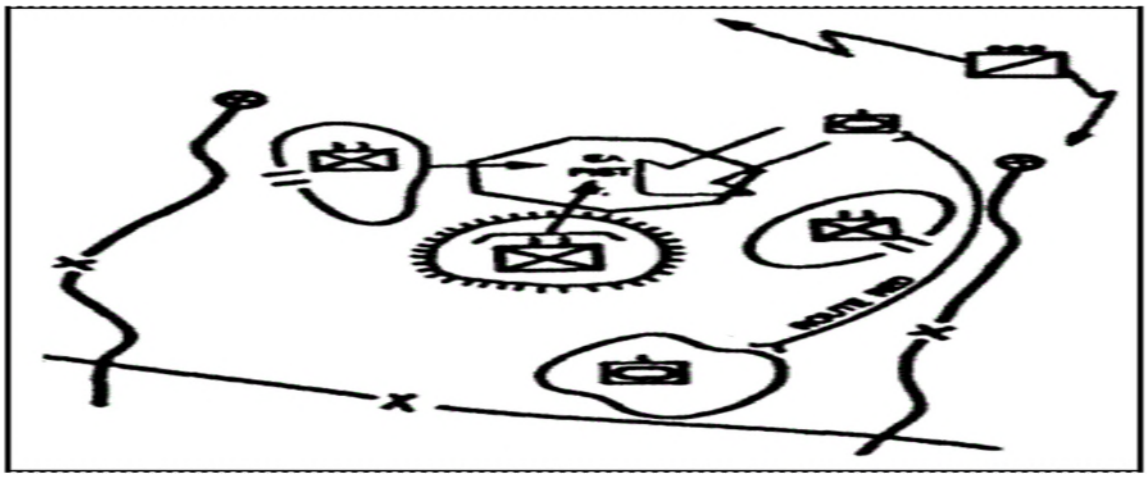


Figure 4.11 – Counter-attack by fire and maneuver.

Tank Company Defense during Limited Visibility

0454. Tank companies can expect to conduct defensive missions in limited visibility whiles supporting infantry. The enemy will attempt to take advantage of the conditions to bypass or get close to defensive positions without being detected. The fundamentals of defensive operations do not change. Infantry units will take special measures during limited visibility, such as increasing the number of patrols between positions. Depending on the situation, commanders can use the following techniques during limited visibility:

a Use tank thermal sights for detection on defined avenues of approach. Integrate their use into the company fire plan.

b Reposition weapons to concentrate on the avenues of approach that the enemy is likely to use during limited visibility.

c Plan and rehearse required movement of units and massing of fires.

d Plan to reposition weapons to compensate for the disparity between day and night acquisitions ranges.

e Reconnoiter the limited visibility positions, mark them, and mark the routes to them.

f Strengthen security with additional OPs and turret personnel.

g Enforce noise and light discipline.

Tank Platoon in Defensive Operations

0455. This section describes the most common defensive missions conducted by the tank platoon while operating with infantry during defensive operations. The tank platoon operates with an infantry battalion in closed terrain lacking in long-range observation and fields of fire. The tank platoon, therefore, may not always be able to defend from BPs. Typically, the tank platoon is used in the infantry defense as a mobile reserve and counterattack unit in the sector or anti-tank defense. Platoon BPs are used in strongpoint and perimeter defenses or in sector defenses when enemy tank or motorized avenues of approach exist.

Direct Fire Planning

0456. The platoon defensive fire plan enables the platoon leader to distribute and control fires in support of the commander’s defensive concept. It assists the commander in preparing his fire plan. If part of the platoon area is threatened, the platoon leader can use the fire plan to determine which weapon can cover the threatened area. Using radio or SOP signals, he can direct fires to destroy the enemy. To develop a defensive fire plan, the platoon leader must:

a Decide where to engage the enemy.

b Assign a location for vehicles and sectors of fire. A sector of fire is an area where a tank has primary responsibility for acquiring and engaging the enemy. Sectors of fire should overlap between individual vehicles and with adjacent elements on the platoon's flanks.

c Designate TRPs and recommend indirect fire targets in the platoon sector. The company or battalion FSO assigns numbers to the indirect fire targets.

d Coordinate with adjacent units.

e Evaluate information from his vehicle commanders to determine if they can effectively observe and engage targets and TRPs within their sectors. Vehicle commanders prepare a sketch card for each position and give a copy to the platoon leader.

f Develop a sketch of the platoon's sector, with a list of direct fire engagements and a legend, for all primary, alternate and supplementary firing positions (see Figure 4.12). The sketch should include— 1) The platoon sector 2) Individual vehicle positions. 3) Ops 4) TRPs and EAs 5) Obstacles 6) Indirect fire targets.

g Give a copy of the platoon fire plan to the infantry commander as well as to each of his MS commanders.

0457. The legend lists targets in the sector and tanks that can fire on those targets. It should also explain the direct fire and indirect fire graphics, as well as obstacles and barriers within the sector of responsibility. The obstacles can assist in the defense by canalizing the enemy into an EA. Standard military symbols are used to depict the obstacles and barriers, which should be covered by direct or indirect fire.

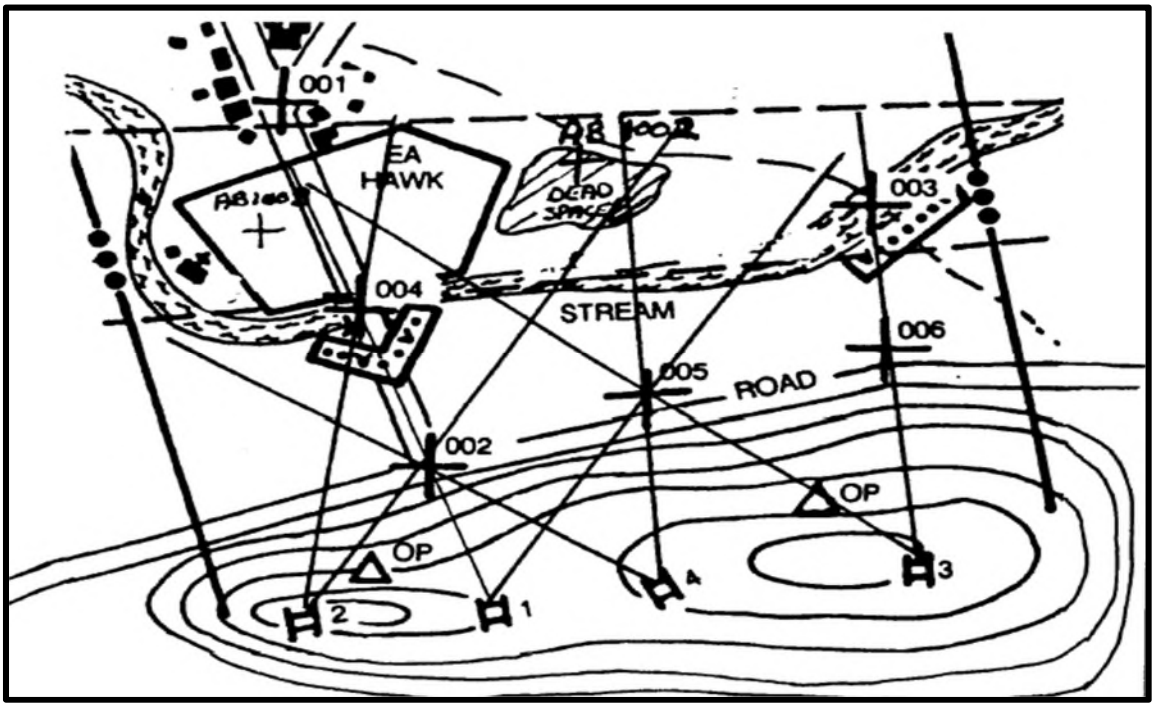


Figure 4.12 – Tank platoon sector sketch.

Defense Preparation

0458. Ideally, the tank platoon will be able to thoroughly prepare for defensive operations. It can conduct reconnaissance, position preparation, fire planning, and rehearsals. If possible, it occupies positions during limited visibility to mask preparations from the enemy.

0459. Many tasks are accomplished concurrently. The infantry battalion commander may give priority to specific tasks based on his defensive plan. He may also need some advice for the employment of the tank platoon. The tank platoon leader is normally the senior tank advisor to the infantry battalion commander. The following are the areas that the tank platoon leader should consider when preparing for the defense. This list also includes issues the infantry battalion commander may ask the platoon leader about.

0460. **Establish Security**. The tank platoon cannot provide its own security in close terrain. Mobility is a major factor in the survivability of the platoon. If the platoon must remain stationary to prepare defensive positions or in reserve, infantry is needed to secure the immediate area. Tank crewmen are vulnerable to sniper attack when placed in closed terrain since the crewmen must usually unbutton to enhance visibility. The three-man crew characteristic of the tanks considerably limits the platoon’s ability to provide local security. Enemy infantry will use stealth, darkness, and restrictive terrain to maneuver around tanked vehicles; the enemy will then attack the vehicles with flank and rear shots using conventional antitank weapons as well as satchel charges, Molotov cocktails, and termite grenades. In continuous operations, a dismounted unit must be augmented to facilitate local security.

0461. **Select Tank Positions**. The platoon leader will be the primary planner in choosing individual tank positions. The platoon leader must understand the infantry commander's defensive plan and knows what his targets are. He selects the location where he wants to kill the enemy and positions his tanks where they can support that location.

0462. **Prioritize Targets**. The tank platoon leader will most likely be responsible for executing a priority target if his position overmatches an obstacle, mounted avenue of approach, or other key point of the defense.

0463. **Disseminate the Final Protective Fires (FPF).** The platoon leader must disseminate the rest of the platoon, the location of indirect targets and the FPF. This is important because the tank platoon will probably have a key role in a counterattack by fire and/or maneuver.

0464. **Distribute the Sector Sketch**. The platoon sector sketch, with individual tank sector sketches, must be compiled and given to the infantry commander (company or battalion) and to adjacent units.

0465. **Clear Fields of Fire.** If clear fields of fire do not exist, the platoon may be required to conduct some clearing on its own, or it may be assisted by engineers or infantry.

0466. **Understand the Obstacle Plan**. Tank platoons will normally not employ or plan the locations of obstacles, but they must have an understanding of the battalion commander's obstacle plan. The tank's long range and lethal weapon system makes it effective in overmatching obstacles (such as during a displacement) and preventing enemy breaching operations.

0467. **Prepare Fighting Positions**. If engineer support is from division engineers, digging assets may be limited, and tank platoons may not be provided with standard fighting positions. Innovation may be necessary in constructing hasty fighting or hide positions; tank may have to use existing terrain, buildings, or other means of protection. Army engineer assets, when available, can construct two-tier fighting positions under the supervision of each tank commander.

0468. **Establish Wire Communications**. Wire communications are maximized during the defense. Radio traffic must be minimized. Each vehicle should be linked by wire, and the platoon should be linked to its higher headquarters by wire.

0469. **Stock Forward Supply Points**. The platoon leader should discuss the resupply plan with the battalion supply officer (S4) to ensure that he can receive critical ammunition and fuel when needed. Caches of bulky supply items typically in tank units should be constructed. The limited transportation assets organic to infantry battalions magnify the need to pre-stock during the preparation phase. The tank platoon may have to help transport critical Class IV and V supplies from the caches to forward supply points or obstacle emplacement sites. Forward supply points should not be directly in front of fighting positions.

0470. **Reconnoiter Movement and Counterattack Routes**. It is critical that the platoon leader, and each tank commander, if possible, reconnoiter the route(s) established for movement or counterattack. Closed terrain may require accompanying infantry for security during movement. Heavy vegetation or defiles may slow the platoon's ability to execute rapid movement to a counterattack position. The leaders should rehearse the movement and determine the time required for the move.

0471. **Prepare Alternate and Supplementary Positions**. Tanks are usually the priority target of the enemy. Platoon leaders must identify alternate and supplementary positions to move when receiving fire or covering other portions of an engagement area.

0472. **Initiate Deception Measures.** Because the tank is a valuable asset in the infantry defense, the enemy will seek to identify these vehicles early. An ambitious deception plan may be needed to deny the enemy information relating to the size, location, and disposition of the tank unit. Techniques for deception includes the use of decoy vehicles or sound effects (Psyops units have loudspeaker sections that can be easily prepared for this purpose) to disguise the actual positions or movement of the tanks. The visual and audible signatures of the tank make it easy for the enemy to identify and locate in an environment that is dominated by dismounted units. Leaders, however, can use this to their advantage as a deceptive measure during the preparation phase by positioning tanks in different areas of the defense.

Defense of a Battle Position

0473. When the terrain gives friendly units the opportunity to defend from a BP, the tank platoon uses the same defense as a tank platoon to control fires and movement. This defense is designed to concentrate direct fires at critical places and times to take advantage of available terrain. The infantry battalion commander will assign the tank platoon a BP when it dominates a tank or motorized avenue of approach. The commander specifies critical tasks for platoons defending from BPs. A minimum level of preparation is assigned at each BP (occupy, prepare, or reconnoiter) to enable the platoon to accomplish its mission. The platoon orients its weapon systems in an enemy avenue of approach using designated EAs and sectors of fire established by TRPs. Other tasks may include:

a Destroying a specified enemy unit, such as an enemy motorized company, from the BP.

b Controlling key terrain or blocking an avenue of approach by holding the BP against a determined assault.

c Reorienting weapon systems on a secondary or flanking avenue of approach from supplemental positions.

d Disengaging and moving to a subsequent BP when the enemy has passed a TRP or EA (break point) with a unit of specified size.

e Assisting in any other task necessary to accomplish the infantry's mission (for example, assisting in passage of lines or reaming contact points).

0474. The platoon positions its elements and maneuvers freely within the limits of the BP to accomplish the commander's intent. Engagement and disengagement criteria are included in the OPORD and must be understood by tank commanders. If the platoon leader must position elements outside the BP to make better use of terrain, increase dispersion, or maximize firepower, he should coordinate locations with the commander (see Figure 4.13).

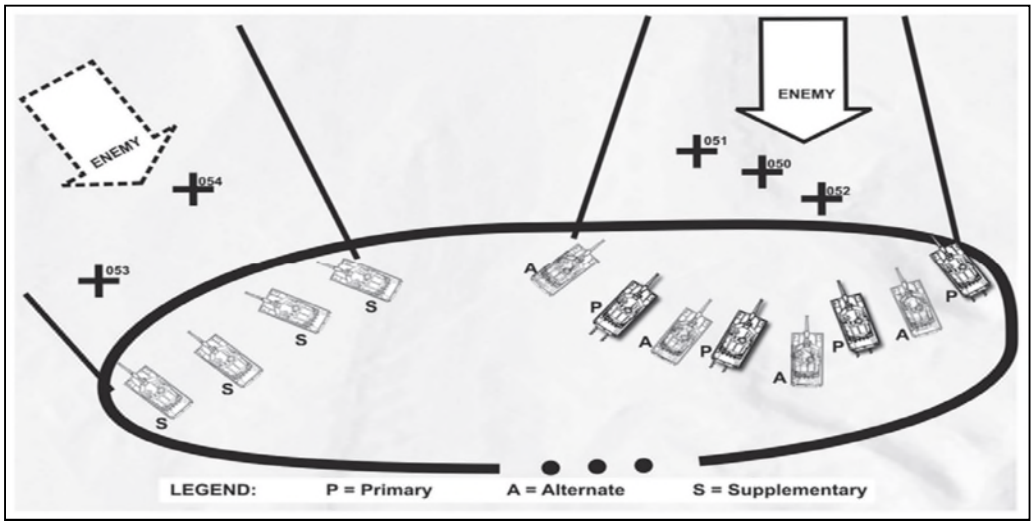


Figure 4.13 - Tank platoon in a battle position.

Defense in Sector

0475. The purpose of a defense in sector is to destroy the enemy unit forward of a rear boundary. A tank platoon is not usually assigned a defense in sector mission, when defending as part of an infantry BCT/TF/Bde. A platoon is assigned a defend in sectors mission as part of a tank company team when:

a The enemy situation is vague.

b Multiple avenues of approach in company areas of responsibility cannot be covered by mutually supporting platoon BPs.

c Retention of terrain is not critical to success of the defense.

d Maximum flexibility is desired.

0476. When defending in sector, the platoon uses mutually supporting fires to destroy the enemy, moves and concentrates fires to disrupt and destroy enemy formations, and counterattacks as needed to accomplish the mission. The platoon leader must correctly identify potential enemy avenues of approach entering his sector from the front, flanks, and rear. As in planning a BP, the company team commander selects tentative weapon positions to cover these avenues with fire and observation. He allocates space to platoons, giving them BPs, sectors, or a combination of both. If absolutely necessary, his task organizes platoons based on his estimate. In defending the sector operation, the anti-tank defense allows for planned penetrations, counterattacks, and ambushes throughout the enemy formation. The anti-tank defense is used against a superior tanks or motorized enemy in close terrain. It is the most offensively oriented defensive technique that infantry can employ. The battalion is assigned a sector by brigade. The battalion commander analyzes his sector according to the estimate process. He may assign some companies sectors and some BPs. The sector is organized to make maximum use of dispersed small-unit tactics (down to squad level) to attack the enemy throughout the depth of his formations. The primary focus of the technique is on the enemy unit; it prevents the attacker from focusing full combat power at one point. Its goal is to reduce enemy units by attrition with a series of anti-tank ambushes. The tank platoon can be used in this mission to concentrate firepower or as a mobile reserve to exploit success or defeat enemy penetrations. Figure 4.14 shows an example of an infantry battalion anti-tank defense.

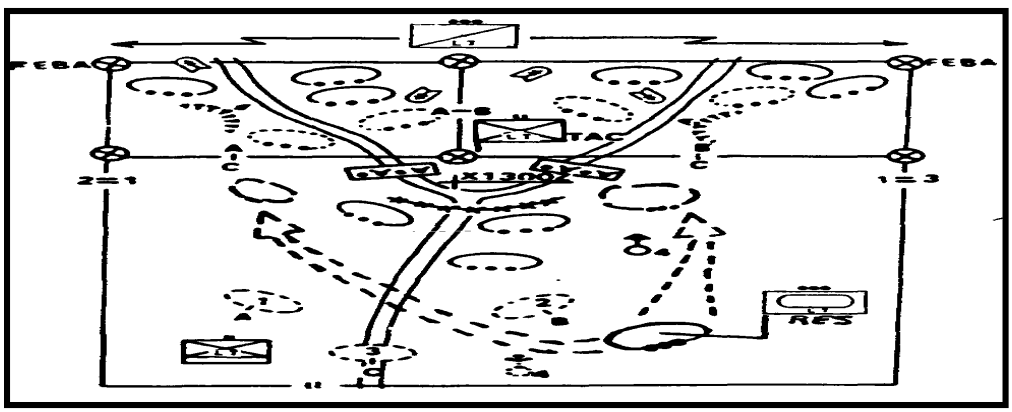


Figure 4.14 – Infantry battalion anti-tank defense.

0477. With some slight adjustments, this technique can be used:

a To deny the enemy the use of a trail or road network in an area of restrictive terrain.

b To deny a chokepoint to the enemy. This could be a mountain pass, a bridge crossing, or a highway through wooded terrain.

c To deny the passage of dismounted infantry or infiltrating guerrilla units through close terrain.

d To defeat a motorized enemy that is attempting a move through restrictive terrain.

Reverse Slope Defense

0478. A reverse slope defense is organized to use a topographical crest to mask the friendly unit from enemy observation and supporting tires. Figure 4.15 illustrates organization of this defense.

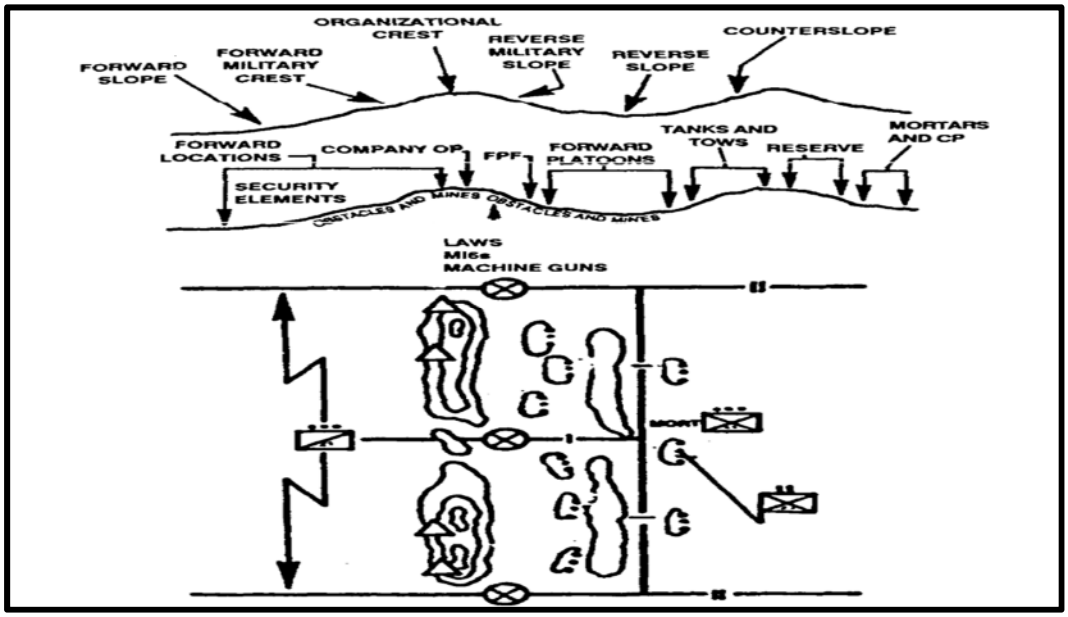


Figure 4.15 – Organization of a reverse slope defense.

0479. The battalion commander may adopt reverse slope positions for defensive elements when:

a Enemy fire or lack of cover and concealment makes occupation of the forward slope dangerous or tactically infeasible.

b The forward slope has been lost or has not yet been gained.

c The forward slope is exposed to enemy direct fires from beyond the effective range of the defender’s weapons. Moving to the reverse slope removes the attacker’s standoff advantage.

d The terrain of the reverse slope affords better fields of fire than are available on the forward slope.

e The defender must avoid creating a dangerous salient or reentrant in friendly lines.

f The commander wants to surprise the enemy or deceive him the true location of the battalion's defensive positions.

0480. Advantages of the reverse slope defense includes the following:

a Enemy ground observation of the battle area is masked, even from surveillance devices and radar.

b Enemy direct fire weapons cannot effectively fire on the position without coming within range of the defender's weapons.

c The enemy is forced to try the breach obstacles on the reverse slope within direct fire range of the defender's weapons. The attacker cannot locate these obstacles until he runs into them.

d The enemy can be deceived as to friendly strength and location.

e Enemy indirect fire is less effective.

f The defender gains tactical surprise.

g The lack of enemy ground observation allows more freedom of movement within the battle area.

h If positions are properly positioned, tanks, artillery, and anti-tank can mass fires on the reverse military crest; infantry small arms can contribute their close fires to the battle.

i The unit can dig in more quickly, even when enemy ground units are approaching, because the slope of the hill covers and conceals the unit from direct tire and observation. Defenders can make more thorough position preparations.

j The terrain protects the unit from the blast and thermal effects of enemy and friendly nuclear weapons.

0481. Disadvantages of a reverse slope defense includes the following:

a Observation of the enemy may be limited, and the defender may be unable to cover obstacles in front with direct fires.

b The range of vital direct fire weapons such as tanks and anti-tank may be limited by the topographical crest; they also have to be positioned away from the infantry to exploit their range.

c The enemy will be able to attack downhill from high ground, while a friendly counterattack will be uphill. This may provide a psychological advantage to the enemy.

d Effectiveness of this defense is reduced in limited visibility because the reverse military crest must be controlled.

0482. The battalion commander organizes the reverse slope defensive position in accordance with procedures and considerations that apply to all defensive techniques. The forward edge of the position should be within small arms range of the crest; however, it should be far enough from the crest that fields of fire allow the defender time to place well-aimed fire on the enemy before he reaches friendly positions. The reverse slope position is most effective when the forward slope can be covered by flanking fires from units on adjacent terrain. A security unit should be established in front of the position to stop or delay the enemy, disorganize his attack, and deceive him as to the location of the position. When the security element is withdrawn, indirect fire, and security must be maintained in front. OPs are established on or forward of the topographical crest. This allows long-range observation over the entire front and makes it possible to cover forward obstacles with indirect fires. OPs are usually provided by the reserve; they may vary in size from a few soldiers to a reinforced squad. They should include forward observers. At night, the number of OPs should be increased to improve security. Conduct of the reverse slope defense closely parallels that forward slope defense. Tanks may be positioned first on the forward slope to engage the enemy at long ranges. As the enemy nears, they move to positions on the reverse slope or on the forward slope of the next hill to the rear (counter slope).

0483. The reverse slope defense can be adapted to fit a particular tactical situation based on the factors of METT-TC; examples are illustrated in Figure 4.16. Possible adaptations include the following:

a Firing positions are prepared on or forward in the topographical crest when the commander wants to use the fields of fire afforded by the forward slope. Most personnel remain on the reverse slope to reduce their exposure to fire; only a skeleton unit is kept forward to slow the attacker while the remainder of the friendly unit occupies reverse slope fighting positions. Reserves (such as a tank platoon) are held in covered positions. These units are used for counterattacks around the flanks of the hill.

b The enemy may be denied the hill or suffer high casualties fighting for it even if neither the forward slope nor the reverse slope is suitable for a BP. The defender can engage the enemy in the reverse slope from positions on other hills. Mortars, artillery, and long-range machine gun fires are targeted on the reverse slope, the crest, and the forward slope. Positioning elements on flanking hills often allows grazing machine gunfire against otherwise protected areas just over the crest.

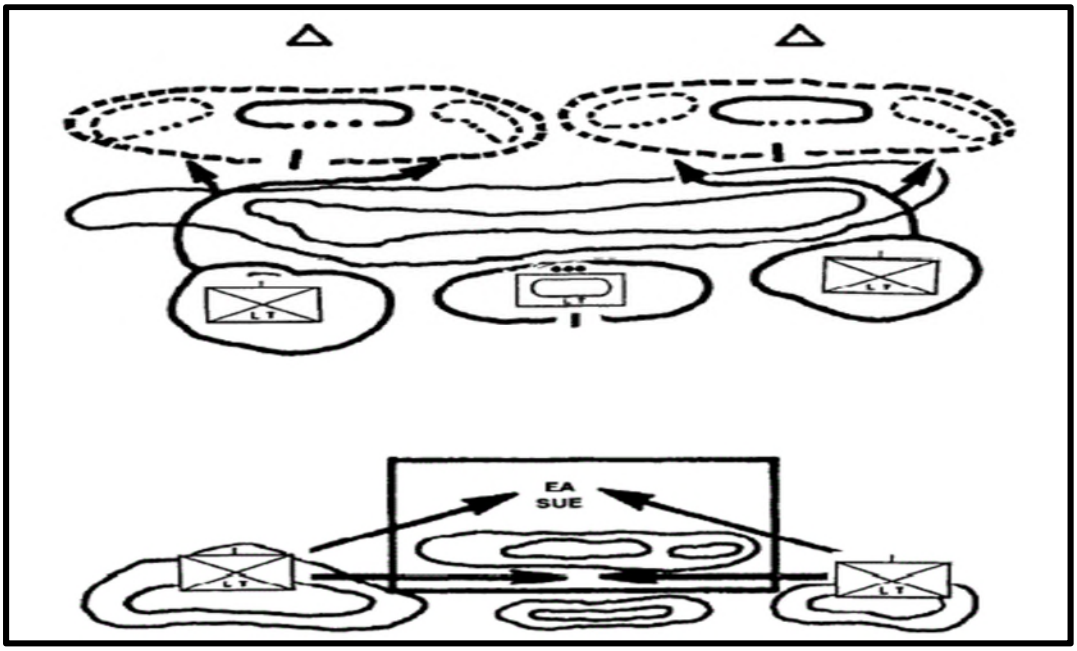


Figure 4.16 – Adaptations of the reverse slope defense.

Perimeter Defense

0484. This is conducted in the same manner as defense of a BP except the perimeter defense orients on 360 degrees. The perimeter defense is often used as an infantry technique. The tank platoon participates in the perimeter defense as part of the infantry BCT/TF/Bde defense. Perimeters may be used to defend:

a Assembly areas

b Specific installations or equipment (TOC, downed aircraft, bridges, airfields, road blocks).

c Key terrain (bridge, hilltop, pickup zone, landing zone, lodgment area).

d As part of a brigade perimeter, airhead, or lodgment.

0485. While a BP can allow some penetration, a perimeter cannot. Perimeter defenses are used to protect the unit, hold specific terrain, or protect a key installation from destruction. Flanks of all units are tied in, to provide mutual support and positions are planned in depth. If the perimeter is penetrated, the tank platoon can be used to block the penetration or to counterattack to restore the perimeter. Tanks are positioned in the most likely enemy mounted avenues of approach. Mortars are usually positioned in the center of the perimeter and can fire 360 degrees. Patrols are used to provide security to the tank platoon. Using the tank platoon in the perimeter defense provides flexibility to counterattack in any direction, allows occupation and control of a specific area, and provides ease of control.

Strongpoint Defense

0486. A strongpoint is a defensive position that is fortified as extensively as time and materials allow. It is used to hold key terrain critical for the defense, to provide a pivot for the maneuver of friendly units, and to canalize the enemy into friendly EAs. A strongpoint may be part of any defensive plan. It may be built to protect vital units or installations, as an anchor or an anvil around which tank units maneuver, or as part of a trap designed to destroy enemy units that attack. It may be in an urban area or in a wilderness. A strongpoint is attacked at the risk of high casualties. It cannot easily be overrun or bypassed. It is tied in with existing obstacles, forcing the enemy to reduce it by dismounted assaults and massive artillery and tactical air concentrations.

0487. While the size and type of unit selected to execute a strongpoint defense will vary according to the situation, an infantry unit is normally used, with tank retained for mobility. A unit required to defend a strongpoint, will need a significant amount of time and engineer resources to construct the position. Defense of a strongpoint is sometimes an extension of a defense of a BP. Depending on the commander's intent, BPs can be developed into strong points if time, terrain, and resources allow.

0488. The strongpoint must be planned so that it can be reduced only with the expenditure of overwhelming units and much time. Each primary, alternate, and supplemental position must be dug in. Positions should be connected by tunnels or trenches if time permits. Each individual vehicle position must be connected by wire to the platoon leader's position, and the platoon and section leaders' positions must be wired into the infantry commander's position. The wire must be dug in to protect it from enemy indirect fires. This can be accomplished by laying wire through sewers and tunnels or by burying it.

0489. Direct fire plans should provide mutual support and overlapping fires to the greatest extent possible. To reduce vulnerability, primary positions and sectors of fire are augmented by alternate and supplemental position in sectors of fire.

0490. Because of the nature of the operation, strongpoints are located in restrictive terrain, such as urban areas, mountains, and thick forests that cannot be easily bypassed. Since the unit must prevent the enemy from bypassing or reducing the strongpoint, priority tasks for engineers are counter mobility and survivability.

0491. Infantry units may be directed to construct a strongpoint as part of a larger overall defensive plan. They must be augmented with extensive engineer support, additional key weapon systems, pioneer tools, additional transportation assets, and CSS resources. To offset some of the support requirements, the commander may decide to take advantage of an existing obstacle, such as a town or village, to reduce the time required to develop a strongpoint.

0492. The following critical aspects of the strongpoint defense should be incorporated into the overall plan:

a Covered and concealed routes are constructed or planned between positions, along routes of supply and communication, and to support counterattacks and maneuver within the strongpoint.

b Food, water, ammunition, pioneer tools, and medical supplies are stockpiled in each fighting position.

c The strongpoint is divided into several independent but mutually supporting positions or sectors. If one of the positions or sectors must be evacuated or is overrun, obstacles and fires limit the enemy penetration and support a counterattack by tanks.

d Obstacles and minefield are constructed to disrupt and canalize enemy formations, to reinforce fires, and to protect the strongpoint from assault. The obstacles and mines are placed as far out as friendly units can observe and cover with fire, within the strongpoint itself, behind the strongpoint, and at points in between where they will be useful.

e Several means of communication within the strongpoint and with higher headquarters are planned and tested. These include radio, wire, messenger, pyrotechnics, and other signals. The strongpoint is improved or repaired until the unit is relieved or withdrawn. Additional positions can be built, tunnels and trenches dug, existing positions improved or repaired, and barriers built or fixed.

f The strongpoint position itself must be an obstacle to enemy mounted movement.

Defense in Built-up Areas

0493. A small, well-organized, determined unit defending a BUA can hold off a much larger attacker for long periods of time. Strongly constructed cities give the defender a decided advantage. Each building or group of buildings is a potential strongpoint for infantry.

0494. The infantry battalion combines BUA techniques with the elastic and strongpoint defensive techniques. The urban defense may unit the enemy into planned EAs that are covered by anti-tank or tank platoon BPs. Forward BPs or covering units may be employed to disorganize and confuse the enemy as to the main defenses. Behind the BPs, defenses are set up to protect the friendly support element and stop the deepest penetration of the attacking unit. Tank may be located within this defense, on the flanks in BPs, or as a mobile counterattack unit.

0495. Infantry battalions are employed in the urban defense to block the penetration of the enemy and to protect friendly logistics elements. The keys to success in this defense are surprise, effective use of terrain, protection, and coordinated massing of fires. An example of an infantry urban defense with tanks is shown in Figure 4.17.

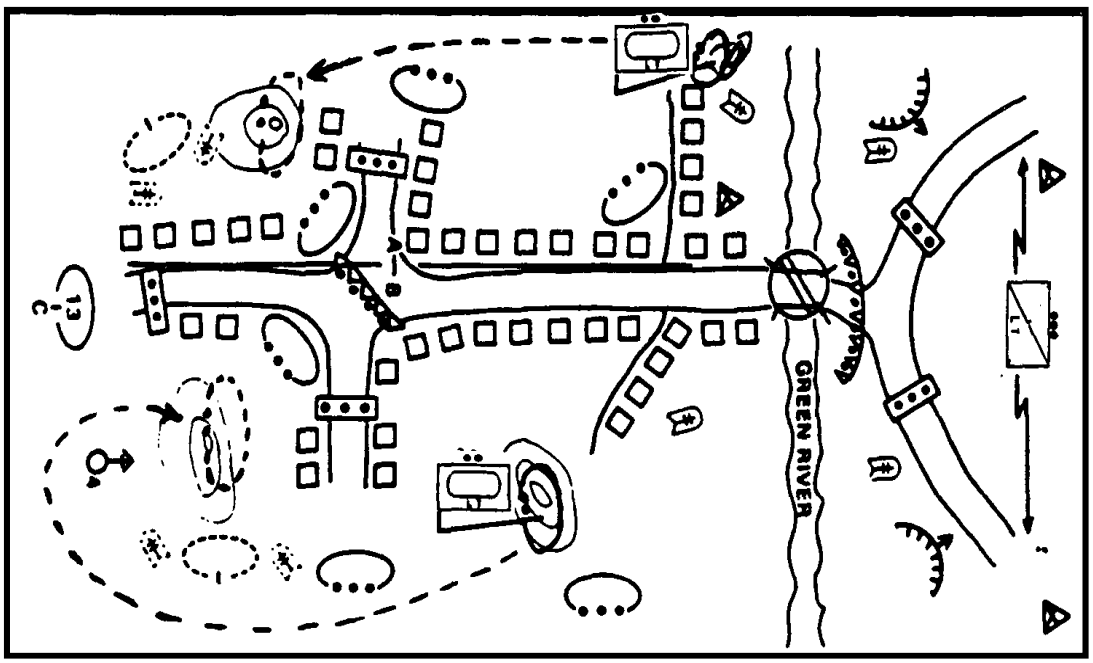


Figure 4.17 – Example of infantry urban defense with tanks.

0496. Infantry commanders should consider the following when integrating tank into the urban defense plan:

a Tank BPs must be positioned to mass their fires on the enemy from multiple directions to maximize destructive capability while minimizing vulnerability to enemy attacks.

b Tank fires must be included in the obstacle plan and counterattack plan.

c Use destroyed BUAs as obstacles when they give significant advantage to the direct fire plan.

d If terrain permits, tank defenses can be established forward of a BUA.

e Adjacent terrain can be used to integrate tanks into the defense.

f Use tank to assist security units in limiting enemy ground reconnaissance and infiltration.

g Use obstacles, mines, and antitank augmentation to prevent enemy tank penetration of BPs.

h Use restrictive missions and detailed control measures to facilitate decentralized execution and prevent fratricide.

i Tank leaders must know what passive resistance measures have been taken along the enemy's avenue of approach. Passive resistance includes removing route indicators and minefield markers and weakening bridges and culverts. Tank commanders must know these plans. Failure to know what passive

measures have been taken may cause unnecessary mistakes and/or injury to tank crewmen.

j Tank leaders must be aware of patrol plans to prevent firing on friendly units. Extensive infantry patrolling is conducted to prevent enemy infiltration. Long- and short-range recognition signals should be incorporated into the security plan.

k Some tanks may be held in reserve to counterattack enemy attempts to envelop the town, while others provide direct FS for the infantry defense.

l LOCs must be controlled to provide the tanks with medical evacuation and resupply of Classes I, III, and V.

0497. Use of tanks in defensive street fighting is limited. Streets and alleys provide limited fields of fire. Restricted observation and the proximity of friendly troops to enemy targets will limit the use of tank fires. However, when the town itself occupies the dominating terrain in the vicinity, it may be organized as a key part of the BP or strongpoint. When the town itself is organized as a defensive position, tank commanders select primary, alternate, and supplementary positions. Because observation in BUAs is greatly restricted, OPs should be set up and communications improvised between them and the tanks. These OPs should not be placed in steeples, prominent towers, or other obvious locations which the enemy is likely to suspect and take under fire.

0498. Tank unit commanders, when reconnoitering for covered routes in advance and withdrawal, it should not overlook the possibility of moving through ground-floor lobbies and corridors of the larger buildings. This type of route requires careful marking, but has the advantage of being largely concealed from aerial observation. A careful reconnaissance, made with engineer assistance, if possible, is necessary to determine whether the floors will support the tank.

Reserve Unit

0499. Early in the planning stage, the commander makes important decisions concerning the size, composition, and mission of the reserve. The primary purpose of the reserve is to retain flexibility, reinforce success, or regain the initiative through counterattacks. Secondary purposes of the reserve are:

a To contain or counterattack enemy units that has penetrated.

b To relieve depleted units and provide for continuous operations.

c To attack enemy units not yet in contact.

d As a last resort, to react to rear area threats.

04100. An infantry battalion conducting a deliberate attack may initially retain a tank platoon in reserve. A reserve is held to exploit success and continue an attack already under way, to maintain momentum of an attack by adding a tank unit at a critical time,

and to provide security. The reserve is an active not reactive unit and is not used to reinforce failure. When employed in a positional defense, such as perimeter defense or a BP, tank reserves can be used to conduct attacks against enemy penetrations by striking a decisive blow against an uncovered enemy flank. Additionally, if the enemy's attack failed, reserves could be used to regain initiative.

Counterattack Unit

04101. The platoon may participate in a counterattack to exploit an existing enemy weakness in the AO. A unit counterattacks to:

a Destroy enemy units.

b Regain freedom of maneuver.

c Regain the initiative.

d Regain key terrain.

e Relieve pressure on an engaged unit.

04102. A platoon executes two types of counterattacks.

a **Counterattack by Fire**. A counterattack by fire is executed to complete the destruction of exposed enemy elements, to free decisively engaged elements, and to regain the initiative (see Figure 4.18). The platoon executes a counterattack by fire by moving on a secured, concealed route to a predetermined BP from which it can engage the enemy in the flank and/or rear while other units hold their positions and continue to engage and maintain contact with the enemy. When necessary, the platoon leader requests permission to maneuver outside the predetermined BP by prior planning, coordination with the commander, or immediate request. If this maneuver influences another unit's mission, the commander is responsible for coordination in that unit.

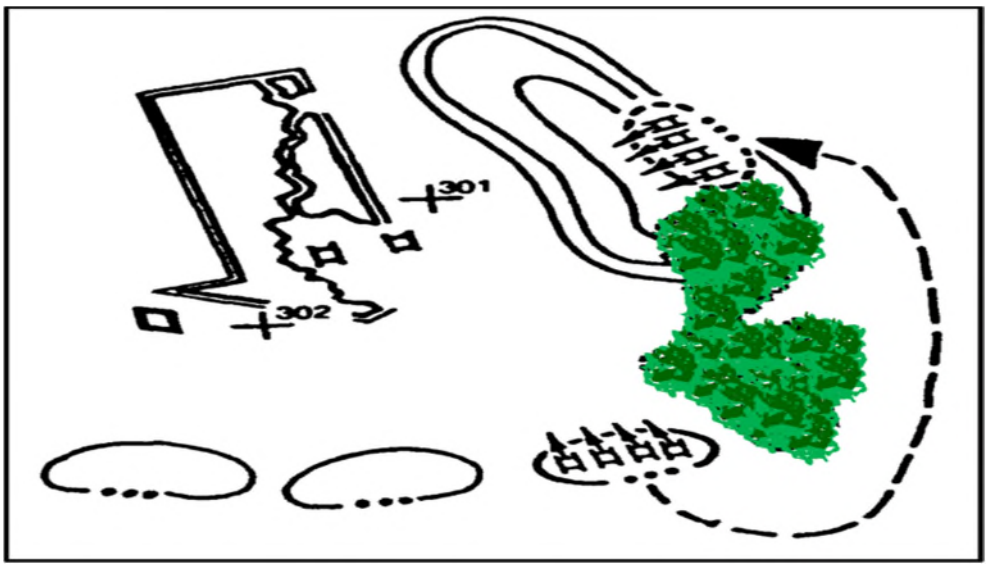


Figure 4.18 – Counter-attack by fire

b **Counterattack by Fire and Maneuver**. A company team normally uses counterattack by fire and maneuver to destroy remaining enemy elements completely, relieve pressure on a friendly unit, or regain key terrain. The counterattack unit hits the enemy on the flank if possible, using fire and maneuver to overwhelm and destroy him. A platoon counterattacks by fire and maneuver in a manner similar to a hasty attack (see Figure 4.19).

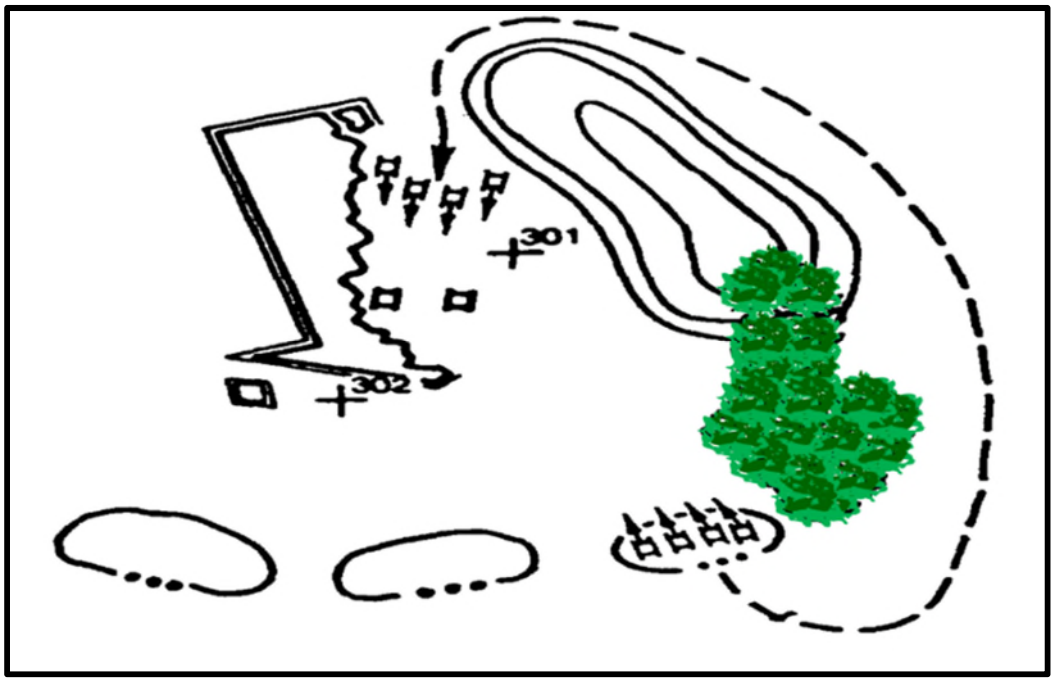


Figure 4.19 – Counter-attack by fire and maneuver

Tank Platoon Defense during Limited Visibility

04103. Although limited visibility creates opportunities for deadly close-range engagements that will achieve surprise, the possibility that the enemy may pass the platoon's position or close to point-blank range must be considered in planning the defense. The fundamentals of defensive operations do not change with limited visibility. Depending on the situation, the tank platoon should:

a Occupy dominating positions along avenues of approach when visibility is good, and reposition closer to or occupy the avenues of approach during periods of limited visibility.

b Reconnoiter the limited visibility positions, mark them, and mark the routes to them.

c Incorporate the use of night-vision devices (NVD) in the platoon fire plan and plan for indirect illumination to supplement passive image intensification or infrared (IR) night sights.

d Request additional security units, such as infantry squads, to compensate for lack of traditional organic manpower because of the tanks' three-man crews.

e Enforce noise and light discipline.

f Ambush as far forward and as many times as METT-T factors allow.

Glossary

Battle Position
Is a general location and orientation of forces on the ground from which units defend. The BP can be used for units from battalion- to platoon-size.

Combined Arms
Is the synchronized or simultaneous application of several Army-specific capabilities represented by military occupational specialties to achieve an effect on the enemy or on a situation that is greater than if each capability was used separately or in sequence.

Command
Is the authority that a commander in the armed forces lawfully exercises over subordinates by virtue of rank or assignment. Command includes the authority and responsibility for effectively using available resources and for planning the employment of, organizing, directing, coordinating, and controlling military forces for the accomplishment of assigned missions.

Command and Control (C2)
It is the exercise of authority and direction by a properly designated commander over assigned and attached forces to accomplish the mission. Command and control functions are performed through an arrangement of personnel, equipment, communications, facilities and procedures employed by a commander in planning, directing, coordinating and controlling forces and operations in the accomplishment of the mission.

Commander's Intent
Is a clear, concise statement of what the force must do and the conditions the force must establish with respect to the enemy, terrain, and civil considerations that represent the desired end state.

Contingency Operations
Involve military forces to achieve operational objectives or protect national interests. They are usually in response to a sudden or short-notice crisis or emergency and occur across the scope of operations

Control
Is the regulation of forces and warfighting functions to accomplish the mission in accordance with the commander's intent.

Coordinates
Numbers and/or letters that designate a location on a gridded map, photograph, or chart by means of a coordinate system.

Counterattack
An attack by part or all of a defending force against an enemy attacking force, for such specific purposes as regaining ground lost or cutting off or destroying enemy advance units, and with the general objective of denying to the enemy the attainment of the enemy's purpose in attacking. In sustained defensive operations, it is undertaken to restore the battle position and is directed at limited objectives.

Defensive Operations
Are combat operations conducted to defeat an enemy attack, gain time, economize forces, and develop conditions favorable for offensive or support to civil governance operations.

Depth
Is the extension of operations in time, space, and resources. Commanders use depth to obtain space for effective maneuver, time to conduct operations, and resources to achieve and exploit success.

Direction
The direction of the spotting line is an angular measurement from a reference north to an imaginary line that passes through the target. Note: It is measured in degrees (to the nearest degree) or mils (to the nearest 10 mils). The unit of measurement and reference north is understood to be mils grid, unless otherwise specified. The unit of measurement (mils or degrees) must remain the same throughout the engagement.

Dispersion
The scattered pattern of hits around the MPI of projectiles fired from one gun or battery of guns under identical conditions.

Exploitation
Is a continuation of an attack to take advantage of and maximize success. It prevents the enemy from reconstituting an organized defense or conducting an orderly withdrawal. It normally follows a successful attack that weakens or collapses enemy defenses. The exploitation focuses on the enemy and destroys his will to resist. Speed is essential

Grid Coordinates
1. Numbers of a coordinated system that designate a point on a gridded map, photograph, or chart. The term “grid” is a shortened form.
2. A target-location method where by the spotter locates his target by the actual grid coordinates of the point on a military map.

Initiative
Is setting or dictating the terms of action throughout an operation.

Insurgency
Is a condition where an organized movement aims to overthrow a constituted government through the use of subversion and armed conflict. It is a condition of politically motivated conflict involving significant intra- or interstate violence but usually short of large-scale operations by opposing forces.

Intelligence
In broad terms, intelligence is knowledge about an enemy, adversary or competitor that is not apparent or evident from the public record or by overt observation. Intelligence activities are designed to elicit information that an enemy, adversary or competitor would prefer not to disclose or to have discovered.

Logistics
The science of planning and carrying out the movement and maintenance of forces. In its most comprehensive sense, those aspects of military operations that deal with: the design and development, acquisition, storage, movement, distribution, maintenance, evacuation, and disposition of materiel; the movement, evacuation, and hospitalization of personnel; the acquisition or construction, maintenance, operation, and disposition of facilities; and, the acquisition or furnishing of services

Main Effort
Is the designated subordinate unit whose mission at a given point in time is most critical to overall mission success. It is usually weighted with the preponderance of Army power.

Movement to Contact
Is an offensive operation designed to gain contact with the enemy.

Observed Fire
Fire for which the points of impact or burst can be seen by the spotter.

Offensive Operations
Are combat operations conducted to defeat and destroy enemy forces and seize terrain, resources, and population centers.

Operational Control
Transferable command authority that may be exercised by commanders at any echelon at or below the level of combatant command. Operational control is inherent in combatant command (command authority) (COCOM). Operational control may be delegated and is the authority to perform those functions of command over subordinate forces involving organizing and employing commands and forces, assigning tasks, designating objectives, and giving authoritative direction necessary to accomplish the mission. Operational control includes authoritative direction over all aspects of military operations and joint training necessary to accomplish missions assigned to the command. Operational control should be exercised through the commanders of subordinate organizations. Normally this authority is exercised through subordinate joint force commanders and Service and/or functional component commanders. Operational control normally provides full authority to organize commands and forces and to employ those forces as the commander in operational control considers necessary to accomplish assigned missions. Operational control does not, in and of itself, include authoritative direction for logistics or matters of administration, discipline, internal organization, or unit training.

Plan
Is a design for a future or anticipated operation. This continuous, evolving frame work of anticipated actions guides subordinates through each phase of the operation. A plan is a frame work from which to adapt rather than a script to follow.

Planning
Is the process by which commanders (and the staff, if available) translate the commander’s visualization into a specific course of action for preparation and execution, focusing on the expected results.

Position Location Device
An aid to navigation and should not take the place of map and compass navigation. Leaders must continue to use map and compass as a primary means in the event position location system signals are interrupted by interference from vegetation, weather, or other masking features. There may also be times when satellites or land-based emitters are inoperative, causing lapses in signals

Rear Area
Is provided primarily for the performance of support functions and where the majority of the echelon's sustaining operations occur. Operations in rear areas assure freedom of action and continuity of operations, sustainment, and C2.

Reconnaissance
Reconnaissance complements surveillance in obtaining, by visual observation or other detection methods, specific information about the activities and resources of an enemy or potential enemy; or in securing data concerning the meteorological, hydrographic, or geographic characteristics of a particular area.

Reference point
Prominent, easily located terrain feature or a recorded target.

Retrograde operations
Are movements to the rear or away from the enemy. The movement may be forced or voluntary, but it must be with the higher commander's approval. Units conduct retrograde operations to gain time, preserve force strength, avoid combat under undesirable conditions, or draw the enemy into an unfavorable position

Rules of Engagement
Rules of Engagement (ROE) are directives to military forces and individuals that define the circumstances, conditions, degree and manner in which the use of force or other action may or may not be applied.

Search and Attack
Technique for conducting a movement to contact that shares many of the characteristics of an area security mission

Security operations
Provide information about the enemy and provide reaction time, maneuver space, and protection to the infantry. When properly task organized, augmented, and supported, tank units may be tasked to perform two primary types of security missions-screen and guard

Shift from a Known Point
A target location method whereby the spotter locates the target in Relation to a known point by means of lateral, range, and vertical shifts from the known point.

Support by Fire
A tactical mission task in which a maneuver force moves to a position where it can engage the enemy by direct fire in support of another maneuvering force.

Surveillance
Surveillance is the function of systematically observing air, surface, or subsurface areas, places, persons, or things, by visual, aural, electronic, photographic, or other means.

Synchronization
Is the arrangement of military actions in time, space, and purpose to produce maximum relative Army power at a decisive place and time.

Target
1. A geographic area, complex, or installation planned for potential engagement by military forces.
2. A spotting made by a spotter when the salvo has effect on the target.

Versatility
Is the ability of Army forces to meet the diverse mission requirements of full spectrum operations in various defense mission areas. Competence in a variety of missions and skills allows Army forces to quickly transition from one type of operation to another with minimal changes to the deployed force structure.

War
Is the sustained use of armed force between nations or organized groups within a nation; it involves regular and irregular forces in a series of connected battles and campaigns to achieve vital national objectives

Abbreviation and Acronym

AH	attack helicopter
AO	area of operations
APC	armored personnel carrier
ATGM	antitank guided missile
AVLB	armored vehicle launched bridge
Bde	brigade
BP	battle position
BUA	built-up areas
C2	command and control
CEOI	Communication and electronics operation instruction
COMMO	communications
CP	command post
CS	combat support
CSS	combat service support
CVC	combat vehicle crewman
DLIC	detachment left in contact
DS	direct support
EA	engagement area
EPW	enemy prisoner of war
FIST	fire support team
FPF	final protection fires
FRAGO	fragmentary order
FS	fire support
FSCL	fire support coordination line
FSO	fire support officer
GPS	global positioning system
GSR	ground surveillance radar
HEAT	high-explosive antitank
HHC	headquarters and headquarters company
HQ	headquarters
IR	infrared
TF	task force
JTF	joint task force
KIA	killed in action
LOA	limit of advance
LOC	lines of communication
MBA	main battle area
MTC	movement to contract
METT-T	mission, enemy, terrain, troops, and time available
METT-TC	mission, enemy, terrain troops, and time available civilians
MICLIC	mine clearing line charge
MOPP	mission-oriented protective posture
NBC	nuclear, biological, chemical
NEO	noncombatant evacuation operation
NVD	night-vision device
OBJ	objective
OPCON	operational control
OPORD	operation order

OPS	operations
OPSEC	operations security
ORP	objective rally point
PEWS	platoon early warning system
Psyops	psychological operation
REDCON	readiness condition
RFL	restrictive fire line
RP	release point
SHELREP	shelling report
SOP	standing operating procedure
SP	start point
TF	task force
TRP	target reference point
XO	executive officer